

Exploring Household Consumption Patterns and Socioeconomic Disparities: Insights from the HCES Survey (2023-24)

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1 Abstract

The Household Consumption Expenditure Survey (HCES) provides a comprehensive dataset on household spending patterns across demographic and socio-economic segments. This study leverages HCES 2023-24 data to analyze trends in Monthly Per Capita Expenditure (MPCE), regional disparities, and the effects of inflation, income distribution, and policy interventions. Findings from this study contribute to economic forecasting and policy-making by providing deeper insights into consumption behavior and welfare analysis.

Acknowledgements

I extend my heartfelt gratitude to all those who supported and guided me in the successful completion of this project titled “*Exploring Household Consumption Patterns and Socioeconomic Disparities: Insights from the HCES Survey (2023–24)*”.

I am thankful to the **Ministry of Statistics and Programme Implementation (MoSPI)**, Government of India, for offering me the opportunity to undertake this internship under the National Internship in Official Statistics (NIOS) 2025 programme. This platform enabled me to engage deeply with real-world official statistics and empirical research using the HCES 2023–24 dataset.

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Certificate

This is to certify that the internship report titled “**Exploring Household Consumption Patterns and Socioeconomic Disparities: Insights from the HCES Survey (2023–24)**” has been successfully completed and submitted by **Ayan Bashir Sheikh**, a student of M.Sc. Statistics at Savitribai Phule Pune University, under the National Internship in Official Statistics (NIOS) 2025, organized by the Ministry of Statistics and Programme Implementation (MoSPI), Government of India.

The study was carried out during the internship period from March 24, 2025 to May 24, 2025. This report is an original piece of research based on empirical analysis using the Household Consumption Expenditure Survey (HCES) 2023–24 dataset.

This certificate is issued as a formal recognition of the successful completion of the internship and submission of the project report.

(Signature)

Authorized Signatory

Ministry of Statistics and Programme Implementation (MoSPI)

Government of India

2 Introduction

The Household Consumption Expenditure Survey (HCES) is a statistical initiative designed to collect data on household consumption patterns and expenditure on goods and services. It serves as a key source for analyzing economic well-being, updating the consumer goods basket, and determining Consumer Price Index (CPI) weights. Additionally, HCES plays a crucial role in assessing welfare, poverty, and inequality, enabling policymakers to formulate and monitor economic and social policies tailored to different population segments.

The data collected in HCES provides insights into household living standards, changes in consumption behavior over time, and disparities among various socio-economic groups. Food consumption data derived from HCES is instrumental in food security and nutrition analysis, making it a valuable resource for public health and economic planning.

Following the recommendations of the National Statistical Commission (NSC), the survey was originally scheduled for 2020-21 but was postponed due to the COVID-19 pandemic. To accommodate the need for updated macroeconomic indicators such as Gross Domestic Product (GDP) and CPI, two consecutive surveys were conducted between August 2022 – July 2023 and August 2023 – July 2024. The first survey's findings were published in February 2024, followed by the release of detailed reports and unit-level data in June 2024.

3 Literature Review

Household Consumer Expenditure Surveys (HCES) play a crucial role in assessing economic conditions, informing policies, and understanding consumption patterns. Studies highlight their importance in analyzing income distribution, poverty, and household well-being.

Research has focused on improving data collection techniques, enhancing accuracy, and ensuring representativeness. The integration of technology has strengthened survey reliability, while comparisons across regions and years provide insights into economic trends and policy impacts.

HCES aligns with economic theories like Engel's Law, exploring the relationship between household expenditures and macroeconomic factors. They also contribute significantly to global development initiatives, particularly in tracking Sustainable Development Goals (SDGs).

Overall, HCES remains essential for economic research and policy-making, with continuous improvements ensuring its relevance in addressing contemporary economic challenges.

4 Data and Methodology

4.1 Data Source

The study utilizes unit-level data from the HCES 2023-24 survey, covering diverse demographic, economic, and consumption variables.

4.2 Statistical Methods Used

The analysis employs descriptive statistics, regression models, and data visualization techniques to explore expenditure patterns and trends. Advanced methods such as clustering and predictive modeling are applied to forecast future consumption behaviors.

5 Focus of the Study

This research delves into key dimensions of household consumption in India, exploring socioeconomic patterns, regional disparities, and the evolving structure of expenditures in response to policy, income, and demographic transitions.

1. Disparities in Household Consumption

Analyzes Monthly Per Capita Expenditure (MPCE) differences between rural and urban households, highlighting how cost of living, access to services, and lifestyle variation influence spending behavior.

2. Impact of Rising Inflation

Explores the burden of inflation on household budgets, with emphasis on low-income groups. Examines changes in both essential and discretionary spending as households adapt to price volatility.

3. Income-Based Consumption Patterns

Studies how different income groups allocate resources across food, housing, education, health, and discretionary items. Highlights contrasts in priorities between low, middle, and high income households.

4. Regional Disparities in Household Spending

Compares MPCE across Indian states to uncover regional inequalities. Links observed patterns to differences in infrastructure, income levels, and local economic conditions.

5. Socioeconomic Determinants of Expenditure

Examines how social identity (caste/community), employment status, and household type shape consumption levels. Provides insight into the intersection of social structure and economic capacity.

6. Digital Payments and Consumption Behaviour

Assesses the rise of digital payment systems in both rural and urban settings, and their influence on how households transact and consume. Reflects broader behavioral shifts linked to financial technology.

7. Welfare Programs and Financial Resilience

Evaluates the impact of key welfare schemes—including the Public Distribution System (PDS), mid-day meals, and health coverage—on household consumption and financial security. Emphasis is placed on ration card coverage patterns, including the growing “No Card” population, urban–rural disparities, and state-level anomalies.

8. Forecasting Household Expenditures

Applies machine learning techniques to predict future MPCE trends. Supports long term planning by identifying emerging patterns in consumption under varied economic scenarios.

9. Food vs. Non-Food Expenditure Trends

Investigates the shift in consumption priorities from food to non-food items, influenced by rising income, changing aspirations, and urbanization. Tracks evolving household demand structures.

10. Prioritization Within the Consumption Basket

Analyzes how households balance essential and non-essential expenditures across income groups. Offers a deeper understanding of spending flexibility and constraints.

11. Urbanization and the Changing Food Share

Explores how urbanization is reshaping food expenditure patterns, particularly among rural households. Captures shifts in dietary preferences and nutritional priorities.

12. Clustering States by Food and Cereal Spending

Uses cluster analysis to group states based on their food consumption profiles, particularly in cereals and processed foods. Highlights regional dietary trends with implications for food security and agricultural policy.

6 Analysis and Findings

State/UT	FSUs Allotted		FSUs Surveyed		Sample Households		
	Rural	Urban	Rural	Urban	Rural	Urban	Total
Andhra Pradesh	361	240	360	239	6,306	4,159	10,465
Arunachal Pradesh	160	80	160	79	2,653	1,420	4,073
Assam	341	140	337	140	6,053	2,519	8,572
Bihar	762	200	759	200	13,656	3,599	17,255
Chhattisgarh	160	120	160	120	2,856	2,112	4,968
Delhi	20	200	20	198	310	3,049	3,359
Goa	20	20	20	20	360	360	720
Gujarat	320	320	320	319	5,699	5,572	11,271
Haryana	160	140	155	139	2,785	2,478	5,263
Himachal Pradesh	82	60	80	60	1,355	1,051	2,406
Jharkhand	221	140	219	140	3,866	2,465	6,331
Karnataka	381	340	380	340	6,728	5,805	12,533
Kerala	220	200	218	200	3,883	3,517	7,400
Madhya Pradesh	482	320	472	319	8,426	5,679	14,105
Maharashtra	660	660	659	652	11,629	11,117	22,746
Manipur	180	160	91	145	1,637	2,607	4,244
Meghalaya	120	60	120	59	2,154	1,062	3,216
Mizoram	80	120	80	120	1,440	2,160	3,600
Nagaland	120	60	109	60	1,962	1,080	3,042
Odisha	381	140	379	140	6,768	2,445	9,213
Punjab	180	160	178	160	3,161	2,810	5,971
Rajasthan	501	260	500	259	8,765	4,378	13,143
Sikkim	80	40	80	40	1,439	719	2,158
Tamil Nadu	420	400	419	397	7,484	7,008	14,492
Telangana	200	180	199	179	3,571	3,215	6,786
Tripura	180	100	180	100	3,240	1,800	5,040
Uttar Pradesh	1,100	600	1,100	600	19,663	10,644	30,307
Uttarakhand	100	60	100	60	1,693	1,056	2,749
West Bengal	600	420	597	419	10,706	7,414	18,120
Andaman Nicobar Islands	40	20	38	20	641	359	1,000
Chandigarh	20	20	20	20	359	359	718
Jammu Kashmir	100	100	99	100	1,782	1,800	3,582
Ladakh	20	20	20	20	360	360	720
Lakshadweep	14	20	14	20	249	356	605
Puducherry	20	40	20	40	359	718	1,077

Table 1: Number of FSUs allotted and surveyed, and number of households surveyed

- **Highest Surveyed State:** Uttar Pradesh stands out with the largest number of surveyed households (around 30,000), predominantly from rural areas.
- **Balanced Surveys:** States such as Maharashtra and Tamil Nadu show a balanced distribution between rural and urban surveys, suggesting a comprehensive approach to capturing demographic data.
- **Smaller Regions:** Union territories like Lakshadweep and Goa feature the smallest surveyed household counts (below 1,000), reflecting their limited population sizes.
- **Trends in Urban Surveys:** Urban areas in states like Delhi demonstrate significant representation compared to rural areas, highlighting urban-centric demographic and development patterns.

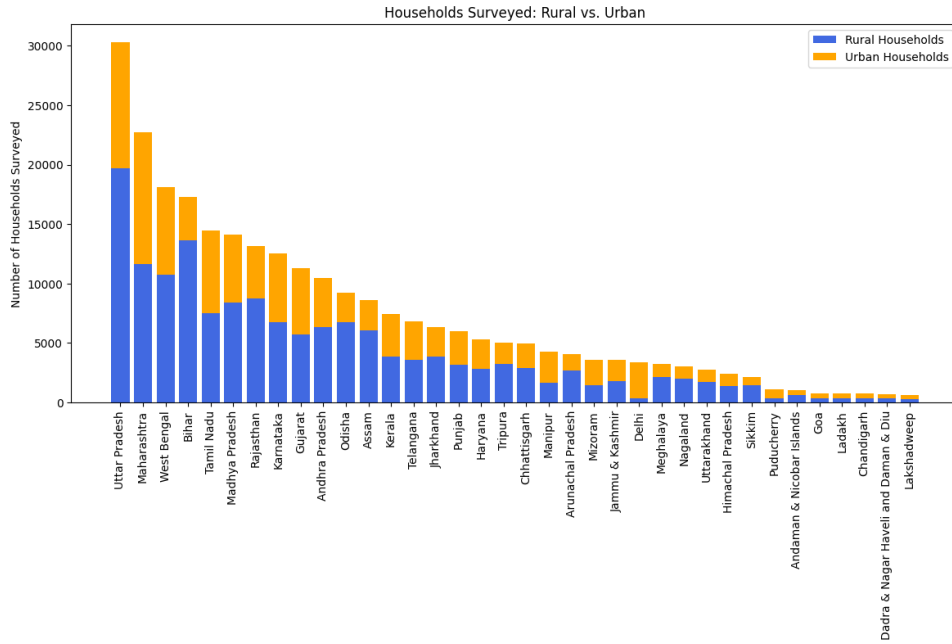


Figure 1: Household Surveyed Rural vs Urban

State/UT	Estimated Rural HH	Estimated Urban HH	Estimated Total HH
Jammu Kashmir	1,799,259	566,668	2,365,927
Bihar	20,151,145	2,167,306	22,318,451
Sikkim	120,725	57,498	178,223
Arunachal Pradesh	217,653	62,305	279,958
Nagaland	276,945	120,338	397,283
Manipur	499,367	183,691	683,058
Mizoram	112,840	102,746	215,586
Tripura	804,896	201,873	1,006,768
Meghalaya	563,295	110,882	674,177
Assam	6,248,139	981,145	7,229,284
West Bengal	16,976,256	8,004,246	24,980,502
Himachal Pradesh	1,557,010	243,984	1,800,994
Jharkhand	5,709,801	1,615,380	7,325,181
Odisha	9,384,147	2,193,468	11,577,615
Chhattisgarh	4,591,188	1,332,890	5,924,078
Madhya Pradesh	11,266,164	4,484,483	15,750,647
Gujarat	7,071,053	6,551,886	13,622,939
Dadra Nagar Haveli	54,438	131,873	186,311
Maharashtra	14,768,694	12,562,136	27,330,830
Andhra Pradesh	9,468,709	4,515,617	13,984,326
Karnataka	9,300,573	6,880,675	16,181,247
Punjab	3,804,113	2,744,353	6,548,466
Goa	134,511	223,872	358,383
Lakshadweep (U.T.)	3,017	9,344	12,361
Kerala	4,886,446	4,925,091	9,811,537
Tamil Nadu	11,176,960	10,686,204	21,863,164
Puducherry (U.T.)	106,244	240,645	346,889
A and N Islands (U.T.)	76,438	50,536	126,974
Telangana	5,689,698	5,407,536	11,097,234
Ladakh (U.T.)	43,519	9,736	53,256
Chandigarh (U.T.)	11,313	242,733	254,046
Uttarakhand	1,856,434	754,955	2,611,389

State/UT	Estimated Rural HH	Estimated Urban HH	Estimated Total HH
Haryana	3,627,657	2,554,196	6,181,853
Delhi	89,607	3,643,667	3,733,274
Rajasthan	10,936,974	4,247,935	15,184,908
Uttar Pradesh	30,398,615	9,075,344	39,473,958
All-India	193,783,843	97,887,235	291,671,078

Table 2: Estimated Household Data

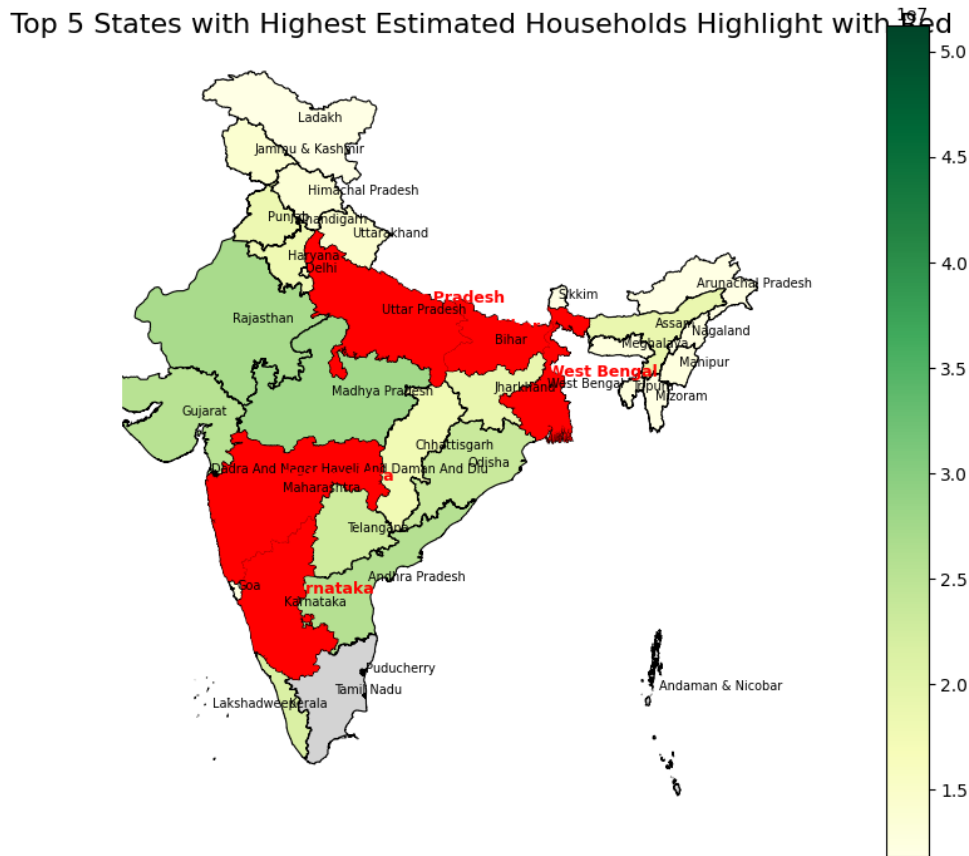


Figure 2: Proportion of Rural Vs Urban Households Surveyed

States with the highest estimated number of households, emphasizing regions of demographic density.

- **Top Highlighted States:** Uttar Pradesh, Bihar, West Bengal, Maharashtra, and Madhya Pradesh stand out due to their large household estimates, indicating their significant share in the national population.
- **Color Gradient Representation:** A gradient from lighter to darker shades represents the variation in household numbers across regions, with the highlighted states falling into the higher bracket.
- **Range of Values:** The estimates range significantly, with the highlighted states surpassing a certain threshold, underlining their demographic prominence.
- **Lowest Household State:** Sikkim stands out as the state with the lowest estimated number of households, which likely correlates with it having the smallest population in the country.

Bottom 5 States with Lowest Estimated Households Highlight with Red

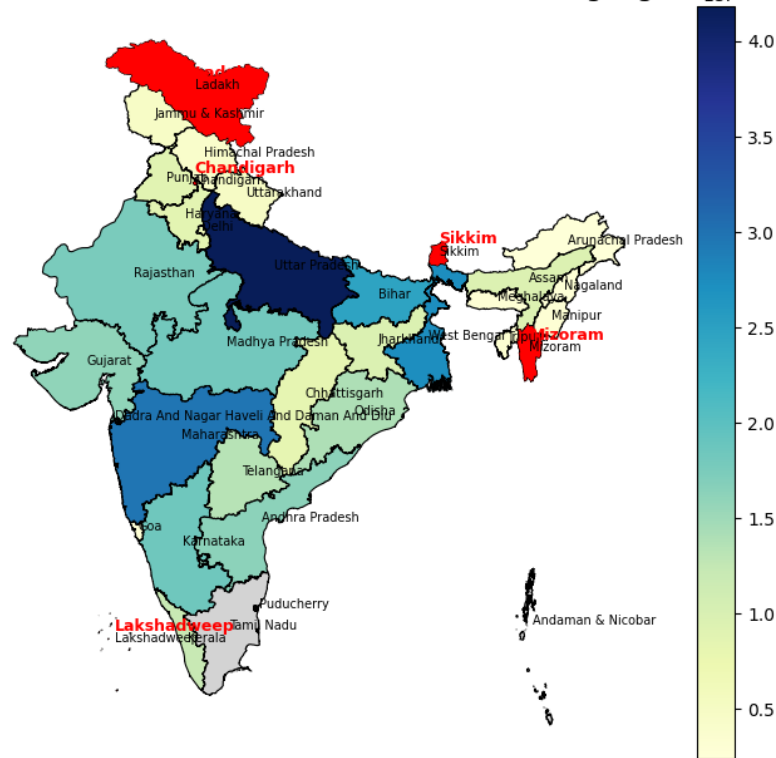


Figure 3: Proportion of Rural Vs Urban Households Surveyed

- **Highlighted States:** Other northeastern states, such as Arunachal Pradesh, Nagaland, Mizoram, and Meghalaya, are also identified with minimal household estimates, reflecting lower population densities.
- **Color Gradient Representation:** The map employs a gradient, transitioning from green (representing higher household counts) to red (indicating the lowest estimates). Sikkim falls at the extreme lower end of the gradient.
- **Implications:** These insights suggest regions with lower demographic density, which can help prioritize resource distribution and infrastructure development in sparsely populated areas.

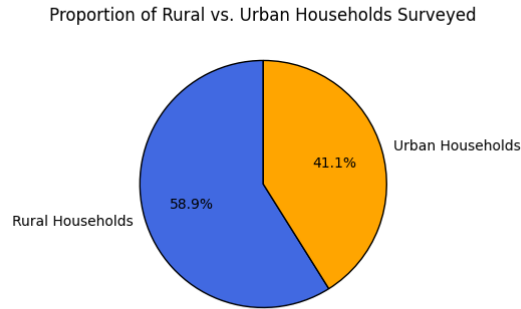


Figure 4: Proportion of Rural Vs Urban Households Surveyed

- **Rural Majority:** The chart shows that 58.9% of the surveyed households are from rural areas, indicated by the blue section. This reflects a higher focus on rural demographics in the survey.
- **Urban Minority:** The orange section, representing urban households, accounts for 41.1%. While smaller in proportion, urban surveys still constitute a significant share.
- **Demographic Insight:** The chart highlights the predominance of rural households in the survey, suggesting that rural areas are prioritized or have higher household counts compared to urban regions.

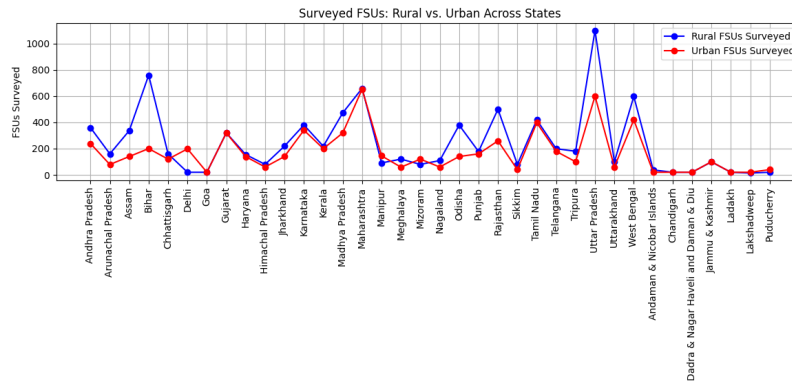


Figure 5: Surveyed FSUs: rural Vs. Urban Across States

- **Rural Dominance:** The graph indicates that in the majority of states, more rural FSUs were surveyed compared to urban FSUs. This highlights a stronger emphasis on rural areas in the survey.
- **State-wise Peaks:** Notable states with significantly high rural FSU surveys include Bihar, Maharashtra, Uttar Pradesh, and West Bengal, reflecting their large rural population bases.
- **Urban Representation:** While urban FSUs are fewer in most states, their consistent inclusion ensures the survey captures urban demographics for a balanced perspective.
- **Regional Variation:** The variation in surveyed FSUs across states emphasizes demographic differences and the tailored approach of the survey to capture diverse data.

Survey Completion Rates

- **High Completion Rates Nationwide:** Most states and union territories exhibit high survey completion rates for both rural and urban areas, reflecting robust data collection efforts.
- **Variations Noted:** Nagaland and Manipur stand out, with noticeable disparities between rural and urban completion rates. Manipur shows a significant gap, with urban areas achieving higher completion rates than rural areas.

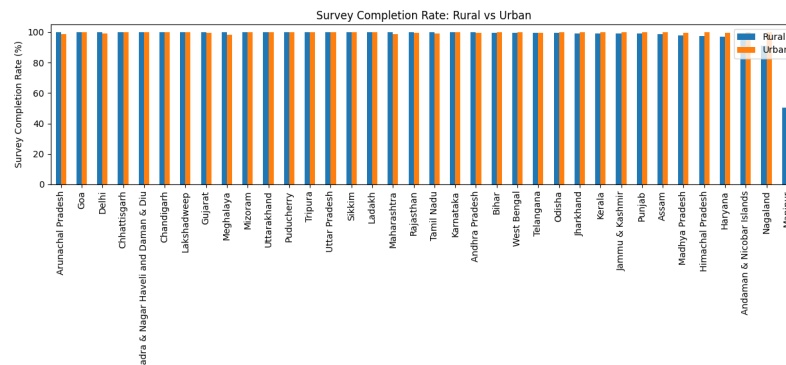


Figure 6: Survey Completion Rates

- **Uniform Patterns Elsewhere:** Other states such as Gujarat, Maharashtra, and Karnataka display almost identical completion rates in both rural and urban surveys, suggesting uniformity in survey execution.

Distribution of Survey Completion Rates

- **Median Completion Rates:** Both rural and urban groups demonstrate high median completion rates close to 100%, indicating successful survey execution in most regions.
- **Outliers in Rural Group:** Rural survey completion rates feature outliers below 90%, with a significant outlier close to 50%. This suggests inconsistencies or challenges in completing surveys in some rural areas.

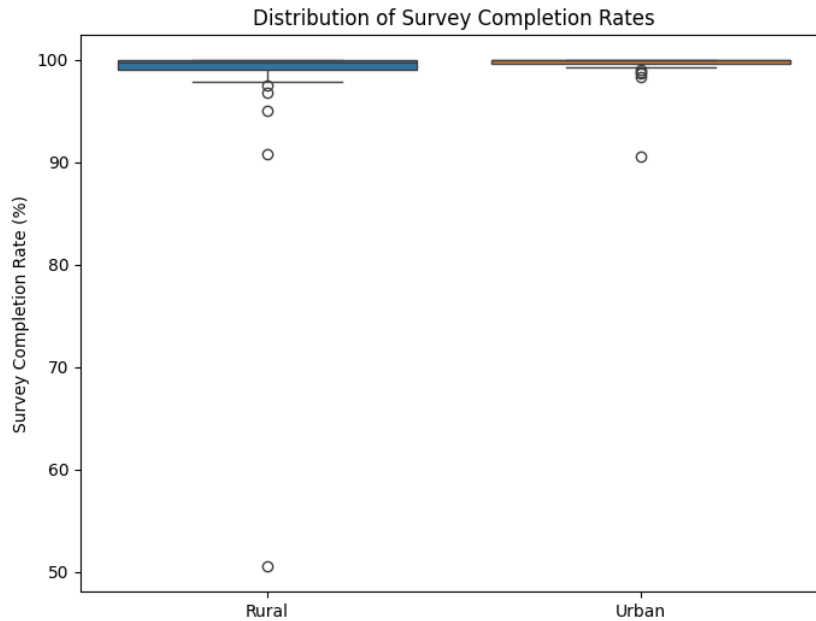


Figure 7: Distribution of Survey Completion Rates

- **Uniformity in Urban Group:** Urban survey completion rates are tightly clustered around the median, with fewer and less extreme outliers compared to rural areas, reflecting more consistency in urban data collection.
- **Comparison:** The plot highlights that while both groups achieve high completion rates overall, rural areas exhibit greater variability and face unique survey completion challenges.

Household Coverage Per FSU - Rural vs Urban

- **Rural Majority in Coverage:** Most states display higher household coverage per FSU in rural areas (blue bars) compared to urban areas (orange bars). This trend aligns with the rural household dominance across the country.
- **State-Level Insights:** Uttar Pradesh, Bihar, and Maharashtra have notably high rural coverage rates. On the urban side, states like Delhi and Goa showcase relatively balanced or elevated coverage.
- **Contrast Across Regions:** Northeastern states (e.g., Arunachal Pradesh, Mizoram) show relatively lower overall coverage, reflecting smaller population sizes and demographic concentration in rural zones.
- **Range of Coverage:** The household coverage per FSU spans from less than 5 in smaller states like Goa to more than 17 in states like Uttar Pradesh, indicating regional variations in survey intensity.

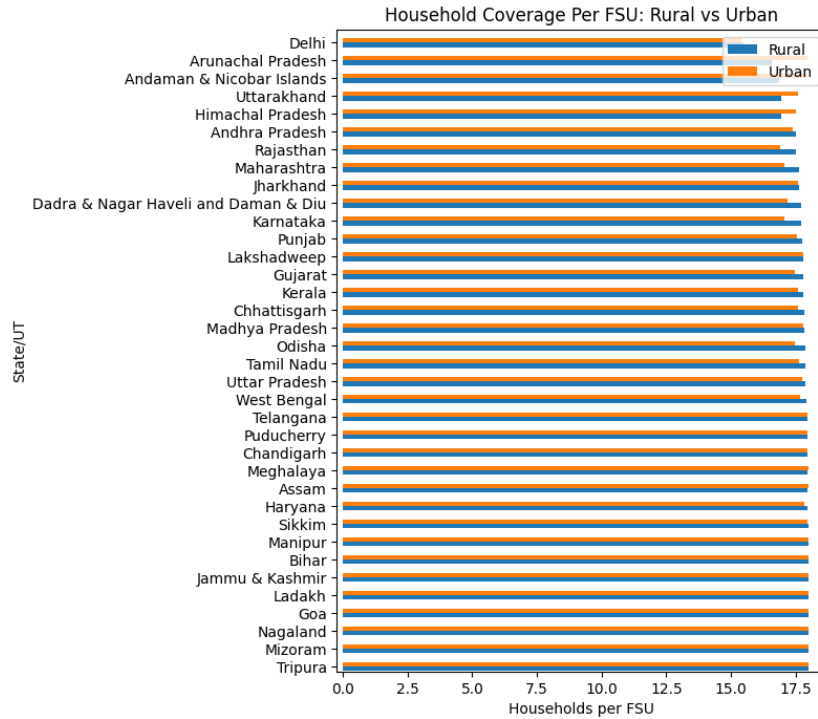


Figure 8: Household Coverage Per FSU - Rural vs Urban

Relationship Between FSUs Surveyed and Households Surveyed

- **Positive Correlation:** The scatter plot demonstrates a clear positive linear relationship between the total FSUs (Field Survey Units) surveyed and the total households surveyed. As the number of FSUs surveyed increases, the households surveyed increase proportionally.
- **textbfClustered Distribution:** The points are well-clustered along the trend line, indicating consistency in data collection efforts across various FSUs.
- **Data Variability:** Some points deviate slightly from the main trend, suggesting variability in household coverage per FSU in certain regions.
- **Categorical Indicators:** The data points are color-coded (e.g., red, blue, light blue), potentially representing different states or territories. These color distinctions might correlate with unique regional characteristics or demographics.

Elbow Method

- **Sharp Decrease in WCSS:** The plot shows a rapid decline in the Within-Cluster Sum of Squares (WCSS) as the number of clusters (k) increases from 1 to 2, and then from 2 to 3, indicating a better-defined clustering structure with additional clusters.
- **Optimal Number of Clusters:** The "elbow" is observed at around $k = 3$ or $k = 4$, where the rate of decrease in WCSS begins to slow down. This suggests that the ideal number of clusters is 3 or 4, as adding more clusters beyond this point results in minimal gains in clustering performance.
- **Diminishing Returns:** Beyond the elbow point, further increases in k lead to diminishing returns, as the reduction in WCSS becomes negligible. This implies that additional clusters would add complexity without significant improvement in model efficiency.

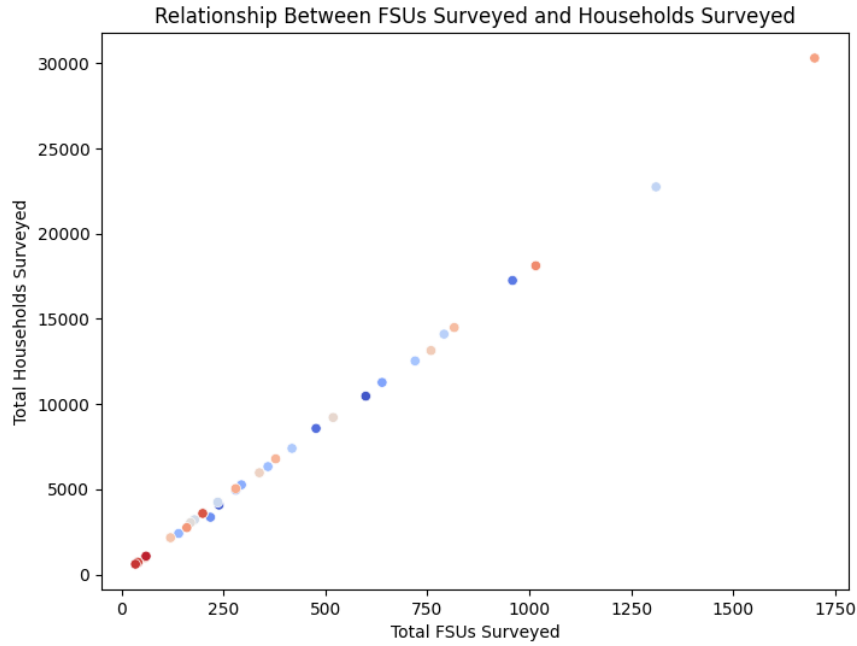


Figure 9: Relationship Between FSUs Surveyed and Households Surveyed

Cluster-wise Average FSUs & Households Surveyed

- **Dominance of Rural Households:** Across all clusters, rural households surveyed consistently demonstrate higher counts compared to urban households. This underscores the survey's focus on rural demographics.
- **Cluster 0 and Cluster 3:** Both clusters exhibit significantly high counts for surveyed rural and urban households, with rural numbers dominating. This indicates high survey activity in these clusters, possibly due to higher population density or prioritization.
- **Cluster 1 - Minimal Survey Coverage:** This cluster displays the lowest counts across all metrics, suggesting regions with limited survey activities.
- **Moderate Survey Coverage in Cluster 2:** This cluster reflects moderate activity with noticeable disparities between rural and urban households surveyed, again emphasizing rural focus.
- **FSUs Patterns:** Similar trends are observed for FSUs surveyed, with rural FSUs showing consistently higher averages compared to urban FSUs across most clusters.

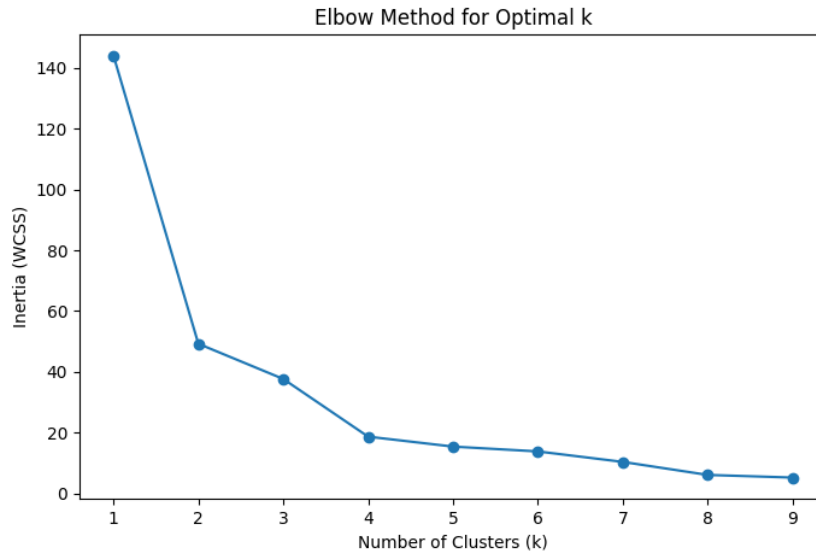


Figure 10: Elbow Method

Cluster-wise Survey Averages

- **Cluster 0 - High Averages:** Rural metrics (492.4 and 8780.6) significantly exceed urban metrics (321.9 and 5636.4), reflecting a predominant rural survey focus. Urban metrics are still notable but consistently lower, indicating demographic differences in surveyed averages.
- **Cluster 1 - Low Coverage:** Both rural (64.2 and 1123.9) and urban (63.6 and 1109.4) averages are the lowest among all clusters. This suggests minimal survey activity, potentially due to sparse demographics or logistical challenges.
- **Cluster 2 - Moderate Metrics:** Rural averages (238.5 and 4248.9) are again higher than urban ones (155.7 and 2752.0), maintaining the trend of rural emphasis. Moderate survey values imply balanced focus but lower priority compared to Cluster 0 and Cluster 3.
- **Cluster 3 - Highest Survey Effort:** The highest averages for both rural (879.5 and 15646.0) and urban (626.0 and 10880.5) metrics, highlighting substantial survey activities. This cluster reflects regions with the densest population or strategic importance in the survey.
- **Overall Trend:** Across all clusters, rural averages consistently exceed urban averages, emphasizing rural demographics as a priority in the survey.

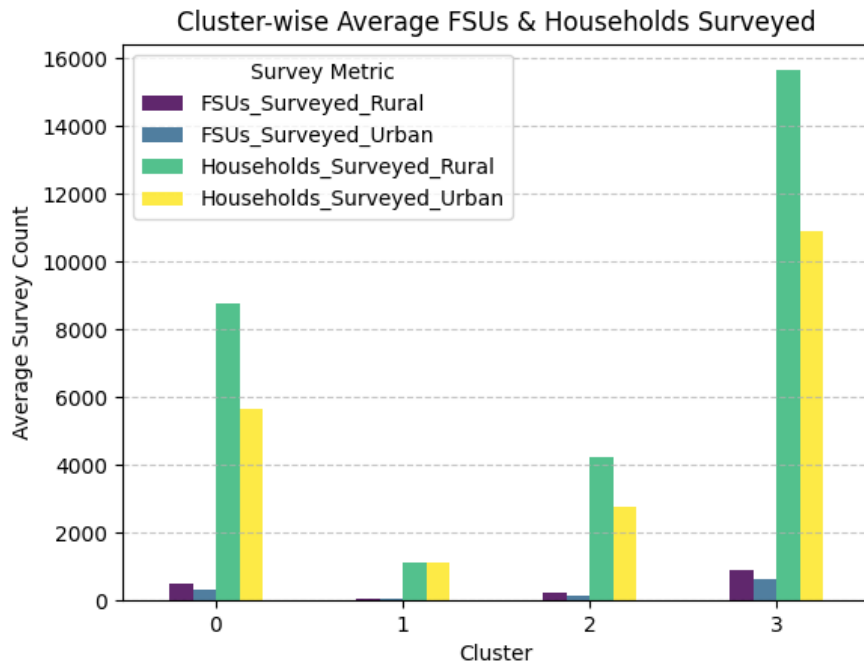


Figure 11: Cluster-wise Average FSUs & Households Surveyed

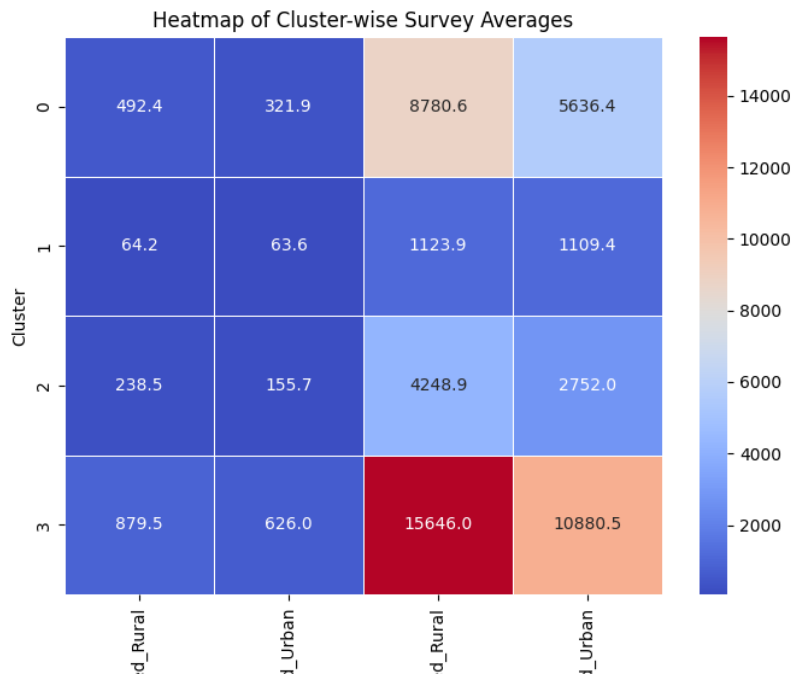


Figure 12: Cluster-wise Survey Averages

Household Surveys Across Clusters

- **Cluster with the Highest Median:** Cluster 3 exhibits the highest median number of households surveyed, indicating a greater concentration of survey efforts in this cluster.
- **Declining Median Values Across Clusters:** After Cluster 3, the median values decrease sequentially for Cluster 0, Cluster 2, and Cluster 1, reflecting varying survey intensities across these

clusters.

- **Cluster 1 - Lowest Activity:** Cluster 1 not only has the lowest median but also a smaller range and interquartile spread, suggesting limited household survey coverage in this cluster.
- **Cluster 3 - Wide Range and High Spread:** Cluster 3 features the widest range and interquartile spread, highlighting greater variability in the number of households surveyed within this cluster.
- **Overall Distribution:** The plot demonstrates significant differences in survey intensities across clusters, with Cluster 3 standing out as a focal point for survey activity and Cluster 1 being relatively less prioritized.

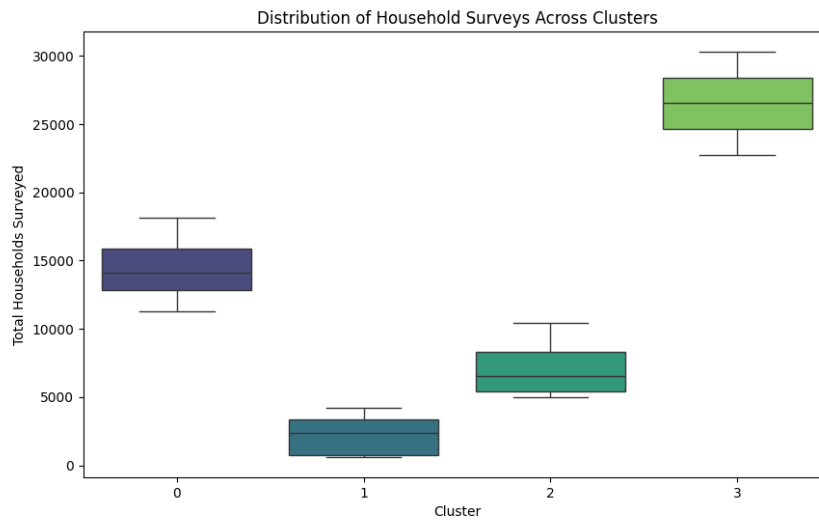


Figure 13: Household Surveys Across Clusters

Cluster-wise Survey Patterns (Radar Chart)

- **Cluster 3 - Dominant Coverage:** The red section of the radar chart, representing Cluster 3, spans the largest area across all axes. This indicates the highest values in surveyed rural households, urban households, and urban FSUs, highlighting extensive survey coverage in this cluster.
- **Cluster 0 - Moderate Coverage:** The blue area, representing Cluster 0, shows moderate values across the axes. While it has a balanced presence, its coverage is smaller compared to Cluster 3, particularly in urban FSUs.
- **Cluster 1 - Minimal Survey Data:** The orange region, representing Cluster 1, is the smallest across the axes, emphasizing the least survey activity in terms of rural households, urban households, and urban FSUs.
- **Cluster 2 - Mixed Patterns:** The green section, representing Cluster 2, reflects mixed patterns with values higher than Cluster 1 but lower than Clusters 0 and 3. It indicates moderate survey efforts.
- **Overall Trends:** The radar chart visually compares survey patterns and highlights significant differences in survey coverage across clusters. Cluster 3 emerges as the focal point with the highest survey intensity, while Cluster 1 exhibits minimal efforts.

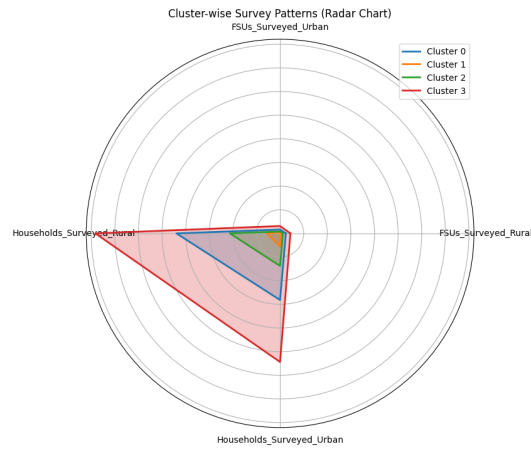


Figure 14: Cluster-wise Survey Patterns (Radar Chart)

7 Monthly Per Capita Expenditure (MPCE)

State Code	State Name	Rural MPCE	Urban MPCE
01	Jammu Kashmir	4774	6327
02	Himachal Pradesh	5825	9223
03	Punjab	5817	7359
04	Chandigarh	8857	13425
05	Uttarakhand	5003	7486
06	Haryana	5377	8427
07	Delhi	7400	8534
08	Rajasthan	4510	6574
09	Uttar Pradesh	3481	5395
10	Bihar	3670	5080
11	Sikkim	9377	13927
12	Arunachal Pradesh	5995	9832
13	Nagaland	5155	8022
14	Manipur	4531	5945
15	Mizoram	5963	8709
16	Tripura	6259	8034
17	Meghalaya	3852	7839
18	Assam	3793	6794
19	West Bengal	3620	5775
20	Jharkhand	2946	5393
21	Odisha	3357	5825
22	Chhattisgarh	2739	4927
23	Madhya Pradesh	3441	5538
24	Gujarat	4116	7175
25	D N. Haveli Daman Diu	4311	6837
26	Maharashtra	4145	7363
27	Andhra Pradesh	5327	7182
28	Karnataka	4903	8076
29	Goa	8048	9726
30	Lakshadweep	6350	6377
31	Kerala	6611	7783
32	Tamilnadu	5701	8165
33	Puduchery	7598	8637
34	Andaman N. Island	7771	10453
35	Telangana	5435	8978
36	Ladakh	5010	7533

Table 3: Monthly Per Capita Expenditure (MPCE) across Indian States (Rural vs. Urban)

7.1 MPCE Distribution in Rural vs. Urban Areas

- **Wider Distribution for Urban Areas:** The orange violin plot for urban areas exhibits a broader range, with MPCE extending beyond 12,000Rs. This indicates a more diverse expenditure pattern among urban households.
- **Higher Median in Urban Areas:** The box plot within the urban violin highlights a higher median MPCE compared to rural areas, suggesting that urban households, on average, spend more.
- **Tighter Distribution in Rural Areas:** The blue violin plot for rural areas shows a narrower range, with MPCE largely concentrated below 8,000Rs. This reflects a more consistent expenditure pattern among rural households.
- **Overlap in Lower Ranges:** Both plots overlap in the lower expenditure ranges (0Rs–4,000Rs), indicating similarity in spending among the lowest-income groups in both demographics.

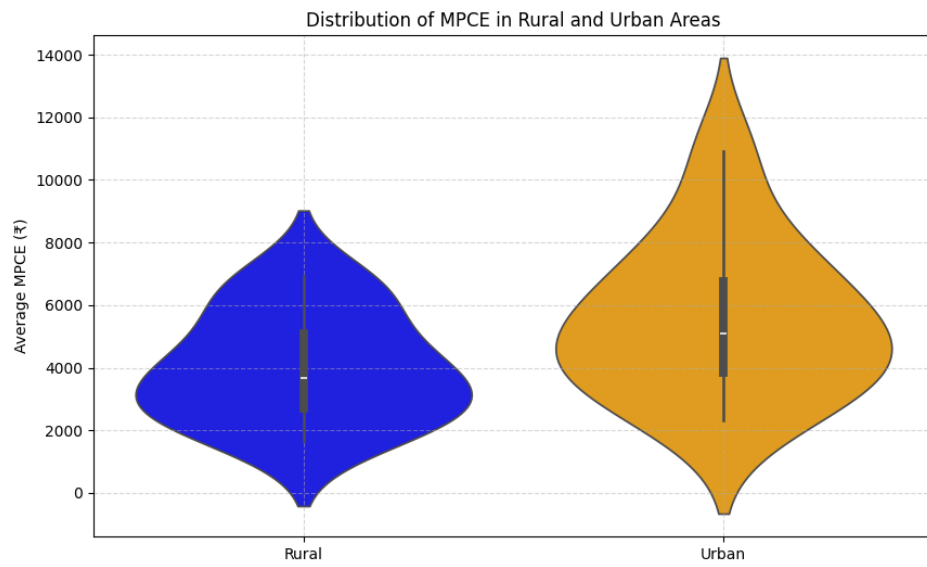


Figure 15: MPCE Distribution in Rural vs. Urban Areas (Violin Plot)

- **Policy Implications:** The distinct patterns emphasize the need for tailored economic policies to address the disparities in expenditure between rural and urban households.

8 Trends in Rural vs. Urban MPCE Across Fractile Classes

Fractile	Total Expenditure	Household Size	MPCE
0-5	71,657,262,868	42,736,481	1677
5-10	90,833,408,181	42,726,577	2126
10-20	211,403,024,803	85,480,531	2473
20-30	242,163,622,787	85,472,417	2833
30-40	270,266,748,526	85,476,775	3162
40-50	298,921,699,477	85,467,313	3497
50-60	330,473,715,843	85,480,935	3866
60-70	367,858,056,668	85,472,525	4304
70-80	417,553,308,080	85,472,218	4885
80-90	492,590,169,238	85,475,974	5763
90-95	296,107,354,997	42,736,651	6929
95-100	433,204,423,242	42,737,029	10137

Table 4: Fractile-wise Distribution of Total Expenditure, Household Size, and MPCE for Rural Sector

Fractile	Total Expenditure	Household Size	MPCE
0-5	43,107,834,397	18,144,642	2376
5-10	56,120,204,278	18,144,637	3093
10-20	133,811,935,916	36,289,896	3687
20-30	157,985,265,930	36,291,215	4353
30-40	180,682,200,884	36,291,854	4979
40-50	204,013,920,066	36,290,695	5622
50-60	229,856,488,092	36,290,184	6334
60-70	261,203,938,231	36,283,291	7199
70-80	303,117,438,687	36,292,326	8352
80-90	367,994,935,413	36,297,493	10138
90-95	232,543,354,883	18,144,072	12816
95-100	368,559,422,756	18,147,728	20309

Table 5: Fractile-wise Distribution of Total Expenditure, Household Size, and MPCE (Urban)

- **General Trend:** Both Rural and Urban MPCE show an increasing trend across the fractile classes. This indicates that as fractile classes increase (representing higher economic strata), the average Monthly Per Capita Expenditure also rises.
- **Urban MPCE Consistently Higher:** The orange line (Urban MPCE) remains consistently above the blue line (Rural MPCE) for all fractile classes. This reflects the higher cost of living and spending capacity in urban areas compared to rural areas.
- **Wide Gap in Higher Fractile Classes:** The gap between Rural and Urban MPCE widens significantly in the higher fractile classes (e.g., 90–100%), signifying greater inequality in expenditure levels among wealthier groups in rural and urban demographics.
- **Policy Implications:** These trends suggest the need for targeted policies addressing the disparity in rural and urban expenditure patterns, especially for high-income groups. Additionally, efforts might focus on improving spending capacity in rural areas to narrow the gap.

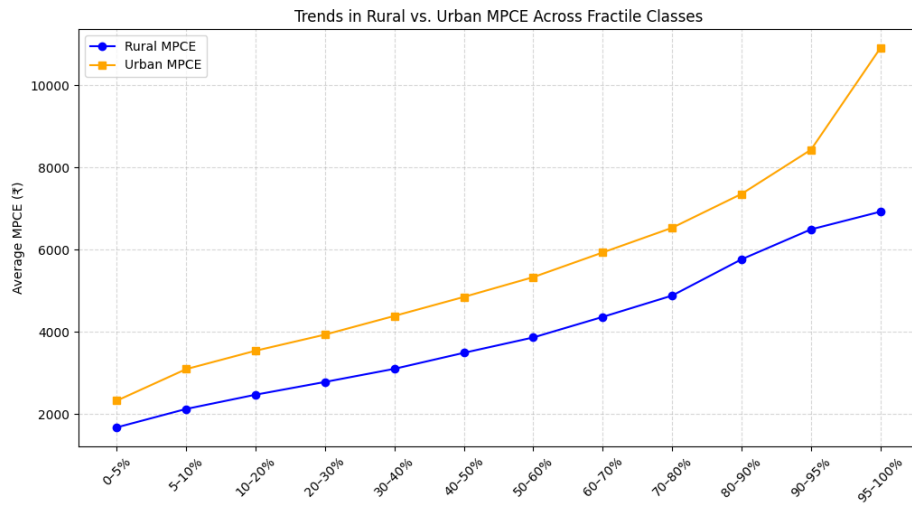


Figure 16: Trends in Rural vs. Urban MPCE Across Fractile Classes

8.1 Rural vs. Urban MPCE

- **Higher Urban Expenditure:** The chart indicates that Urban MPCE (Monthly Per Capita Expenditure) exceeds Rural MPCE across most percentage ranges, with the orange dashed line extending further outward compared to the blue solid line in the radar chart.
- **Expenditure Patterns:** Urban areas demonstrate a consistently higher spending pattern, reflecting differences in cost of living, consumption habits, and availability of goods and services compared to rural areas.
- **Wide Gap in Higher Ranges:** The disparity becomes more prominent in higher percentage ranges (70%-90%), indicating that the urban population in these ranges spends substantially more than their rural counterparts.

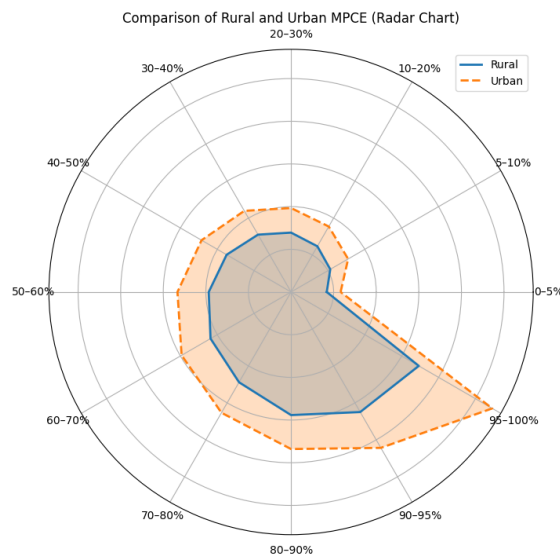


Figure 17: Rural vs. Urban MPCE (Radar Chart)

- **Narrower Gap in Lower Ranges:** At lower percentage ranges (0%-10%), the gap between Rural and Urban MPCE is relatively narrow, suggesting similar expenditure levels among lower-income groups in both regions.

- **Policy Implications:** This comparison underscores the need for tailored policy interventions to address the unique expenditure patterns and cost of living disparities between rural and urban populations.

8.2 Share of Rural and Urban MPCE Across Fractile Classes

- **Increasing MPCE Across Fractile Classes:** The bar chart shows a steady increase in total Monthly Per Capita Expenditure (MPCE) as we move from lower fractile classes (0–5%) to higher ones (95–100%). This suggests that households in higher economic strata spend significantly more on average.
- **Urban MPCE Dominance:** For every fractile class, the urban MPCE (orange segments) surpasses the rural MPCE (blue segments), reflecting higher expenditure patterns in urban households due to factors like increased income, cost of living, and diverse consumption.

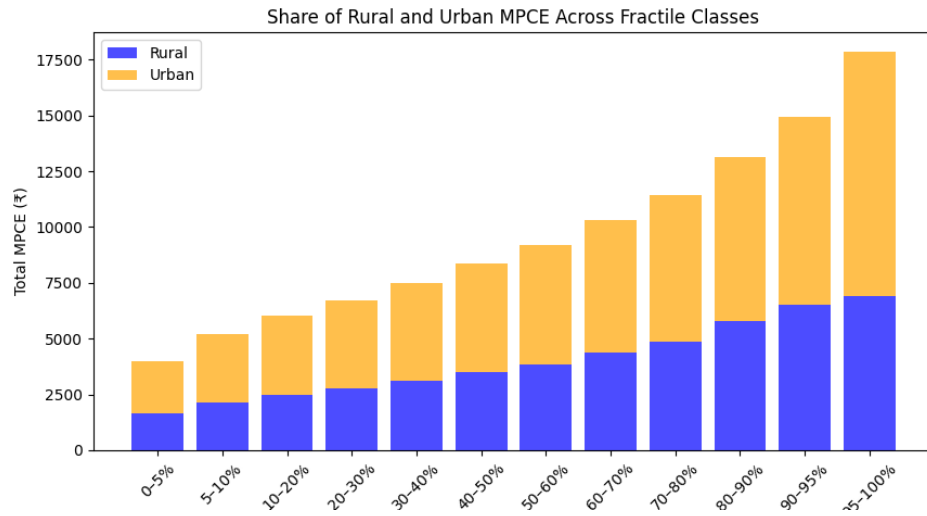


Figure 18: Share of Rural and Urban MPCE Across Fractile Classes

- **Proportional Increase in MPCE:** Both rural and urban MPCE rise proportionally across fractile classes, but the gap between urban and rural MPCE widens as we move toward higher classes, emphasizing greater inequality in expenditure among wealthier households.
- **Fractile Representation:** Each fractile class is well-represented, providing insights into expenditure trends across various economic segments of the population.
- **Policy Implications:** The consistent gap between rural and urban MPCE suggests the need for targeted policies aimed at reducing expenditure disparity and improving economic opportunities in rural areas.

MPCE Distribution Across Fractile Classes

- **Increasing Trend Across Fractile Classes:** The box plots indicate that the Monthly Per Capita Expenditure (MPCE) values rise steadily from lower fractile classes (0–5%) to higher ones (95–100%). This trend reflects increasing expenditure with higher income levels.
- **Urban MPCE Dominance:** Yellow box plots (urban sectors) consistently display higher MPCE values across all fractile classes compared to blue box plots (rural sectors). This highlights the disparity in spending between rural and urban households.
- **Greater Variability in Urban Areas:** Urban MPCE values exhibit wider boxes and whiskers, especially in higher fractile classes, suggesting greater variability in spending patterns among urban households compared to rural ones.

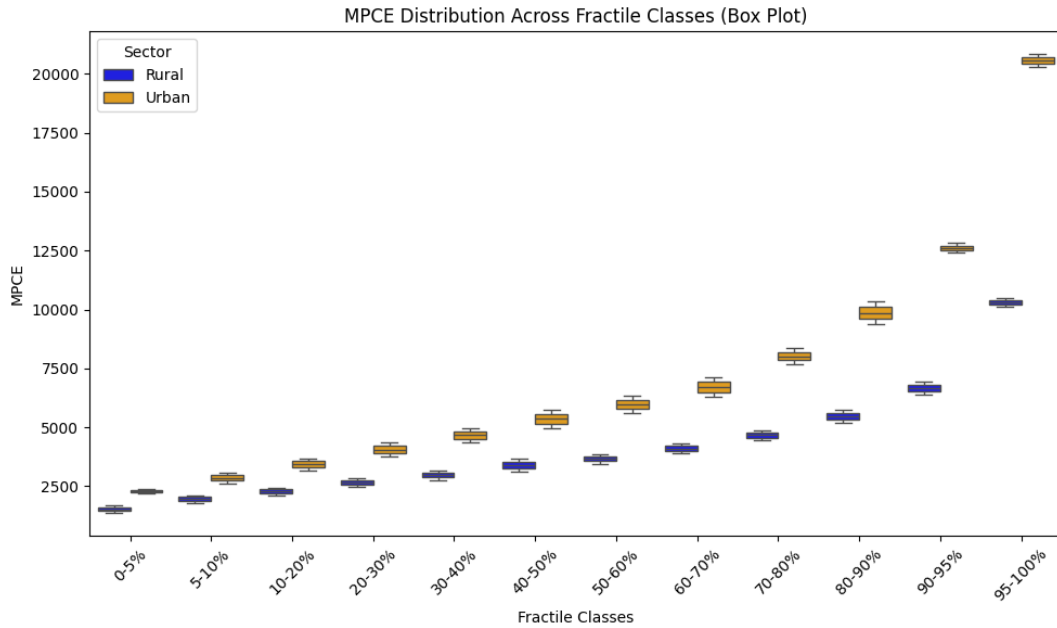


Figure 19: MPCE Distribution Across Fractile Classes

- **Smaller Range in Rural Areas:** Rural MPCE values are more concentrated with narrower boxes and whiskers, reflecting less variability in expenditure across different income groups in rural areas.
- **Policy Implications:** These findings emphasize the need to address expenditure disparities through policies aimed at improving economic conditions in rural areas and reducing inequality between rural and urban households.

Cumulative Distribution of MPCE for Rural vs. Urban Areas

- **Distinct Expenditure Patterns:** The blue dashed line (representing rural MPCE) and the orange solid line (representing urban MPCE) highlight distinct expenditure patterns between rural and urban areas.
- **Urban Areas Show Higher MPCE Values:** The urban CDF (orange line) consistently reaches higher MPCE values than the rural CDF (blue line), reflecting greater per capita spending in urban areas across most income levels.
- **Steeper Growth in Urban MPCE at Lower Values:** At MPCE values below 5,000Rs, the urban CDF exhibits steeper growth compared to the rural CDF, suggesting a larger proportion of urban households in this expenditure range.
- **Converging Trends at Higher MPCE Values:** Both distributions converge as MPCE values approach 30,000Rs, indicating similar expenditure behaviors among the wealthiest households in both demographics.
- **Policy Implications:** These findings underscore the need for policies targeting expenditure disparities, with a focus on uplifting rural household spending capacity while addressing income inequality.

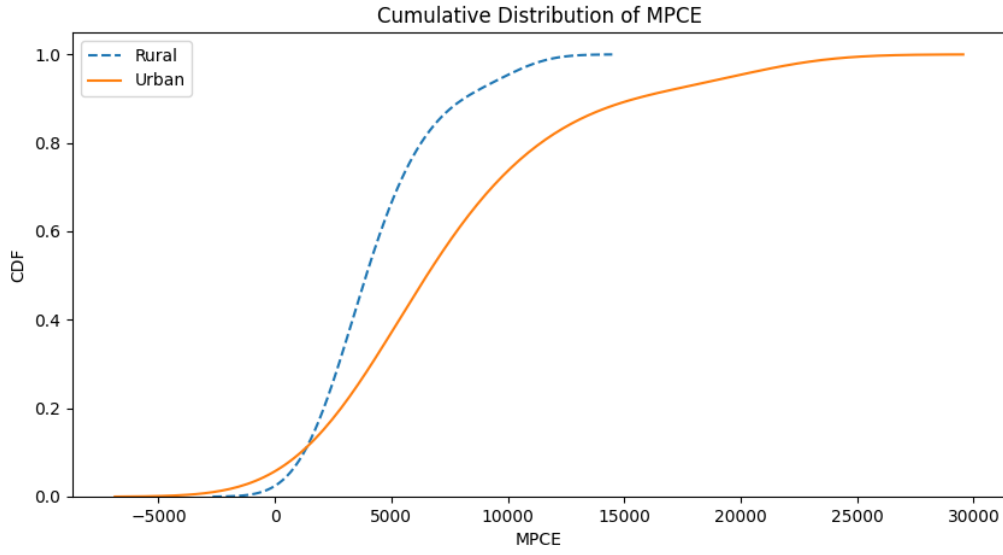


Figure 20: Cumulative Distribution of MPCE for Rural vs. Urban Areas

9 MPCE Growth Rate (Rural vs Urban)

- **Higher Growth in Rural Areas at Extremes:** Rural areas (blue bars) show a higher MPCE growth rate for the lowest income percentiles (0–5%) and the highest income percentiles (95–100%). This indicates targeted economic interventions or natural growth patterns benefiting these specific segments.
- **Urban Areas Dominate Middle Percentiles:** Urban areas (orange bars) exhibit higher MPCE growth rates in the middle-income percentiles (10–90%), which may reflect increased opportunities and consumption trends in urban centers for this demographic.

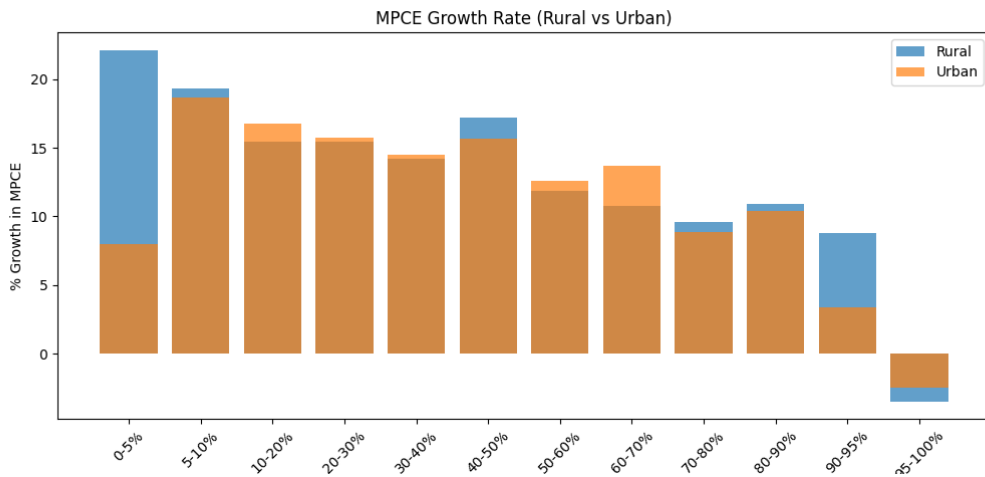


Figure 21: MPCE Growth Rate (Rural vs Urban)

- **Broad Trends in Rural and Urban Growth:** Rural MPCE growth displays a more polarized distribution, with significant growth at the extremes, while urban MPCE growth is more consistent across the income percentiles.
- **Policy Implications:** The observed trends suggest the need for strategies aimed at sustaining rural growth across all income percentiles and managing urban growth disparities, especially for low-income and high-income households.

9.1 High-Growth vs. Low-Growth Fractile Classes

- **Distinct Patterns Between Rural and Urban Growth:** Blue dots (Rural Growth) and orange dots (Urban Growth) exhibit distinct growth patterns across fractile classes. Urban Growth generally surpasses Rural Growth for mid-range fractile classes (10–90%), reflecting disparities in growth dynamics.
- **High-Growth Fractile Classes:** Rural and Urban Growth rates are significantly higher in the lowest (0–5%) and highest (95–100%) fractile classes, possibly driven by targeted economic interventions or natural economic expansions at these extremes.

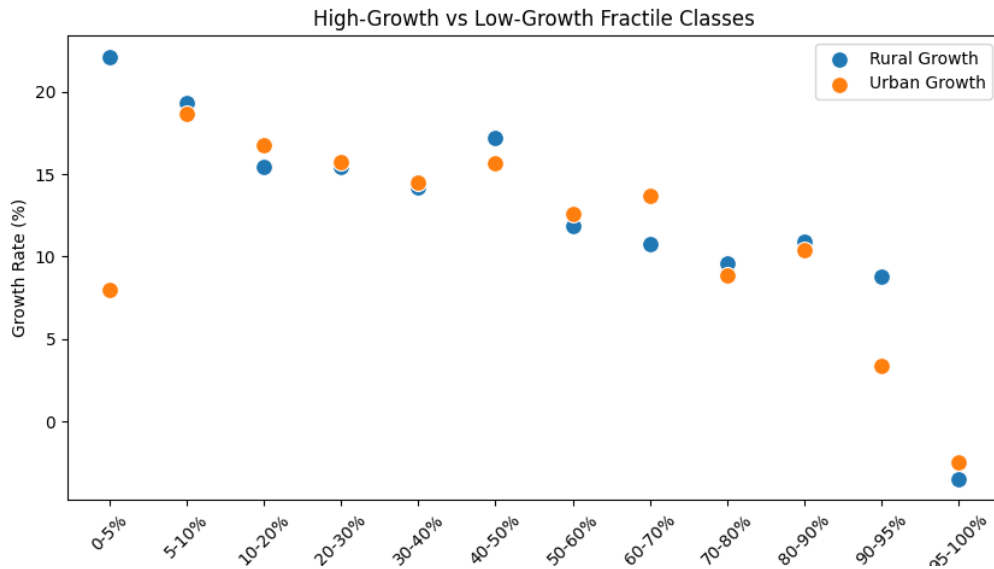


Figure 22: High-Growth vs. Low-Growth Fractile Classes

- **Mid-Range Fractile Classes:** Urban Growth maintains relatively steady and dominant growth rates in the middle fractile classes, whereas Rural Growth fluctuates, highlighting greater variability and challenges in sustaining consistent rural growth.
- **Policy Implications:** The analysis underscores the need for policies that address growth variability in rural areas while sustaining urban growth. Interventions should focus on improving economic opportunities in rural mid-range fractile classes to bridge the growth gap.

9.2 MPCE Growth Rate (Rural vs Urban)

- **Regional Differences in Growth:** States like Odisha, Rajasthan, and Chhattisgarh show exceptionally high MPCE growth rates in rural areas, reflecting economic improvements or increased spending capacity among rural households in these regions. Conversely, states like Punjab exhibit negative rural growth, suggesting challenges or reduced expenditure trends in rural sectors.
- **Urban Growth Dominance in Some States:** Urban growth rates surpass rural growth rates in states such as Andhra Pradesh, Gujarat, Maharashtra, and Haryana. This trend highlights urban economic dynamism and the potential impact of urban-focused policies.
- **All-India Trends:** At the "All-India" level, urban MPCE growth outpaces rural growth, indicating overall stronger expenditure trends in urban areas.
- **Policy Implications:** The wide disparities in rural and urban growth across states suggest the need for customized policy interventions. Regions with declining rural growth, such as Punjab, may require targeted support, while high-growth states might focus on sustaining momentum.

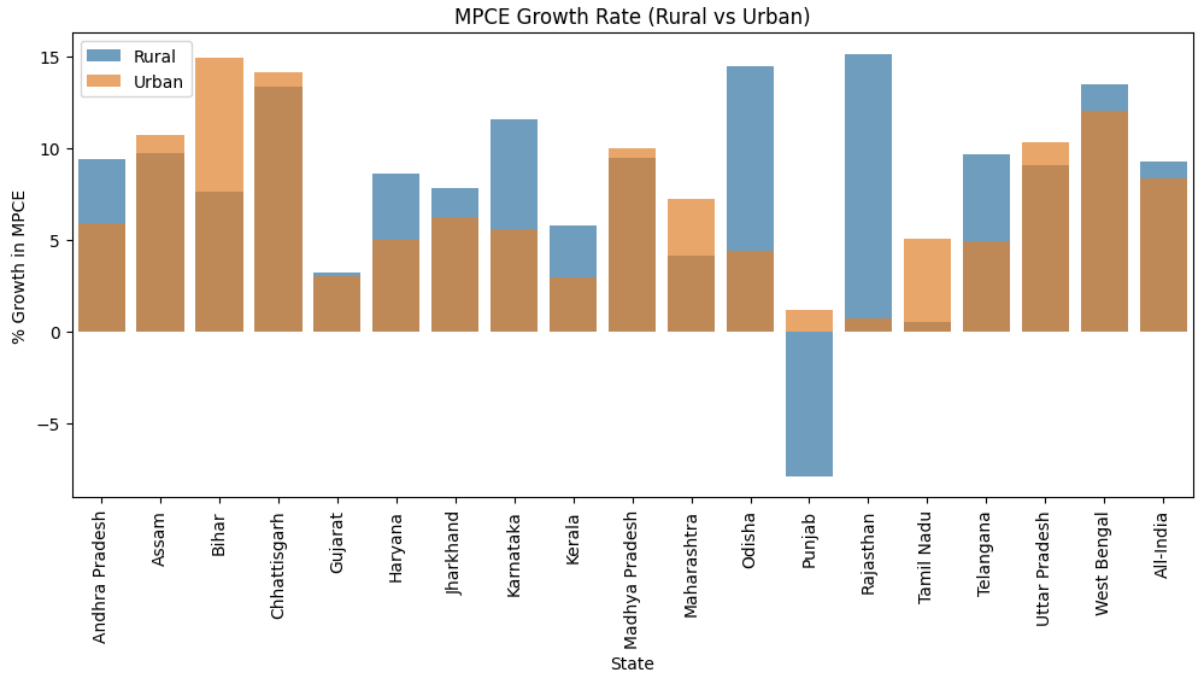


Figure 23: MPCE Growth Rate (Rural vs Urban)

9.3 Urban-to-Rural MPCE Ratio Analysis

- **Kerala Leads in Disparity:** With a ratio of approximately 2.6, Kerala shows the most significant difference in Monthly Per Capita Expenditure (MPCE) between urban and rural areas. This indicates that urban households in Kerala spend considerably more than their rural counterparts.
- **Higher Ratios Across Developed States:** States like Maharashtra, Gujarat, and Tamil Nadu also exhibit higher ratios (above 2.0), reflecting similar urban-rural disparities, likely driven by economic opportunities and cost of living in urban areas.
- **Relatively Balanced Ratios in Some States:** States like Bihar, Chhattisgarh, and Uttar Pradesh show ratios closer to 1.5, suggesting more balanced expenditure patterns between rural and urban households.
- **All-India Average:** The All-India Urban-to-Rural MPCE ratio lies around 2.0, indicating that, on average, urban households spend twice as much as rural ones across the country.
- **Policy Implications:** The data highlights the need for targeted policies aimed at reducing urban-rural expenditure disparities, particularly in states with higher ratios like Kerala, while promoting rural development and economic opportunities.

MPCE Comparison Heatmap (2023-24)

- **Highest Urban MPCE (Kerala):** Kerala's urban MPCE value of Rs.9302 significantly exceeds its rural MPCE value of Rs.3611, showcasing the most pronounced urban-rural expenditure disparity across states.
- **Balanced States:** Bihar and Chhattisgarh display relatively balanced MPCE values, with rural MPCE values closer to urban ones compared to other states. Bihar's values are Rs.3426 (rural) and Rs.4792 (urban), while Chhattisgarh's are Rs.2796 (rural) and Rs.4781 (urban).
- **Higher Urban MPCE Nationwide:** Across all states, urban MPCE consistently surpasses rural MPCE, indicating higher expenditure trends in urban areas due to economic opportunities and cost of living differences.

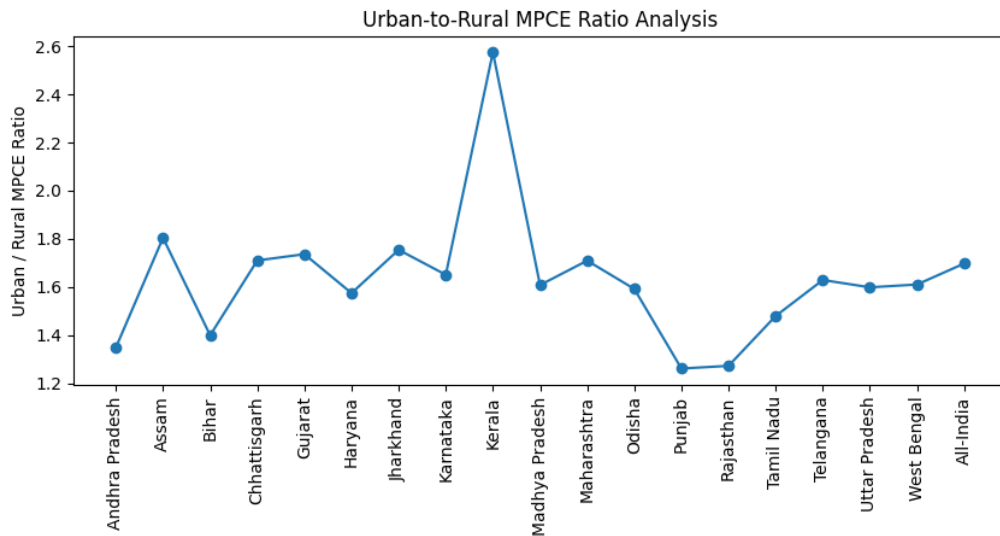


Figure 24: Urban-to-Rural MPCE Ratio Analysis

- **All-India Average:** The All-India average for rural MPCE is Rs.4122, while urban MPCE stands at Rs.6996, highlighting a nationwide disparity between rural and urban expenditures.
- **Policy Implications:** These patterns suggest the need for targeted policies aimed at reducing urban-rural disparities and improving economic conditions in rural areas, particularly in states with significant gaps like Kerala and Haryana.

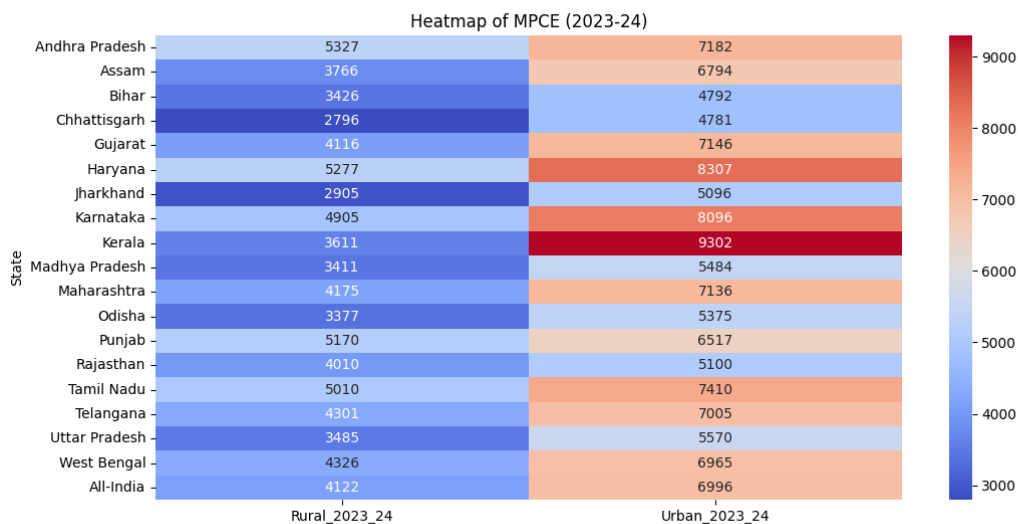


Figure 25: Urban-to-Rural MPCE Ratio Analysis

Predicted MPCE for 2024-25 (Rural vs Urban)

- **Higher Urban MPCE Nationwide:** The orange line, representing Urban MPCE, is consistently higher than the blue line for Rural MPCE across all states, indicating higher spending trends in urban areas due to factors such as cost of living and economic opportunities.

- **Kerala Leads in Urban MPCE:** Kerala shows the highest urban MPCE value among all states, reflecting higher living standards and expenditure patterns in urban households.
- **Narrower Gap in States like Bihar and Chhattisgarh:** States such as Bihar and Chhattisgarh exhibit relatively smaller gaps between rural and urban MPCE, indicating more balanced spending trends.
- **Consistent Growth Across States:** Both Rural and Urban MPCE display an increasing trend across states with higher expenditure levels observed in economically advanced states like Maharashtra, Gujarat, and Tamil Nadu.
- **All-India Trends:** The "All-India" data point reflects an average MPCE value where urban MPCE dominates rural MPCE, highlighting the nationwide urban-rural expenditure disparity.
- **Policy Implications:** These trends emphasize the need for targeted rural development policies to reduce disparities and promote balanced economic growth across states.

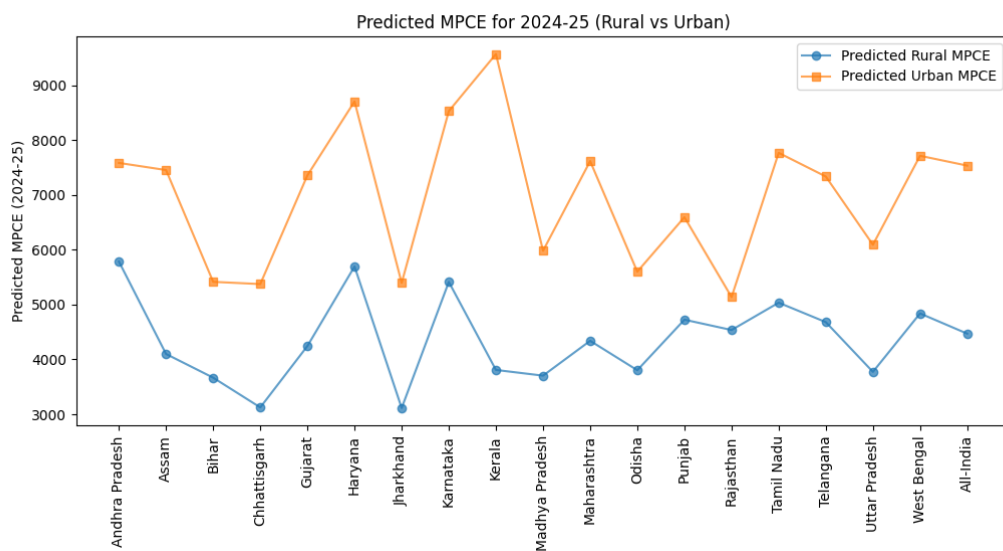


Figure 26: Predicted MPCE for 2024-25 (Rural vs Urban)

Urban-to-Rural MPCE Ratio Trend Analysis

- **Peak in Nominal MPCE Ratio (2009-10):** This marked the period of the highest urban-to-rural expenditure difference, illustrating a significant disparity in nominal spending.
- **Real MPCE Ratio Stability and Decline:** Initially steady, the Real MPCE Ratio displayed balanced inflation-adjusted spending patterns. The subsequent decline suggests narrowing economic disparities over time.

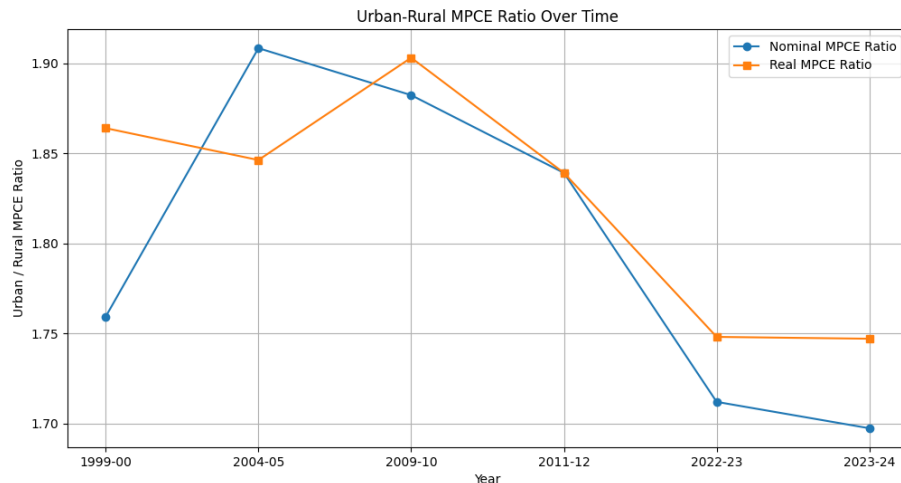


Figure 27: Urban-to-Rural MPCE Ratio Trends Over Time

- **Divergence Between Nominal and Real Ratios:** While the Nominal MPCE Ratio highlights temporal spending disparities, the Real MPCE Ratio points to longer-term economic convergence trends.
- **Implications for Policy Development:** These trends indicate potential benefits from targeted rural development policies, further emphasizing the necessity for balanced economic growth across urban and rural sectors.

Projected Future MPCE Analysis Using Exponential Trend

- **Exponential Growth in MPCE:** Both rural and urban MPCE show exponential growth trends. By 2030, the projected MPCE for urban areas is expected to exceed Rs 12,000, while for rural areas, it is projected to cross Rs 8,000.
- **Persistent Urban-Rural Gap:** Despite growth in both sectors, urban MPCE consistently remains higher than rural MPCE, indicating a persistent disparity in expenditure levels.
- **Convergence or Stability?:** The proportional increase in rural MPCE appears to outpace urban MPCE slightly, suggesting a potential narrowing of relative disparities in the future.
- **Policy Implications:** These projections underscore the need for sustained policy focus on rural development to bridge expenditure gaps while supporting sustainable urban growth.

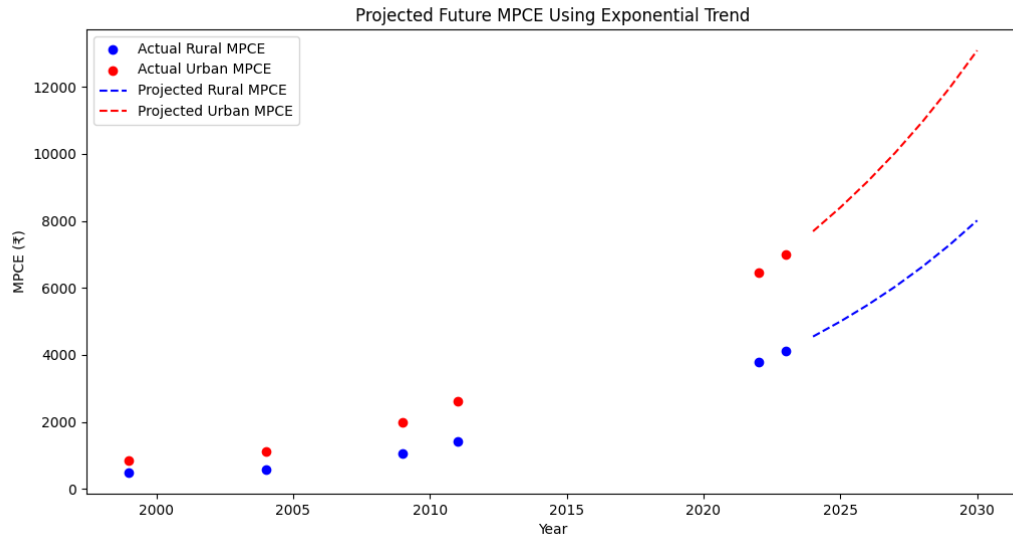


Figure 28: Projected Future MPCE Using Exponential Trend

10 Rural vs Urban MPCE Spending Patterns

- **Significant Gap in Food Total:** Urban households exhibit much higher MPCE for the "Food Total" category, indicating diverse and extensive food consumption compared to rural areas.
- **Higher Urban Spending Across Most Categories:** Urban MPCE surpasses rural MPCE significantly in:
 - Milk and Milk Products
 - Vegetables
 - Egg, Fish, and Meat
 - Beverages and Processed Foods
- **Narrower Gap in Basic Necessities:** Spending differences in categories like Cereals and Cereal Substitutes and Pulses and Their Products are smaller, reflecting similarities in essential food consumption.
- **Similar Spending in Pan, Tobacco, and Intoxicants:** Both rural and urban households display comparable expenditure in this category, suggesting similar consumption habits.
- **Higher Fuel and Light Costs for Urban Areas:** Urban households spend more on Fuel and Light, likely due to higher energy demands and costs associated with urban living.

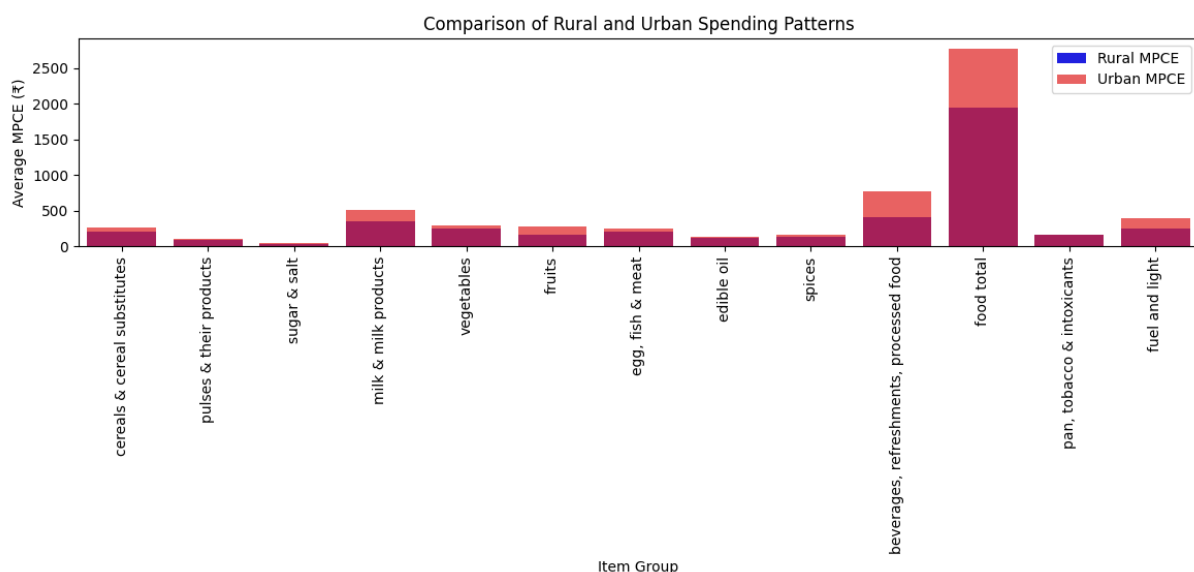


Figure 29: Rural vs Urban MPCE Spending Patterns

11 Rural vs Urban Expenditure Share Analysis

- **Rural Areas:**
 - Food Share: 57.0% of total expenditure.
 - Non-Food Share: 43.0% of total expenditure.
- **Urban Areas:**
 - Food Share: 47.6% of total expenditure.
 - Non-Food Share: 52.4% of total expenditure.
- **Urban-Rural Comparison:**
 - Rural households prioritize food, allocating a higher percentage of their expenditure to essentials.
 - Urban households have a more balanced spending pattern, with a larger proportion dedicated to non-food items like housing and education.
- **Policy Implications:**
 - Policies aimed at increasing rural discretionary income can help shift spending patterns and reduce dependency on food expenditures.
 - Enhancing access to non-food resources in rural areas might bridge the urban-rural divide.

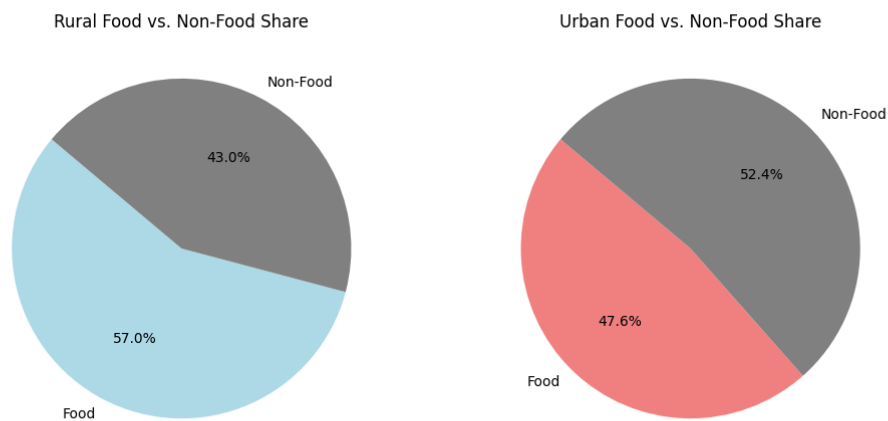


Figure 30: Rural vs Urban Expenditure Share

12 Healthy vs Processed Food Consumption Analysis

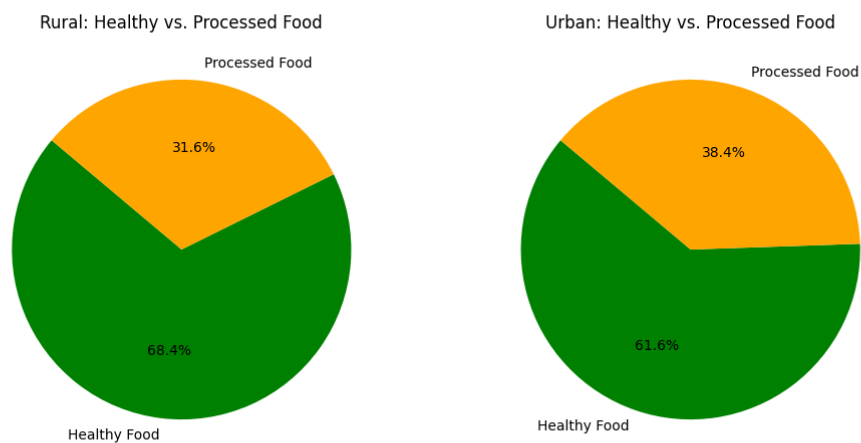


Figure 31: Healthy vs Processed Food Consumption in Rural and Urban Areas

- **Rural Areas:**
 - Healthy Food Share: 68.4% of total food consumption.
 - Processed Food Share: 31.6% of total food consumption.
- **Urban Areas:**
 - Healthy Food Share: 61.6% of total food consumption.
 - Processed Food Share: 38.4% of total food consumption.
- **Urban-Rural Comparison:**

- Rural households consume a higher proportion of healthy food compared to urban households.
- Urban areas exhibit greater processed food consumption, potentially reflecting urban lifestyle and accessibility.

- **Policy Implications:**

- Encouraging urban populations to shift towards healthier options through awareness campaigns and incentives.
- Supporting rural access to diverse and balanced food choices while preserving traditional eating habits.

12.1 Comparison of Rural vs Urban MPCE in Non-Food Categories

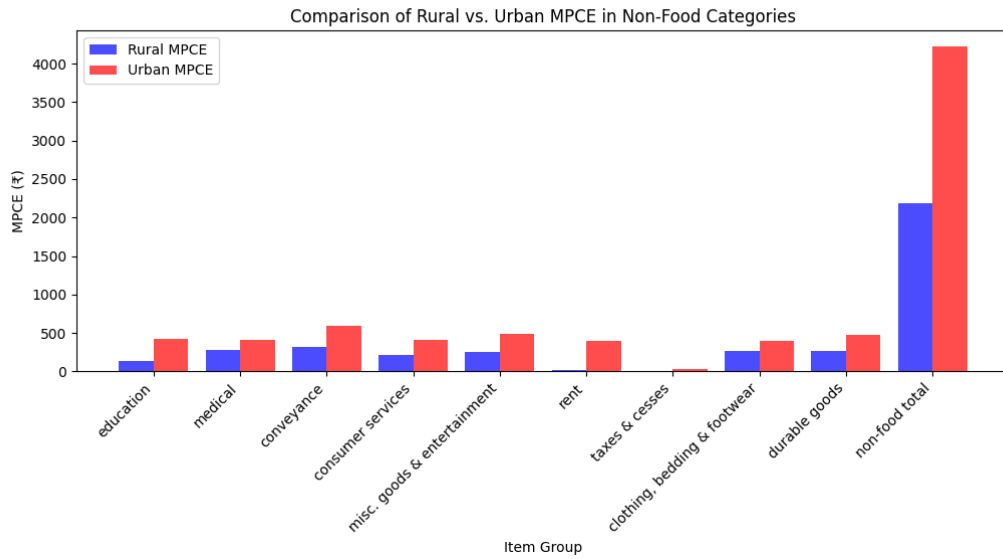


Figure 32: Comparison of Rural vs Urban MPCE in Non-Food Categories

- **General Observation:** Urban MPCE is consistently higher than Rural MPCE across all non-food categories.
- **Significant Gap in Non-Food Total:** Urban areas show the highest MPCE in the Non-Food Total category, highlighting broader spending capabilities in non-food sectors.
- **Key Categories with Notable Differences:**
 - **Education:** Urban MPCE is significantly higher, indicating greater expenditure on educational services.
 - **Medical:** Urban households spend noticeably more, reflecting either higher costs or more frequent access to healthcare services.
 - **Conveyance:** Urban spending outpaces rural, aligning with increased reliance on transport due to urban living conditions.
 - **Rent, Taxes, and Cesses:** Urban households allocate more funds in this category, consistent with higher costs of living in urban areas.
- **Smaller Disparities in Other Categories:**
 - Categories such as Clothing, Bedding and Footwear and Durable Goods show smaller MPCE differences, indicating relatively closer spending patterns between rural and urban populations.
- **Policy Implications:**

- The data underscores the need for tailored policies to address disparities in education and healthcare accessibility in rural areas.
- Enhanced rural spending capacity and infrastructure development could help bridge urban-rural gaps in key non-food sectors.

12.2 Percentage Share of MPCE in Rural Areas

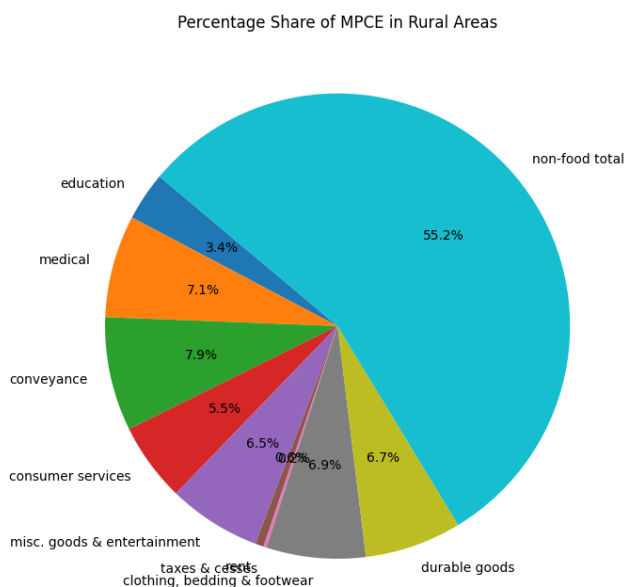


Figure 33: Percentage Share of MPCE in Rural Areas

- **Major Share in Non-Food Total:** 55.2% of rural expenditure is directed toward non-food items, reflecting significant allocation across various categories.
- **Durable Goods:** Constitutes 6.7% of total expenditure, representing investments in essential long-lasting items.
- **Clothing, Bedding, and Footwear:** Accounts for 6.9%, indicating spending on textiles and accessories.
- **Miscellaneous Goods and Entertainment:** Captures 6.5% of total expenditure, showcasing some discretionary spending.
- **Essentials:**
 - Conveyance: 7.9%
 - Medical: 7.1%
 - Consumer Services: 5.5%
 - Education: 3.4%
- **Taxes and Cesses:** Minimal expenditure at 0.6%, reflecting limited tax obligations in rural households.

12.3 Urban MPCE Spending Patterns

- **Major Share in Non-Food Total:** Non-food total accounts for 53.3% of urban expenditure, highlighting significant spending beyond essential food items.
- **Key Expenditure Categories:**
 - Conveyance: 7.5%
 - Education: 5.3%
 - Medical: 5.2%
 - Rent: 5.5%
 - Durable Goods: 5.8%
 - Taxes and Cesses: 5.3%
- **Miscellaneous Spending Patterns:**
 - Consumer services (5.1%) and miscellaneous goods and entertainment (6.1%) reflect discretionary expenditures within urban households.
 - Clothing, Bedding and Footwear: 6.1%, indicating moderate spending in these areas.
- **Policy Implications:**
 - These spending patterns underline the economic diversity of urban areas, with investments across essential and discretionary sectors.
 - Policymakers may aim to optimize urban resource allocations by improving access to education, healthcare, and conveyance.

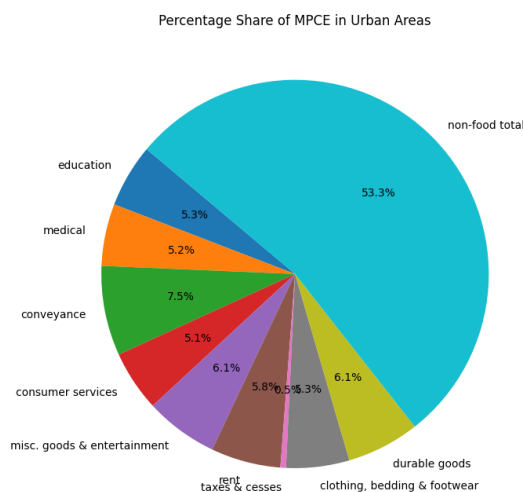


Figure 34: Percentage Share of MPCE in Urban Areas

12.4 Rural vs Urban MPCE in Non-Food Categories

- **Education:** Urban MPCE (**418**) is significantly higher than Rural MPCE (**133**), reflecting greater investment in education in urban households.
- **Medical Expenditure:** Urban households spend **409**, almost 50% more compared to rural households (**282**), highlighting higher access or cost of healthcare.

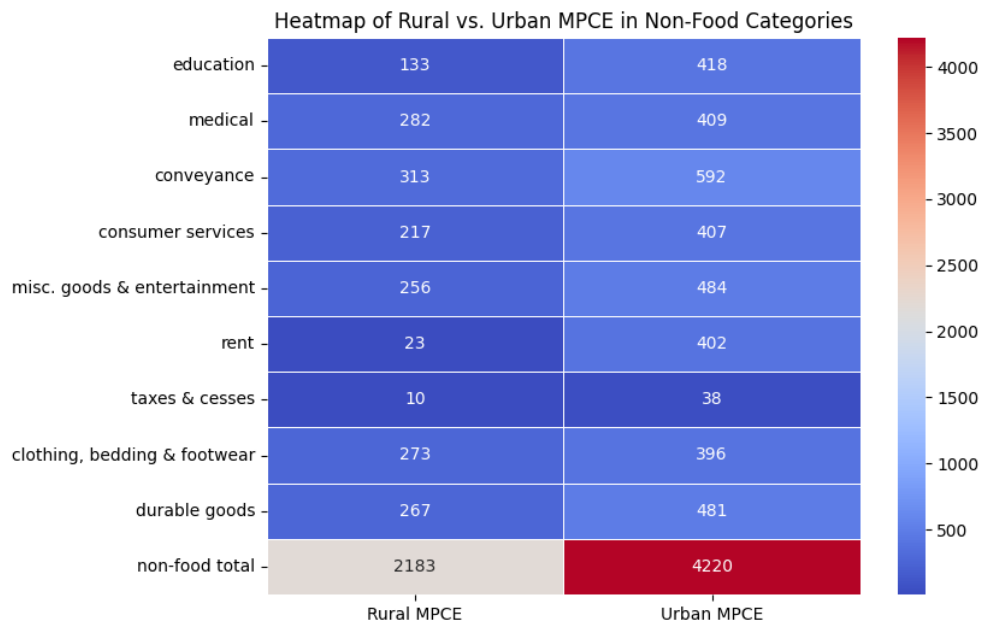


Figure 35: Rural and Urban MPCE Comparison Across Non-Food Categories

- **Conveyance:** Urban MPCE is nearly double (**592**) of rural areas (**313**), likely due to increased transportation needs in urban environments.
- **Non-Food Total:** The most significant disparity, with Urban MPCE at **4220** compared to Rural MPCE of **2183**, indicating broader urban expenditure capacities.
- **Smaller Disparities:**
 - Categories such as **Clothing, Bedding, and Footwear**, and **Durable Goods** show relatively closer spending patterns, suggesting common necessities across both sectors.
 - **Taxes and Cesses:** Minimal contributions, but urban households (**38**) still outpace rural (**10**) by a wide margin.
- **Policy Implications:**
 - Enhanced rural investment in education, healthcare, and transport infrastructure could help bridge these stark divides.
 - Targeted subsidies in essential sectors (e.g., durable goods, clothing) may alleviate economic burdens in rural areas.

13 Urbanization Impact on Food Share

- **Higher Food Share in Rural Areas:**
 - Food expenditure in rural areas consistently accounts for a larger percentage compared to urban counterparts across all states.
 - Represents a focus on subsistence-based consumption and dependency on agricultural produce.
- **Regional Trends:**
 - States like Kerala and Punjab exhibit narrower disparities, possibly due to economic and cultural factors.

- Larger gaps are observed in states like Bihar and Madhya Pradesh, where rural food share is significantly higher, indicating economic constraints.

- **Economic Implications:**

- Lower food share in urban areas reflects diversified expenditure patterns, allocating more resources to non-food items such as education, housing, and healthcare.
- Rural dependency on food expenditure underscores the need for infrastructure and development initiatives.

- **Policy Implications:**

- Targeted policies can help bridge the economic disparity by diversifying spending in rural areas.
- Development in healthcare, education, and transport infrastructure could facilitate balanced urban-rural growth.

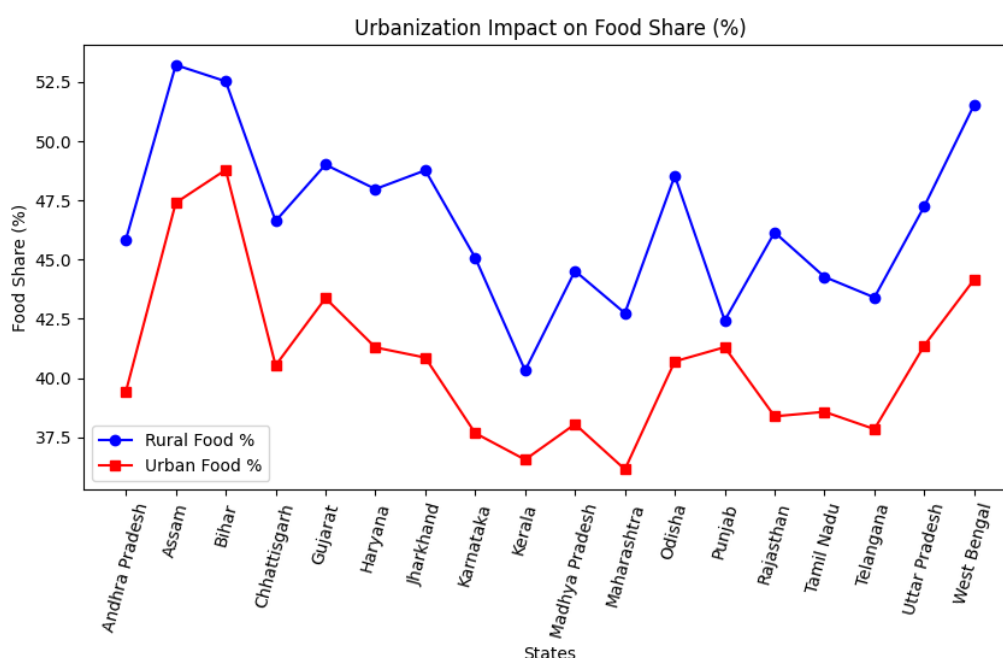


Figure 36: Urbanization Impact on Food Share (%) Across Indian States

Rural Food Expenditure by State in India

- **Top-Spending State (Assam):**
 - Assam ranks highest in rural food expenditure, potentially due to regional dietary preferences and reliance on agricultural staples.
- **Lowest-Spending State (Kerala):**
 - Kerala records the lowest expenditure, which may indicate more diversified spending patterns or differences in food prices and access.
- **Middle-Tier States:**
 - States like Jharkhand, Odisha, and Haryana reflect moderate food expenditures, showcasing varying levels of economic activity and consumption behaviors.
- **Regional Trends:**
 - Northern states such as Haryana and Rajasthan exhibit relatively lower food expenditures compared to northeastern states like Assam and Bihar, highlighting economic and cultural disparities.
- **Policy Implications:**
 - Addressing economic disparity through targeted food subsidy programs in high-spending states.
 - Promoting infrastructure and economic development in lower-spending states to improve overall expenditure and quality of life.

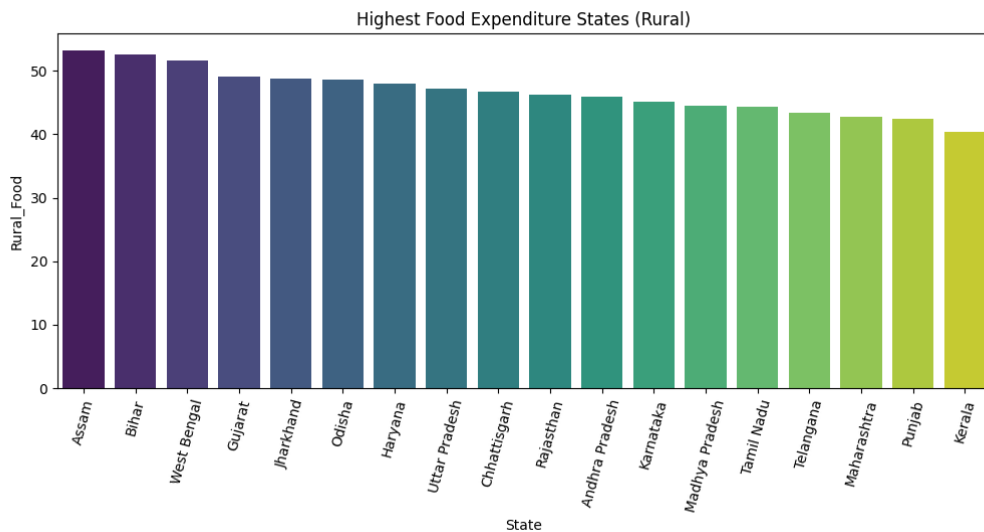


Figure 37: Highest Food Expenditure States (Rural)

13.1 Cereal % vs. Food % (Rural vs Urban)

- **Plot Overview:** The scatter plot compares the percentage share of cereal consumption (x-axis) against the percentage share of food consumption (y-axis) in rural and urban areas.
- **Rural vs Urban Distribution:**
 - Rural data points (blue dots) are more concentrated toward higher cereal and food share percentages, showcasing a higher dependency on cereals.
 - Urban data points (red dots) are distributed toward lower cereal and food share percentages, reflecting a diversified diet with lesser emphasis on cereals.
- **Trends and Observations:**
 - There is a visible correlation where higher cereal share corresponds to higher food share, especially prominent in rural areas.
 - Urban areas show less reliance on cereals, with data points skewed toward the lower end of the x-axis.
- **Economic Implications:**
 - Higher cereal share in rural areas could indicate limited access to diverse food items or a focus on subsistence farming.
 - Urban dietary diversity highlights the impact of higher incomes and better access to varied food sources.
- **Policy Insights:**
 - Policies targeting food security in rural areas could promote access to diverse and nutritious food.
 - Urban health policies should address the balance between dietary diversity and nutritional adequacy.

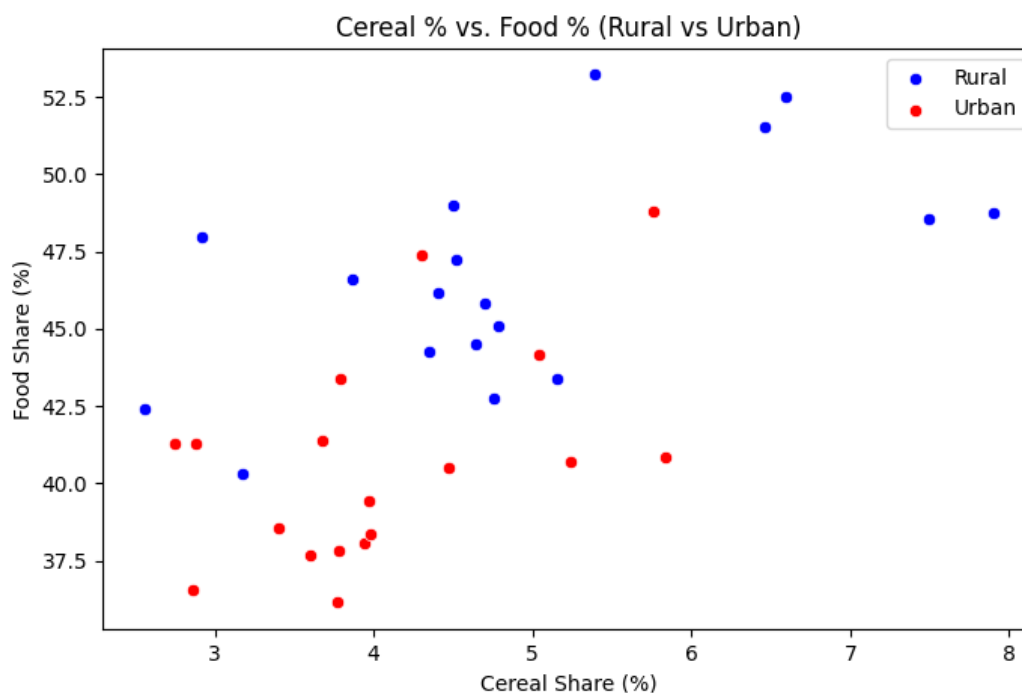


Figure 38: Cereal % vs. Food % (Rural vs Urban)

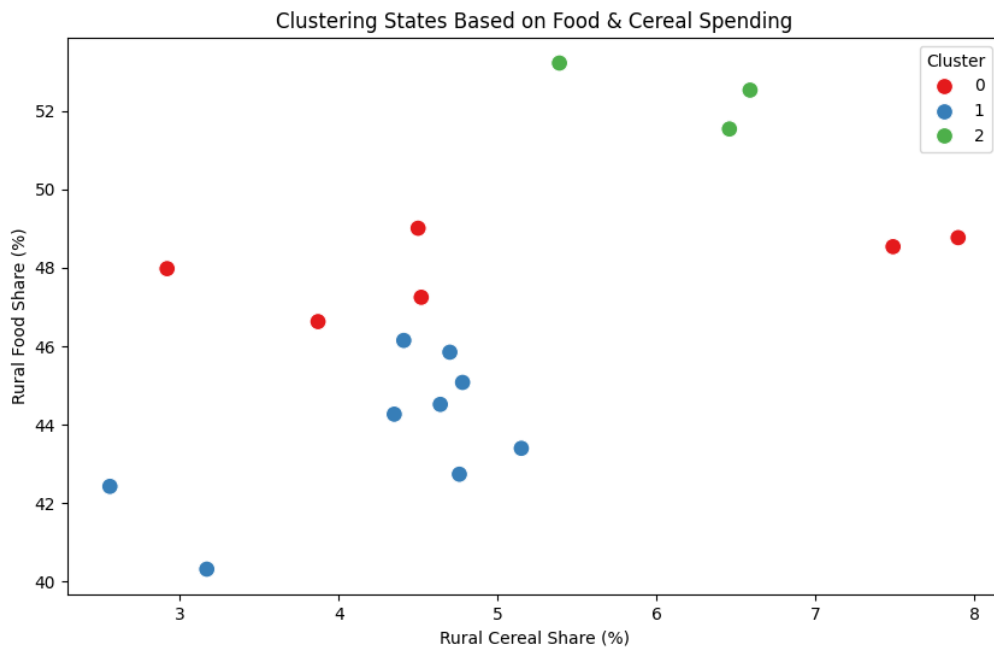


Figure 39: Clustering States Based on Food and Cereal Spending

13.2 Clustering States Based on Food and Cereal Spending

- **Visualization Summary:**

- The scatter plot maps states based on their Rural Cereal Share (%) against Rural Food Share (%).
- States are grouped into three clusters represented by different colors: Cluster 0 (red), Cluster 1 (blue), and Cluster 2 (green).

- **Cluster Analysis:**

- **Cluster 0 (Red):**
 - * Includes states with both low Rural Cereal Share and Rural Food Share.
 - * May represent states where diversified food consumption patterns are observed in rural areas.
- **Cluster 1 (Blue):**
 - * Contains states with higher Rural Cereal Share and moderate Rural Food Share percentages.
 - * Reflects a dependency on cereals as a major component of food expenditure.
- **Cluster 2 (Green):**
 - * Represents states with the highest Rural Food Share and Rural Cereal Share.
 - * Indicates reliance on agricultural staples and limited dietary diversity.

- **Policy Implications:**

- Cluster 0 states may benefit from policies promoting cereal consumption or access to staple foods.
- Cluster 1 states might require initiatives that encourage dietary diversification in rural areas.
- Cluster 2 states could benefit from programs enhancing economic growth and access to diverse food items.

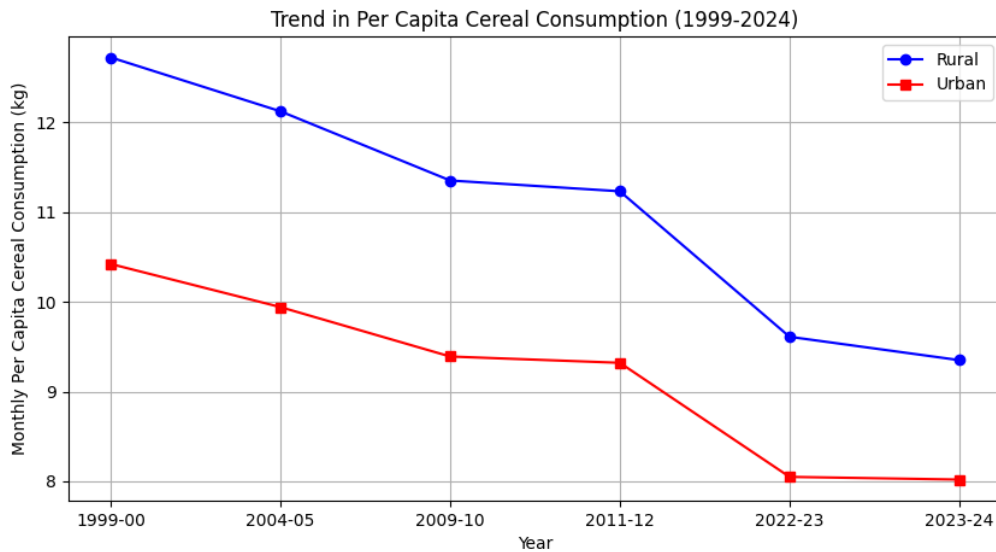


Figure 40: Trend in Per Capita Cereal Consumption (1999–2024)

13.3 Analysis of Per Capita Cereal Consumption Trend (1999–2024)

- **Decline in Consumption for Both Sectors:**
 - Rural areas show a steeper decline compared to urban areas, indicating changing consumption patterns or reduced reliance on cereals.
 - Urban areas also witness a gradual drop, though the extent is less pronounced than in rural regions.
- **Comparison Between Rural and Urban Trends:**
 - Rural areas start with higher cereal consumption in 1999–00 but converge closer to urban levels by 2024.
 - The disparity between rural and urban cereal consumption narrows over time, reflecting shifts in dietary habits or economic factors.
- **Year-Wise Observations:**
 - 1999–2000: Cereal consumption is at its peak, with rural households consuming significantly more than urban counterparts.
 - 2011–2012: A steady decline is observed across both rural and urban sectors.
 - 2022–2024: The lowest recorded cereal consumption is projected, highlighting ongoing changes in food preferences and economic scenarios.
- **Possible Contributing Factors:**
 - Dietary Diversification: Increased consumption of other food groups like proteins and processed foods.
 - Economic Growth: Enhanced income levels leading to diverse eating habits in urban areas.
 - Lifestyle Changes: Rising preference for ready-to-eat foods, especially in urban households.
- **Policy Implications:**
 - Rural areas may require interventions to ensure nutritional adequacy amid declining cereal consumption.
 - Strategies to promote balanced diets and support for agricultural production might address these trends effectively.

13.4 Analysis of Total Cereal Consumption by State (kg per month)

- **Highest Consumption (Maharashtra):**
 - Maharashtra records the highest cereal consumption per month among Indian states, reflecting substantial dietary reliance on cereals.
- **Lowest Consumption (Kerala):**
 - Kerala has the lowest cereal consumption, potentially due to dietary preferences or availability of alternative staple foods.
- **Moderate Consumption States:**
 - States like Andhra Pradesh, Tamil Nadu, and Gujarat show moderate consumption levels, suggesting diverse diets with a balanced reliance on cereals and other food groups.
- **Regional Patterns:**
 - Northeastern states such as Assam exhibit relatively high cereal consumption, indicative of staple food habits in these regions.
 - Southern states, particularly Kerala, appear to have more diversified consumption patterns with lower dependency on cereals.
- **Policy Implications:**
 - High cereal consumption in states like Maharashtra and Assam may require support for sustainable agricultural production and storage.
 - States with lower cereal consumption, such as Kerala, might benefit from nutritional awareness programs to ensure balanced diets.

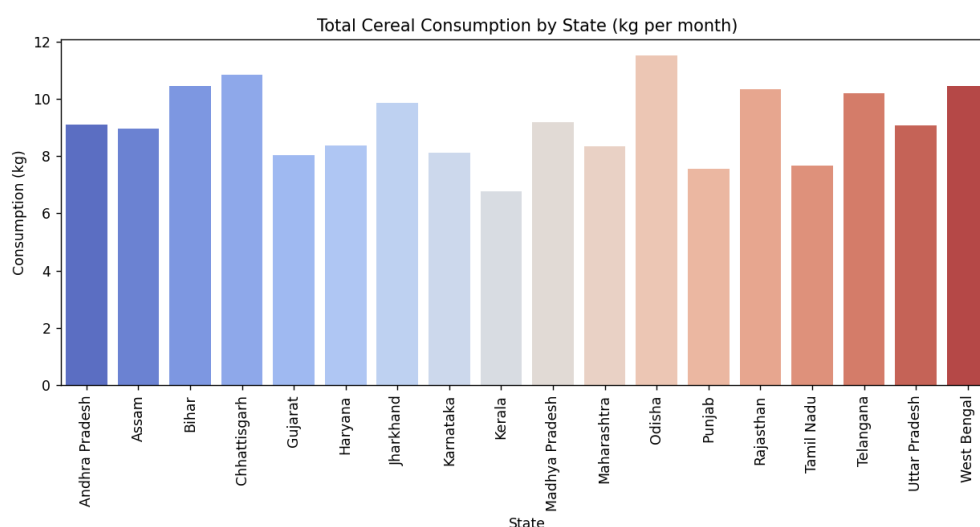


Figure 41: Total Cereal Consumption by State (kg per month)

13.5 Correlation of Cereal Consumption

- **Correlation Highlights:**
 - **Monthly Per Capita Cereal Consumption (kg):**
 - * Positively correlated with % Share of Rice (0.23), indicating rice consumption contributes slightly to higher overall cereal consumption.

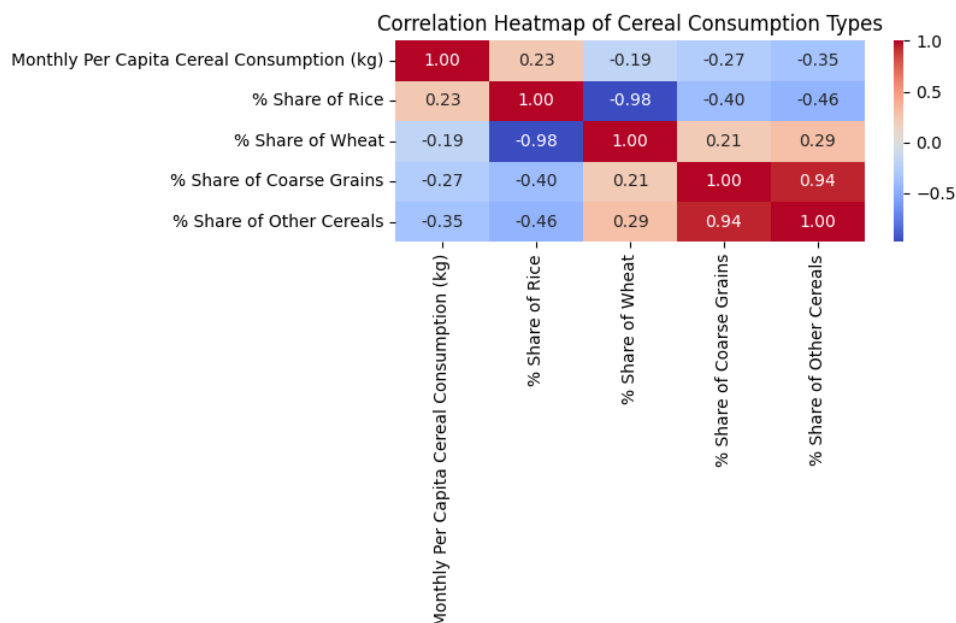


Figure 42: Correlation Heatmap for Cereal Consumption and Monthly Per Capita Cereal Consumption (kg)

- * Negatively correlated with % Share of Wheat (-0.19), % Share of Coarse Grains (-0.27), and % Share of Other Cereals (-0.35), suggesting these cereals are less commonly associated with higher total cereal consumption.
- **% Share of Rice:**
 - * Strongly negatively correlated with % Share of Wheat (-0.98), implying that regions with high rice consumption tend to have low wheat consumption.
 - * Moderately negatively correlated with % Share of Coarse Grains (-0.40) and Other Cereals (-0.46).
- **% Share of Coarse Grains and Other Cereals:**
 - * Strong positive correlation (0.94) between these two categories, highlighting co-consumption patterns.

• Policy Implications:

- Regions with high reliance on rice or wheat may need diversification programs to include coarse grains and other cereals for balanced nutrition.
- Addressing the negative correlations between rice/wheat consumption and coarse grains could promote greater cereal diversity.

13.6 Cereal Consumption Clustering

• Scatter Plot Overview:

- The scatter plot compares Rice Consumption (%) on the x-axis with Wheat Consumption (%) on the y-axis.
- States are clustered into three distinct groups based on their cereal consumption patterns:
 - * **Cluster 0 (Red):** Low Rice Consumption, High Wheat Consumption.
 - * **Cluster 1 (Blue):** Balanced Rice and Wheat Consumption.
 - * **Cluster 2 (Green):** High Rice Consumption, Low Wheat Consumption.

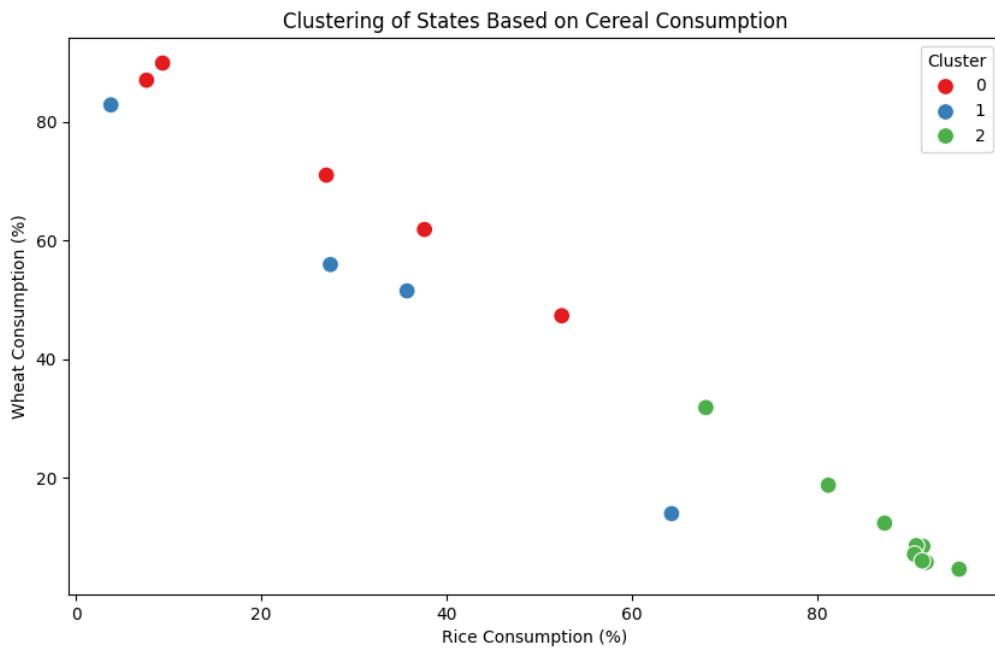


Figure 43: Clustering of States Based on Cereal Consumption Patterns

- **Key Observations:**

- States in Cluster 2 show a strong preference for rice, typical in regions where rice is a staple, e.g., southeastern states.
- Cluster 0 highlights states reliant on wheat, commonly seen in northern regions.
- Cluster 1 demonstrates balanced cereal consumption, indicating dietary diversity in these states.

- **Policy Implications:**

- Promoting dietary diversity through awareness programs in regions overly reliant on one type of cereal.
- Ensuring access to a variety of cereals, particularly in states that display skewed preferences.
- Aligning agricultural strategies with regional dietary patterns to optimize resource use.

14 Analysis of Clustering of States Based on Food Expenditure(Rural)

- **Plot Overview:**

- The scatter plot visualizes the relationship between Cereal Expenditure (%) and Processed Food Expenditure (%) across states in India.
- States are categorized into four clusters, each represented by a different color:
 - * **Cluster 0 (Red):** Odisha, Jharkhand.
 - * **Cluster 1 (Teal):** Gujarat, Chhattisgarh, Uttar Pradesh, Madhya Pradesh, Maharashtra, Telangana, West Bengal.
 - * **Cluster 2 (Green):** Kerala, Andhra Pradesh, Assam, Karnataka, Tamil Nadu.
 - * **Cluster 3 (Purple):** Punjab, Haryana, Rajasthan.

- **Cluster Analysis:**

- **Cluster 0 (Red):**
 - * States in this cluster have relatively high Cereal Expenditure but low Processed Food Expenditure.
 - * Likely represents rural or agrarian economies with traditional dietary patterns.
 - **Cluster 1 (Teal):**
 - * Includes states with moderate values for both Cereal and Processed Food Expenditure.
 - * Reflects transitional economies moving toward diversified food habits.
 - **Cluster 2 (Green):**
 - * Features states with low Cereal Expenditure and high Processed Food Expenditure.
 - * Indicates urbanized or economically advanced regions with modernized diets.
 - **Cluster 3 (Purple):**
 - * States exhibit a balance between Cereal and Processed Food Expenditure.
 - * Suggests regions with varied diets that include both traditional and processed food options.
- **Policy Implications:**
 - Cluster 0 states may benefit from initiatives encouraging dietary diversification and improved access to processed or fortified foods.
 - Cluster 1 states could emphasize infrastructure for food distribution to support economic and dietary transitions.
 - Cluster 2 states might focus on promoting balanced consumption of traditional staples alongside processed foods for nutritional adequacy.
 - Cluster 3 regions could leverage their diversity in food habits as a model for balanced regional diets.



Figure 44: Clustering of States Based on Food Expenditure

14.1 Clustering of States Based on Urban Food Expenditure

- **Scatter Plot Overview:**

- This scatter plot represents the relationship between Cereal Expenditure (%) and Processed Food Expenditure (%) in urban areas across Indian states.
- States are grouped into four distinct clusters, visually represented by different colors:
 - * **Cluster 0 (Red):** States such as Tamil Nadu, Telangana, Assam, Karnataka, and Maharashtra.
 - * **Cluster 1 (Blue):** Includes states like Kerala, Andhra Pradesh, Gujarat, and Chhattisgarh.
 - * **Cluster 2 (Green):** States such as Haryana, Punjab, and Rajasthan.
 - * **Cluster 3 (Purple):** Odisha, West Bengal, Bihar, and Jharkhand.

- **Cluster Characteristics:**

- **Cluster 0 (Red):**
 - * Shows higher reliance on processed food expenditure with comparatively lower cereal spending.
 - * Urbanization and lifestyle shifts likely influence these patterns.
- **Cluster 1 (Blue):**
 - * Represents a balance between cereal and processed food expenditure.
 - * May include transitioning urban economies with evolving dietary patterns.
- **Cluster 2 (Green):**
 - * Exhibits moderate cereal expenditure with diverse urban food habits including processed options.
 - * Indicates relative economic affluence and access to modern food markets.
- **Cluster 3 (Purple):**
 - * Skewed heavily towards cereal expenditure with lower processed food costs.
 - * Likely reflects the influence of traditional eating habits in less urbanized states.

- **Policy Implications:**

- For states in Cluster 3, urban food programs could promote dietary diversity and better access to processed food markets.
- Cluster 0 and 1 regions might benefit from awareness campaigns on balanced diets to ensure nutritional adequacy alongside lifestyle-driven food consumption.
- Cluster 2 urban areas could explore strategies to optimize food production and distribution systems for sustainable urban diets.

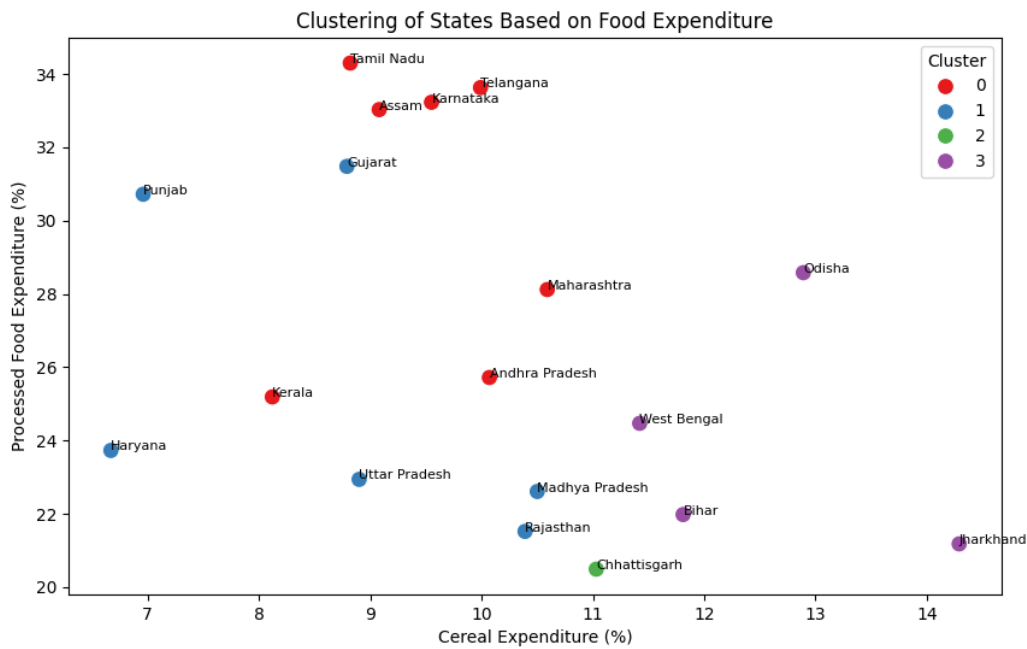


Figure 45: Clustering of States Based on Urban Food Expenditure

14.2 Analysis of State-wise Food Category Distribution

- **Cereal as a Staple:**
 - Cereals dominate expenditure across most states. For example, Bihar, Odisha, and West Bengal allocate a significant percentage to cereals, emphasizing their importance as dietary staples.
- **Milk & Milk Products:**
 - States like Haryana, Punjab, and Gujarat exhibit a larger proportion of expenditure on milk and dairy products, reflecting regional dietary preferences.
- **Fruits & Vegetables:**
 - Kerala and Tamil Nadu show higher expenditure on fruits and vegetables, showcasing dietary diversity and better access to fresh produce.
- **Protein Sources (Egg, Fish & Meat):**
 - Assam and Odisha highlight notable spending on protein-rich foods, driven by cultural and regional dietary habits.
- **Processed Foods & Beverages:**
 - Urbanized states like Maharashtra and Karnataka reveal increased spending on processed foods and beverages, influenced by lifestyle and convenience.
- **Other Expenditure:**
 - States such as Telangana and West Bengal include spending in miscellaneous categories, indicating a variety of food preferences unique to these regions.

Policy Implications

- **Cereal-centric Diets:** States heavily dependent on cereals could benefit from programs promoting nutritional diversity.
- **Regional Specialization:** Dairy-rich states like Punjab might focus on enhancing dairy production, while fish-consuming states like Odisha could support aquaculture initiatives.
- **Urban vs. Rural Patterns:** Rising processed food consumption in urban areas suggests a need for health awareness campaigns encouraging balanced diets.

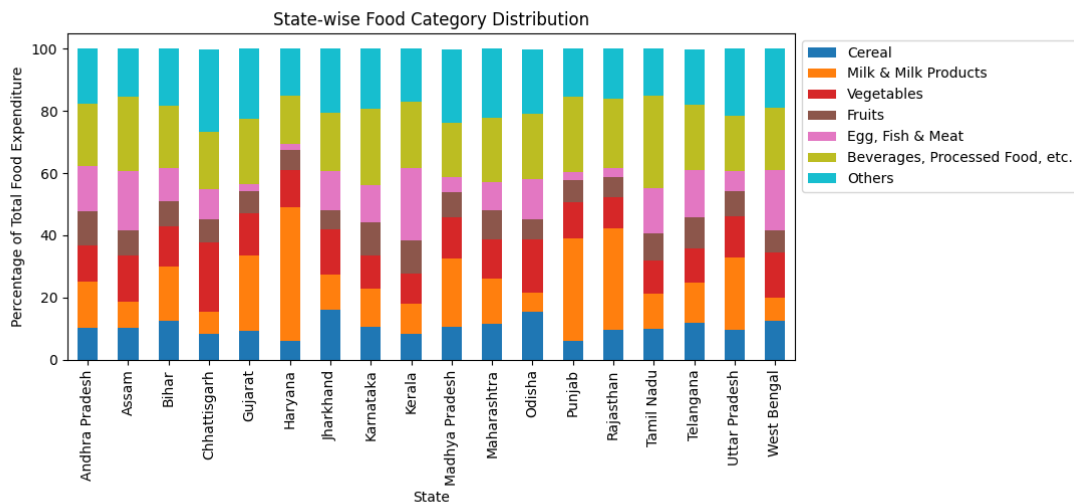


Figure 46: State-wise Food Category Distribution in India

15 Average MPCE by Social Group in Rural Areas

State	Scheduled Tribe (ST)	Scheduled Caste (SC)	Other Backward Class (OBC)	Others
Jammu Kashmir	3948	4806	3866	5092
Himachal Pradesh	5406	5250	5815	6190
Punjab	5150	5136	5921	6867
Chandigarh	20299	7643	8674	9473
Uttarakhand	4687	4213	4664	5617
Haryana	5688	4756	5529	5786
Delhi	6974	6862	6552	8901
Rajasthan	3384	4195	4854	5238
Uttar Pradesh	2980	3219	3480	3955
Bihar	3334	3374	3733	4017
Sikkim	9318	8079	9564	10170
Arunachal Pradesh	6048	5634	7815	5568
Nagaland	5151	5487	5064	—
Manipur	3997	4109	4688	5102
Mizoram	5948	9733	—	5901
Tripura	5803	6414	6637	6678
Meghalaya	3820	3043	4227	5174
Assam	3774	3758	3823	3785
West Bengal	3077	3507	3630	3791
Jharkhand	2497	2827	3260	3361
Odisha	2800	3235	3618	3924
Chhattisgarh	2471	3047	2902	3329
Madhya Pradesh	3004	3416	3582	3964
Gujarat	3690	3825	4008	5137
D N.Haveli Daman Diu	3943	5052	5147	6902
Maharashtra	3103	3986	4242	4599
Andhra Pradesh	4162	5145	5384	5715
Karnataka	4590	4620	4926	5566
Goa	6006	7623	7940	8638
Lakshadweep	6263	5394	16056	16624
Kerala	4908	5449	6262	7738
Tamil Nadu	4831	5340	5835	6202
Puducherry	—	6749	8013	16785
Andaman Nicobar Islands	5688	4133	7649	8307
Telangana	4981	5162	5482	6449
Ladakh	5010	—	—	—

Table 6: Average MPCE by Social Group in 2023-24 Across Indian States

- **Social Group Comparison:**

- **Others (Purple Bar):** Represents the highest average MPCE, indicating relatively better economic conditions among this group in rural areas.
- **OBC (Green Bar):** Second highest MPCE, showcasing moderate spending ability, potentially influenced by access to resources and income levels.
- **SC (Red Bar):** Third position in MPCE, reflecting limited economic expenditure compared to OBC and Others.
- **ST (Blue Bar):** The lowest average MPCE, highlighting economic constraints faced by this group in rural areas.

- **Observations:**

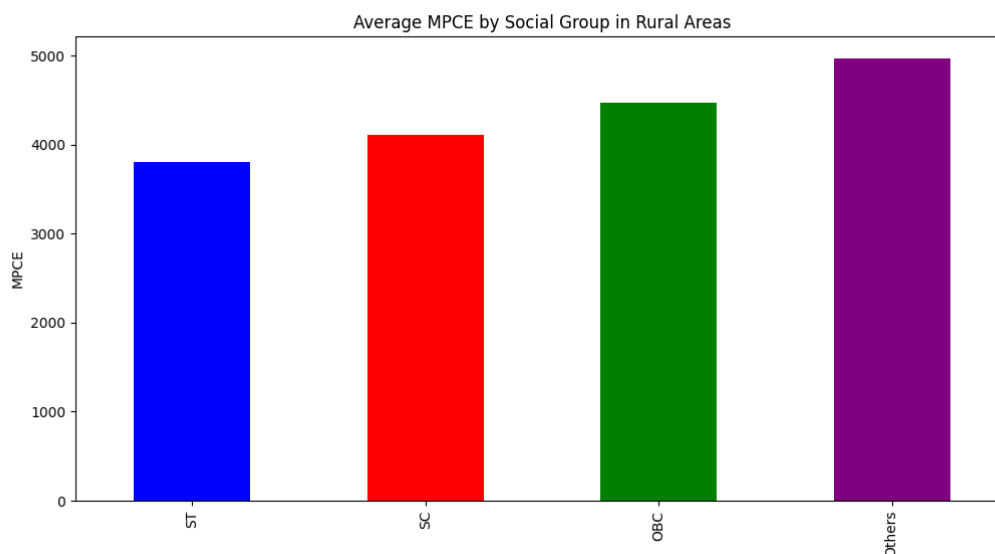


Figure 47: Average MPCE by Social Group in Rural Areas

- The disparity in MPCE across social groups underscores existing socio-economic inequities in rural areas.
- Differences in access to income, education, and resources are key contributors to this variation.

- **Policy Implications:**

- **Focus on ST and SC Groups:** Targeted policies aimed at improving income levels, education, and resource accessibility can help reduce MPCE disparities.
- **Strengthening Rural Economies:** Investment in infrastructure, skills development, and job creation in rural areas can benefit all social groups.
- **Inclusive Growth:** Programs promoting equity and inclusion across social groups can address systemic economic challenges.

15.1 Average MPCE by Social Group in Urban Areas

State	Scheduled Tribe (ST)	Scheduled Caste (SC)	Other Backward Class (OBC)	Others
Jammu Kashmir	4872	6605	6155	6372
Himachal Pradesh	12124	7669	7835	9996
Punjab	4039	6160	7054	8290
Chandigarh	—	9140	11601	14740
Uttarakhand	8513	6283	6306	8329
Haryana	11188	6080	7465	10113
Delhi	7116	7459	7855	9475
Rajasthan	6065	5050	6051	8011
Uttar Pradesh	5383	4918	5030	6522
Bihar	4208	4174	5120	5650
Sikkim	14160	11886	14298	13522
Arunachal Pradesh	10115	10560	8879	9376
Nagaland	8161	6835	6456	8638
Manipur	6406	5200	5929	6113
Mizoram	8707	9944	7721	—
Tripura	8714	7630	7617	8461
Meghalaya	7657	7763	9612	9180
Assam	6914	5918	7026	6825
West Bengal	4761	4704	4976	6172
Jharkhand	4761	3969	5321	6543
Odisha	4650	4611	5785	6807
Chhattisgarh	3988	4337	4596	6179
Madhya Pradesh	4445	4711	5221	6571
Gujarat	5837	5955	6415	7971
D N. Haveli Daman Diu	5804	6517	6912	7324
Maharashtra	5377	5956	6580	8250
Andhra Pradesh	6244	6686	7102	7543
Karnataka	5908	6584	7971	9614
Goa	9231	6772	8465	10089
Lakshadweep	6262	7262	12688	12931
Kerala	7688	6609	7165	9708
Tamil Nadu	8050	6662	8338	9139
Puducherry	14265	7140	8840	9976
Andaman Nicobar Islands	7218	11687	12439	10011
Telangana	9065	7680	8421	10893
Ladakh	7034	6157	—	9752

Table 7: Average MPCE by Social Group Across Indian States (2023-24)

• Social Group Comparison:

- **Others (Purple Bar):** Highest average MPCE, approximately 7500, showcasing better economic conditions and access to resources in urban settings.
- **OBC (Green Bar):** Slightly above 6000 MPCE, reflecting moderate expenditure levels and economic stability.
- **SC (Red Bar):** Slightly below 6000 MPCE, highlighting limited economic expenditure compared to OBC and Others.
- **ST (Blue Bar):** Lowest average MPCE, around 6000, indicating economic disparities faced by Scheduled Tribes in urban areas.

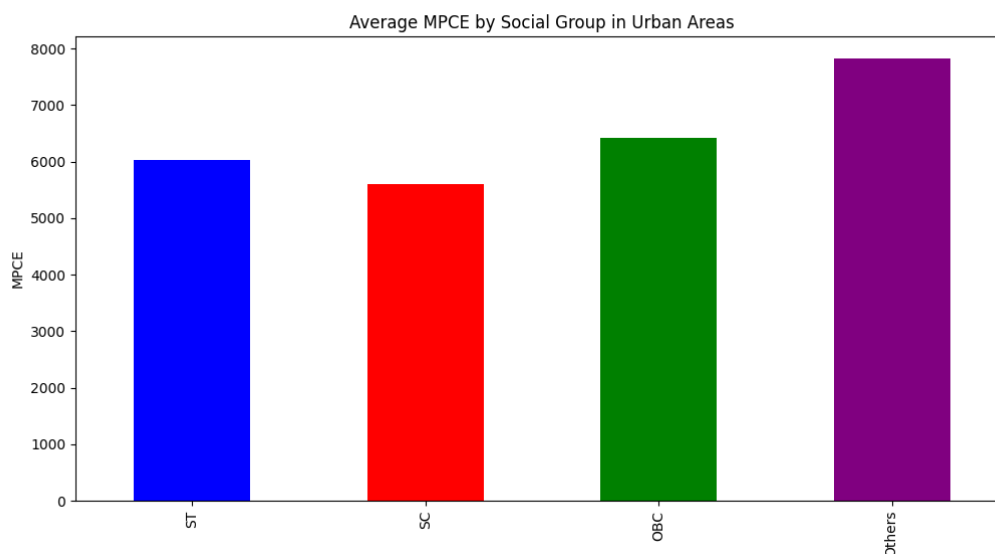


Figure 48: Average MPCE by Social Group in Urban Areas

- **Observations:**

- The urban MPCE disparities among social groups reveal ongoing socio-economic inequities despite better access to urban infrastructure.
- Factors such as income levels, education, and job opportunities contribute significantly to this variation.

- **Policy Implications:**

- **Support for ST and SC Groups:** Targeted policies for skill development and resource accessibility to uplift these social groups.
- **Inclusive Urban Growth:** Urban development strategies focused on equitable access to employment, education, and housing.
- **Economic Empowerment Programs:** Initiatives aimed at reducing income disparities across urban social groups.

16 Analysis of MPCE by Household Type Across Indian States (2023-24)

State	Self-Employment in Agriculture	Self-Employment in Non-Agriculture	Regular Wage/Salary in Agriculture	Regular Wage/Salary in Non-Agriculture	Casual Labour in Agriculture	Casual Labour in Non-Agriculture	Others
Jammu Kashmir	4785	5087	4502	5302	3213	3635	6091
Himachal Pradesh	5534	6156	10127	6391	4430	4663	7424
Punjab	6971	5772	4677	5776	5020	4738	7238
Chandigarh	9940	8291	6507	9234	6503	7974	10519
Uttarakhand	5124	5011	5511	5407	4341	4167	5352
Haryana	5829	5136	5102	5397	5055	4879	5841
Delhi	7467	7204	8176	7545	7565	6526	6823
Rajasthan	4509	4848	4645	5008	3760	3908	5553
Uttar Pradesh	3618	3513	3688	3769	3161	3131	3779
Bihar	3694	3954	3027	3878	3327	3478	4031
Sikkim	8240	9914	9378	10324	5044	8191	12471
Arunachal Pradesh	5288	6967	6085	7151	4629	5351	9590
Nagaland	4796	5661	5018	5919	4625	4388	6419
Manipur	3995	4686	4289	5265	4733	4443	5557
Mizoram	5035	6482	6449	6869	4344	5646	5275
Tripura	5823	6463	5780	7419	5690	5672	8119
Meghalaya	3684	4154	3550	4608	3177	3766	5482
Assam	3677	3907	3236	4505	3403	3587	4835
West Bengal	3522	3886	3618	4235	3223	3387	4156
Jharkhand	2674	3335	3058	3428	2743	2830	3240
Odisha	3181	3837	3844	4336	2976	3008	3236
Chhattisgarh	2622	2957	3369	3435	2606	2698	3066
Madhya Pradesh	3470	3778	3456	4201	3269	3072	4121
Gujarat	4207	4394	4287	4589	3441	3687	5725
D N Haveli Daman Diu	3979	5353	3964	4544	3111	2889	5459
Maharashtra	4099	4691	4184	5261	3298	3628	5013
Andhra Pradesh	5368	5525	6339	5797	4944	5097	5308
Karnataka	4670	5220	4590	6899	4097	4746	5822
Goa	8446	8543	7962	8219	5518	6596	7454
Lakshadweep	4626	6188	3596	6936	10225	5724	6524
Kerala	7272	6672	6072	6914	5137	5712	7506
Tamil Nadu	5521	5984	6131	6381	5069	5264	5860
Puducherry	7115	8241	10097	8285	6716	6696	8934
Andaman Nicobar Islands	7264	7681	9080	8394	4821	5461	8174
Telangana	5423	5724	4638	6056	4955	5082	5549
Ladakh	4687	5306	5433	5429	4501	6304	4492

Table 8: Average MPCE by Employment Type Across Indian States for Rural Sector (2023-24) Rural

State	Self-Employment in Agriculture	Self-Employment in Non-Agriculture	Regular Wage/Salary in Agriculture	Others
Jammu & Kashmir	6142	6689	4846	7459
Himachal Pradesh	8285	9216	5836	12367
Punjab	7624	7424	5618	8858
Chandigarh	12131	13703	8240	23047
Uttarakhand	7434	7764	5264	7462
Haryana	9184	8748	5250	10252
Delhi	8346	8573	7032	10821
Rajasthan	6204	7018	4267	10543
Uttar Pradesh	5015	6296	4080	7394
Bihar	5002	5428	4198	6298
Sikkim	12693	13799	12155	21921
Arunachal Pradesh	9035	10271	8476	14299
Nagaland	7652	8314	6313	9131
Manipur	5711	6388	5082	7882
Mizoram	8530	8937	7918	8164
Tripura	7603	8776	6195	10582
Meghalaya	7096	8548	5752	9603
Assam	6267	7468	4662	9433
West Bengal	5440	6348	3868	7898
Jharkhand	4883	5927	3440	7658
Odisha	5379	6563	3742	7193
Chhattisgarh	4987	5204	3478	5664
Madhya Pradesh	5380	6077	3973	7667
Gujarat	7337	7296	4856	8796
D N. Haveli Daman Diu	6572	6899	5411	10061
Maharashtra	7117	7551	4487	10170
Andhra Pradesh	6942	7687	5752	8535
Karnataka	6905	9800	5653	10108
Goa	9798	10036	7334	8500
Lakshadweep	5660	6970	6565	6038
Kerala	7420	8537	6357	8693
Tamil Nadu	8181	8581	6184	9693
Puducherry	9016	8815	7073	8701
Andaman Nicobar Islands	9687	10718	9055	12630
Telangana	7965	9217	6535	12619
Ladakh	7089	8356	8800	6496

Table 9: Average MPCE by Employment Type Across Indian States (2023-24) Urban

- The chart compares the **Monthly Per Capita Expenditure (MPCE)** across various household types in urban areas for different Indian states and All-India.
- Household types include:
 - * Self-Employment (blue bar)
 - * Regular Wage/Salary (orange bar)
 - * Casual Labour (green bar)
 - * Others (red bar)
 - * All (purple bar)
- **Key Observations:**
 - **Highest MPCE Category:**
 - * Regular Wage/Salary households exhibit the highest MPCE across most states, indicating economic advantages tied to steady income sources.

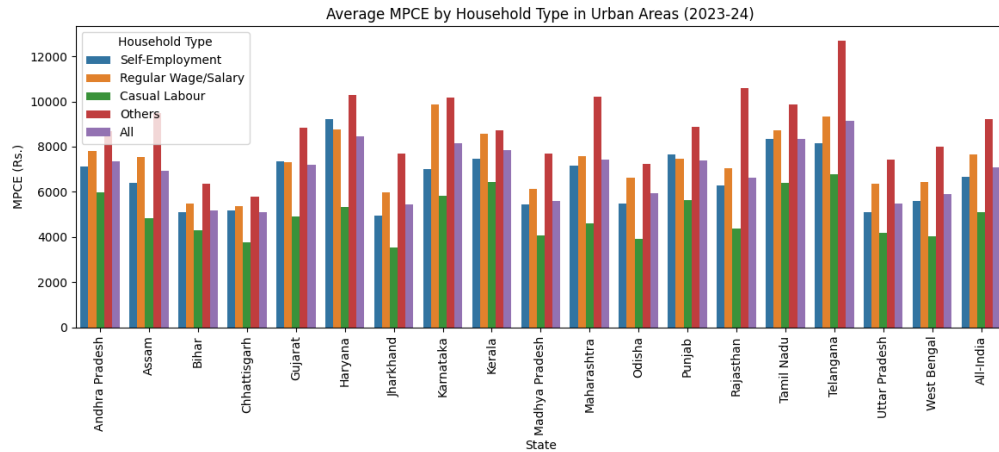


Figure 49: Average MPCE by Household Type in Urban Areas (2023-24)

- **Lowest MPCE Category:**
 - * Casual Labour households have the lowest MPCE, reflecting economic vulnerability and irregular income patterns.
- **All-India Level:**
 - * Regular Wage/Salary and Self-Employment categories show significantly higher MPCE compared to others.
- **State-Level Observations:**
 - **Kerala:** Exhibits the highest MPCE across all categories, highlighting robust urban economic conditions.
 - **Bihar:** Has the lowest MPCE, suggesting economic challenges in urban household expenditures.
 - **Gujarat and Maharashtra:** Demonstrate balanced MPCE distributions among household types.

Policy Implications

- **Support for Casual Labour Households:**
 - Targeted policies addressing job security and income stabilization can uplift casual labour households.
- **State-Specific Interventions:**
 - States like Bihar need focused development programs to enhance urban income levels.
 - Kerala's model of balanced urban economic growth could serve as a benchmark for other states.
- **Encouraging Self-Employment:**
 - Promoting micro-enterprises and entrepreneurship can boost MPCE for self-employed households.

- **Household Types Represented:**
 - Categories are: **Self-Employment** (blue), **Regular Wage/Salary** (light blue), **Casual Labour** (gray), **Others** (orange), and **All Categories** (red).
 - Each household type reflects the MPCE (Monthly Per Capita Expenditure) distribution in urban areas, measured in Indian Rupees.
- **Key Observations by States:**
 - Kerala:
 - * Highest MPCE across most household types, with the **Regular Wage/Salary** category standing out significantly.
 - Bihar:
 - * Exhibits the lowest MPCE across household types, highlighting significant economic disparities compared to other states.
 - Gujarat and Maharashtra:
 - * Showcase relatively balanced MPCE distributions among all household categories, suggesting stable urban income patterns.
- **Category Insights:**
 - **Regular Wage/Salary:**
 - * This category consistently records the highest MPCE in most states, signifying the economic advantages of steady employment.
 - **Casual Labour:**
 - * Lowest MPCE, illustrating economic vulnerability due to irregular income patterns.
 - **Self-Employment:**
 - * Falls between the extremes of Regular Wage/Salary and Casual Labour, reflecting variability in income levels within this group.

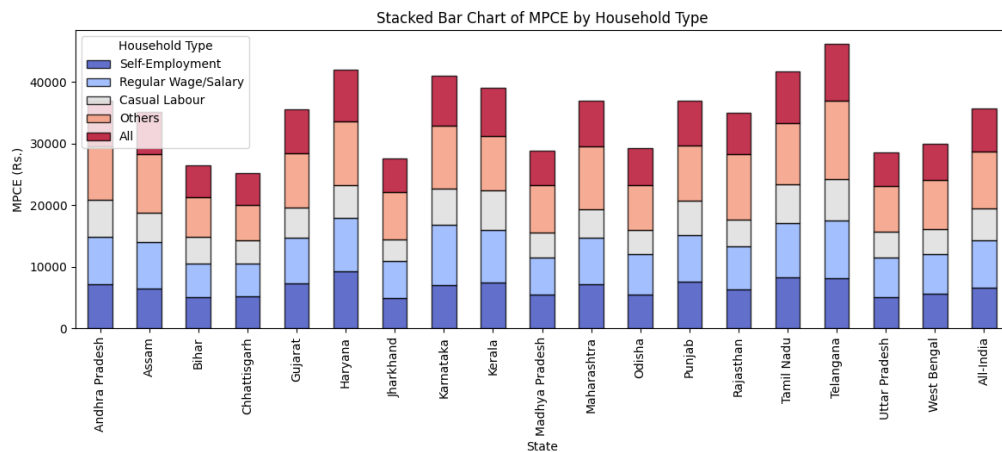


Figure 50: Stacked Bar Chart of MPCE by Household Type Across Indian States

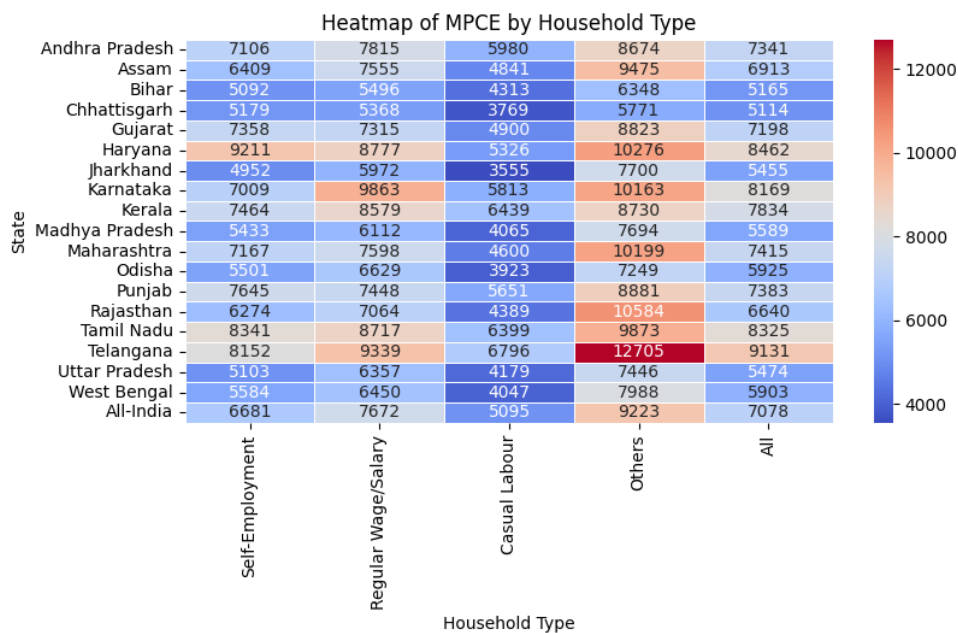


Figure 51: Heatmap of Monthly Per Capita Expenditure (MPCE) by Household Type Across Indian States

Policy Implications

- **Support for Casual Labour Households:**
 - Policies aimed at stabilizing income and expanding job opportunities can uplift this vulnerable segment.
- **State-Specific Interventions:**
 - Development programs in low-MPCE states like Bihar are crucial for reducing regional disparities.
- **Encouraging Self-Employment:**
 - Promoting small businesses and micro-enterprises can significantly boost income in the self-employed category.
- **Role Models:**
 - High-performing states like Kerala provide valuable lessons in achieving balanced urban economic growth.

16.1 MPCE by Household Types

- **Chart Overview:**
 - The box plot presents the **Monthly Per Capita Expenditure (MPCE)** in Indian Rupees (**Rs.**) across different household types in urban areas.
 - The y-axis denotes MPCE ranging from 4000 to 12000 Rs.
 - Categories analyzed: **Self-Employment**, **Regular Wage/Salary**, **Casual Labour**, **Others**, and **All Types**.
- **Key Insights:**
 - **Median MPCE Comparison:**
 - * **Regular Wage/Salary households** exhibit the highest median MPCE, indicating economic benefits from steady employment.
 - * **Casual Labour households** have the lowest median MPCE, reflecting income vulnerability and inconsistent earnings.
 - **Interquartile Range (IQR):**
 - * The widest IQR is observed in the **Self-Employment category**, indicating high variability in expenditure patterns within this group.
 - * **Regular Wage/Salary** has a relatively narrow IQR, suggesting consistent expenditure among salaried households.
 - **Whiskers and Outliers:**
 - * **Casual Labour** and **Others** exhibit lower whiskers, reflecting minimal MPCE in some households.
 - * Outliers in **Self-Employment** and **Regular Wage/Salary** indicate households with exceptionally high expenditure.

Policy Implications

- **Support for Vulnerable Groups:**
 - Policies aimed at stabilizing the incomes of **Casual Labour households** are vital for reducing expenditure disparities.
- **Encourage Entrepreneurship:**

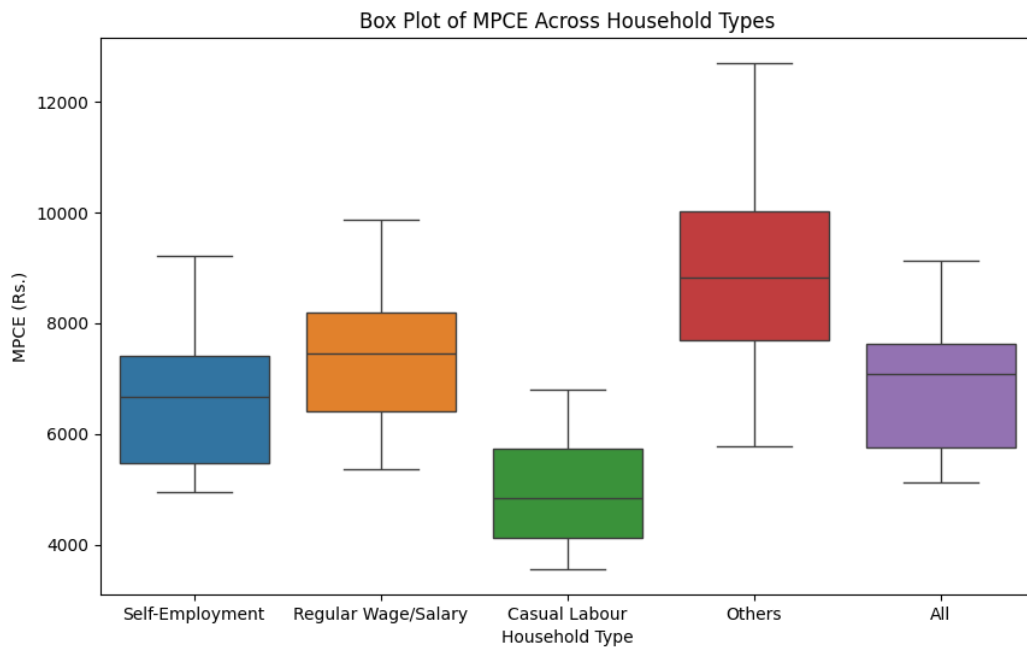


Figure 52: Box Plot of MPCE Across Household Types in Indian Urban Areas

- Supporting **Self-Employment** through skill training and financial inclusion can enhance economic stability in this group.
- **Inclusive Urban Growth:**
 - Leveraging the steady expenditure patterns of salaried households can guide urban economic planning and investment strategies.

16.2 Analysis of Top 5 States by MPCE in Self-Employment

- **Overview of the Chart:**

- The horizontal bar chart compares the Monthly Per Capita Expenditure (MPCE) for self-employment across five states in India.
- States listed include **Haryana, Tamil Nadu, Telangana, Punjab, and Kerala.**

- **Key Observations:**

- **Highest MPCE (Haryana):**
 - * Haryana leads with the highest MPCE value (close to Rs. 9000), indicating significant expenditure capacity among self-employed households.
- **Lowest MPCE (Kerala):**
 - * Kerala ranks fifth, with MPCE slightly below Rs. 7000, despite its reputation for strong economic growth and urban infrastructure.
- **Regional Patterns:**
 - * Southern states (Tamil Nadu, Telangana, Kerala) display relatively uniform MPCE values, suggesting consistent economic conditions across the region.
 - * Northern states (Haryana and Punjab) show higher expenditure levels, which may reflect varying income distributions or economic dynamics.

Policy Implications

- **Boost Self-Employment in Lower-MPCE States:**

- Kerala could benefit from targeted programs supporting self-employment ventures, aiming to increase income levels and expenditure capacity.

- **Learn from High-MPCE States:**

- Haryana's strong performance in self-employment expenditure can serve as a model for designing supportive policies in other states.

- **Regional Equity:**

- Promoting balanced regional development to address disparities between northern and southern states could enhance overall economic stability.

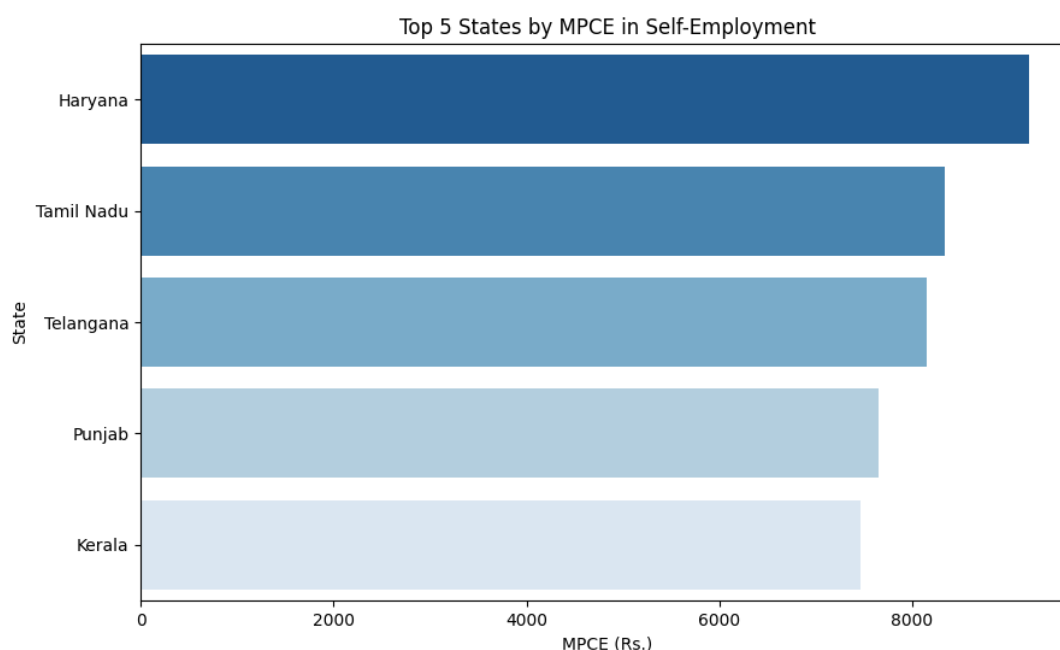


Figure 53: Top 5 States by MPCE in Self-Employment

16.3 Analysis of Bottom 5 States by MPCE in Self-Employment

- **Overview of the Chart:**

- The horizontal bar chart represents the Monthly Per Capita Expenditure (MPCE) of self-employed households for the bottom five states in India.
- States listed from top to bottom are: **Madhya Pradesh, Chhattisgarh, Uttar Pradesh, Bihar, and Jharkhand.**
- The x-axis denotes MPCE values ranging from 0 to 5000 Rs., with bars shaded from dark red (highest MPCE) to light red (lowest MPCE).

- **Key Observations:**

- **Lowest MPCE State (Jharkhand):**
 - * Jharkhand exhibits the lowest MPCE, with values barely crossing the 3000 Rs. mark.
- **Highest MPCE among Bottom 5 (Madhya Pradesh):**
 - * Madhya Pradesh leads among the bottom five states, with MPCE values nearing 5000 Rs., indicating relatively better spending capacity.
- **Regional Trends:**
 - * Eastern states like **Bihar** and **Jharkhand** consistently rank lower in MPCE, highlighting economic vulnerabilities in the region.
 - * Central states such as **Madhya Pradesh** and **Chhattisgarh** show slightly better MPCE values, suggesting moderate expenditure levels in these areas.

Policy Implications

- **Economic Support for Vulnerable States:**

- Targeted interventions in Bihar and Jharkhand can help boost incomes and improve expenditure levels among self-employed households.

- **Strengthening Regional Economies:**

- Focused development programs in eastern and central regions, such as infrastructure investments and market access, can help reduce disparities.

- **Promoting Self-Employment Ventures:**

- Incentives such as microfinance schemes and skill development initiatives can empower self-employed individuals and enhance spending capacity.

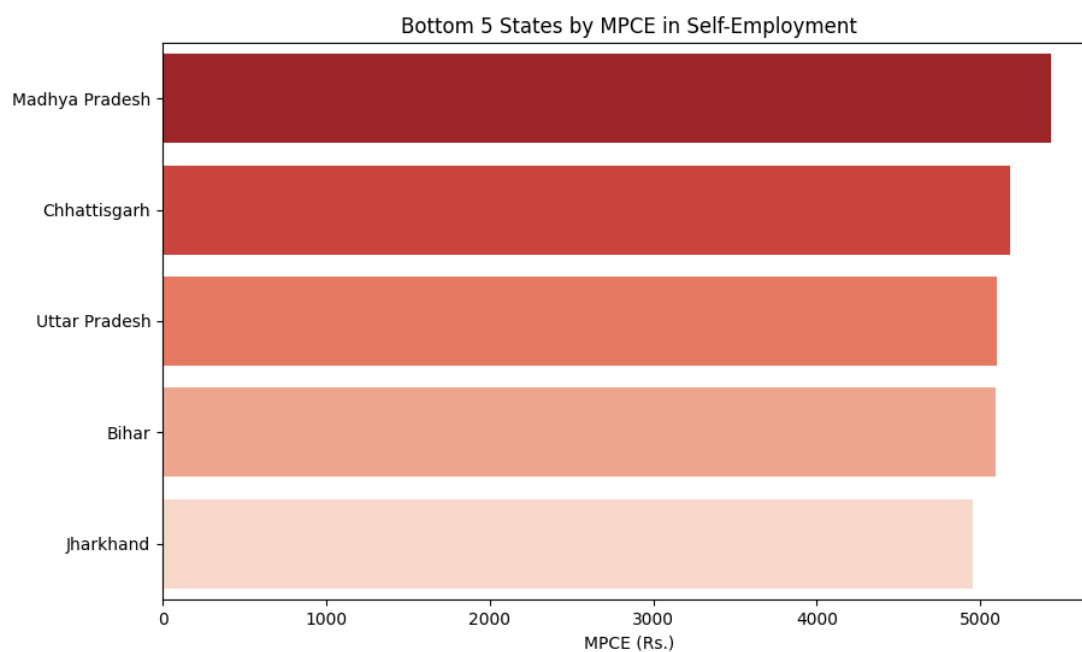


Figure 54: Bottom 5 States by MPCE in Self-Employment

16.4 Analysis of Top 5 States by MPCE in Regular Wage/Salary

- **Chart Overview:**

- The horizontal bar chart depicts the **Monthly Per Capita Expenditure (MPCE)** for regular wage/salary households in the top five states of India: **Karnataka, Telangana, Haryana, Tamil Nadu, and Kerala.**
- MPCE values range from 0 to 10,000 Rs. on the x-axis, represented by bars in varying shades of blue.

- **Key Observations:**

- **Highest MPCE State (Karnataka):**
 - * Karnataka leads with the highest MPCE, indicating strong economic conditions for salaried households.
- **Lowest MPCE among Top 5 (Kerala):**
 - * Kerala, while still among the top five, has the lowest MPCE of the group, suggesting relatively modest expenditure compared to the other states.
- **Regional Insights:**
 - * Southern states like Karnataka, Tamil Nadu, and Kerala dominate the list, showcasing robust economic activity in the south.
 - * Telangana and Haryana reflect significant urban infrastructure and economic opportunities supporting salaried households.

Policy Implications

- **Promote Wage/Salaried Employment:**

- States with lower MPCE in wage/salary households could benefit from skill development programs and education initiatives to support steady employment opportunities.

- **Regional Development Focus:**

- Lessons from Karnataka's strong economic performance can be utilized to strengthen salaried employment in other regions.

- **Investment in Urban Infrastructure:**

- Enhancing infrastructure and employment ecosystems in lower-ranking states can bridge regional disparities in expenditure levels.

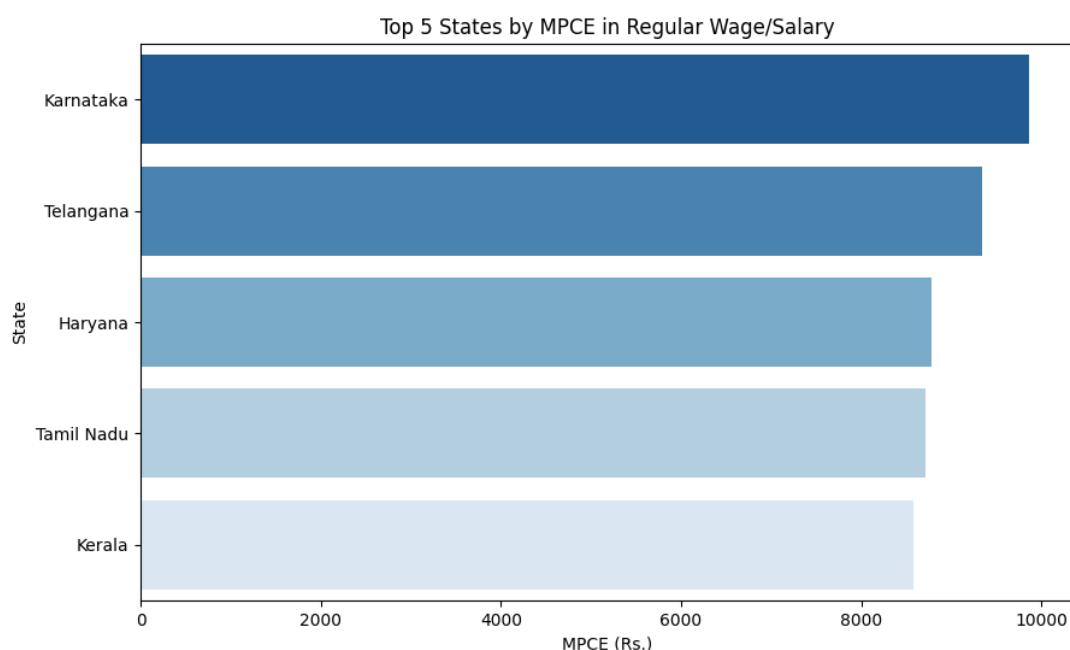


Figure 55: Top 5 States by MPCE in Regular Wage/Salary

16.5 Analysis of Bottom 5 States by MPCE in Regular Wage/Salary

- **Overview of the Chart:**

- This bar chart compares the Monthly Per Capita Expenditure (MPCE) of regular wage/salary households across the five states with the lowest MPCE: **Uttar Pradesh, Madhya Pradesh, Jharkhand, Bihar, and Chhattisgarh**.
- The x-axis displays the MPCE in Indian Rupees (Rs.), ranging from 0 to 6000, while the y-axis lists the states.

- **Key Observations:**

- **Highest MPCE among Bottom 5 (Uttar Pradesh):**
 - * Uttar Pradesh has the highest MPCE in this group, nearing 6000 Rs.
- **Lowest MPCE (Chhattisgarh):**
 - * Chhattisgarh records the lowest MPCE, approximately around 4000 Rs., indicating relatively limited expenditure capacity.
- **Regional Trends:**
 - * Northern and central states (e.g., Uttar Pradesh and Madhya Pradesh) show slightly better expenditure patterns compared to eastern states like Bihar and Jharkhand.

Policy Implications

- **Focus on Vulnerable States:**

- Bihar and Chhattisgarh could benefit from targeted economic policies that boost wages and provide better employment opportunities for regular wage/salary households.

- **Support Infrastructure Development:**

- Investment in infrastructure and urban development in these states can help improve MPCE values.

- **Skill Development Programs:**

- Introducing training and education initiatives to increase skill levels among workers may raise wage levels and, consequently, MPCE values.

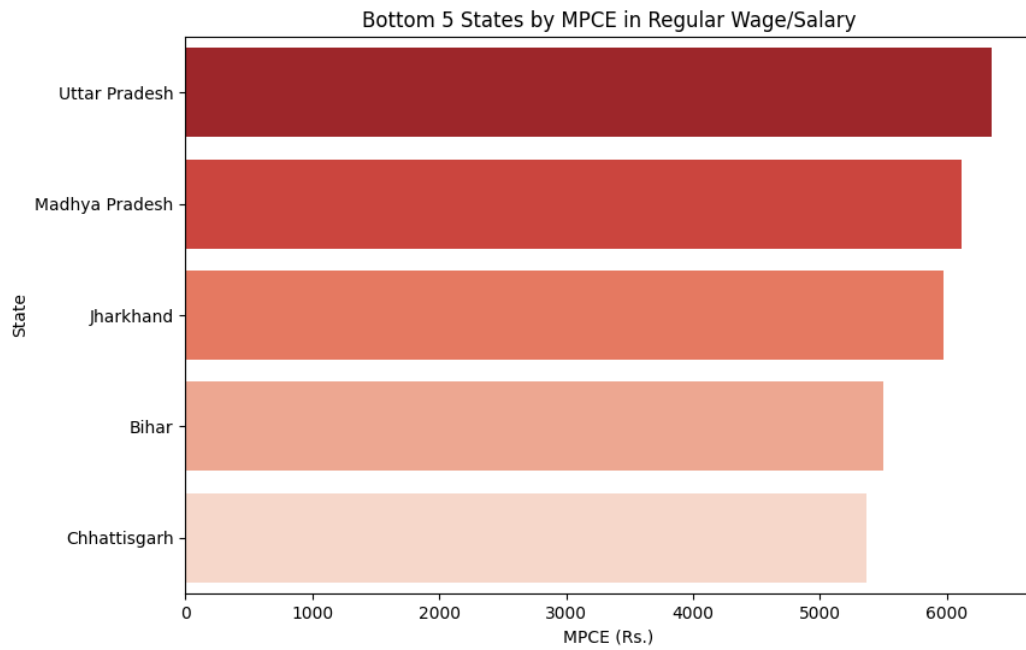


Figure 56: Bottom 5 States by MPCE in Regular Wage/Salary

16.6 Analysis of Top 5 States by MPCE in Casual Labour

- **Chart Overview:**

- The chart presents the Monthly Per Capita Expenditure (MPCE) for casual labor households in the top five states of India: **Telangana, Kerala, Tamil Nadu, Andhra Pradesh, and Karnataka**.
- The x-axis represents MPCE values in Indian Rupees (**Rs.**), while the y-axis lists the states in descending order based on expenditure.

- **Key Observations:**

- **Highest MPCE (Telangana):**
 - * Telangana leads with the highest MPCE among casual labor households, reflecting better economic conditions and spending capacity.
- **Lowest MPCE (Karnataka):**
 - * Karnataka ranks fifth, indicating relatively lower expenditure levels compared to other states in the list.
- **Regional Trends:**
 - * Southern states dominate the list, showcasing robust economic activity and support for casual laborers in the region.

Policy Implications

- **Strengthening Casual Labor Support:**

- Targeted policies in Karnataka could aim to increase incomes and improve living conditions for casual labor households.

- **Leveraging High-MPCE States:**

- States like Telangana and Kerala could serve as benchmarks for developing welfare programs in other regions.

- **Encouraging Regional Development:**

- Expanding economic opportunities for casual laborers across states can help bridge disparities in expenditure levels.

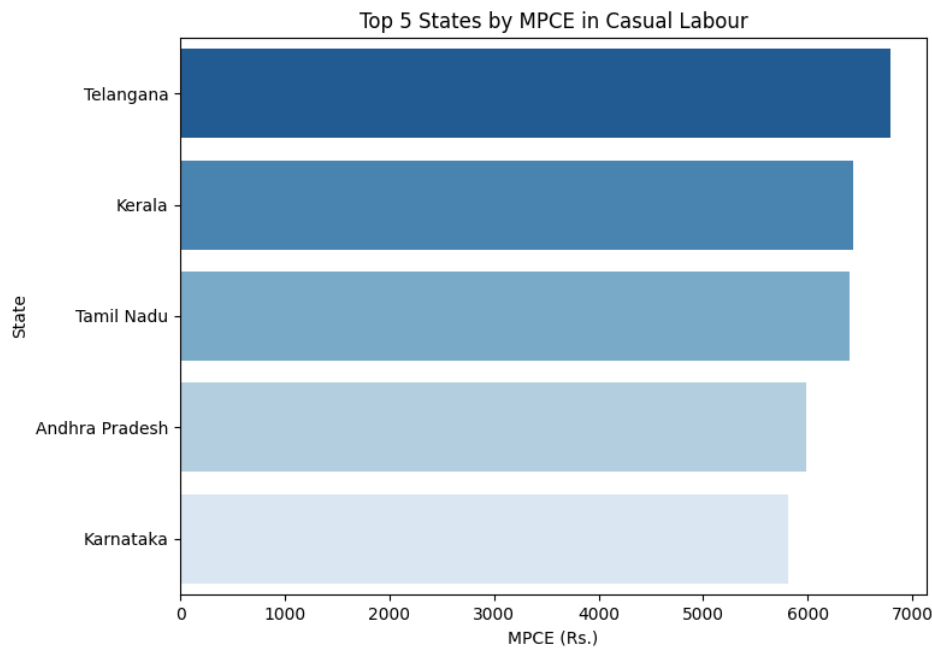


Figure 57: Top 5 States by MPCE in Casual Labour

16.7 Analysis of Bottom 5 States by MPCE in Casual Labour

- **Overview of the Chart:**

- This horizontal bar chart displays the **Monthly Per Capita Expenditure (MPCE)** for casual labour households across five states with the lowest MPCE.
- States listed from top to bottom are: **Madhya Pradesh, West Bengal, Odisha, Chhattisgarh, and Jharkhand.**
- MPCE values are shown on the x-axis in Indian Rupees (**Rs.**), ranging from 0 to 4000.

- **Key Observations:**

- **Highest MPCE among Bottom 5 (Madhya Pradesh):**
 - * Madhya Pradesh has the highest MPCE among the bottom 5 states, nearing 3500 Rs.
- **Lowest MPCE (Jharkhand):**
 - * Jharkhand records the lowest MPCE, barely exceeding 2500 Rs., indicating significant economic challenges for casual labour households.
- **Regional Trends:**
 - * Eastern states like **West Bengal, Odisha, and Jharkhand** consistently rank lower, underscoring economic vulnerabilities in the region.
 - * Central states such as **Madhya Pradesh** and **Chhattisgarh** show slightly better MPCE values but still fall below the national average.

Policy Implications

- **Economic Support for Vulnerable States:**

- Policies targeting income stabilization and welfare for casual labourers in Jharkhand and Odisha can reduce disparities.

- **Infrastructure Development:**

- Investments in rural and urban infrastructure in these states can provide better employment opportunities and improve household expenditures.

- **Skill Development and Education Initiatives:**

- Training programs aimed at enhancing skill levels of casual labourers can help elevate income levels and, consequently, MPCE values.

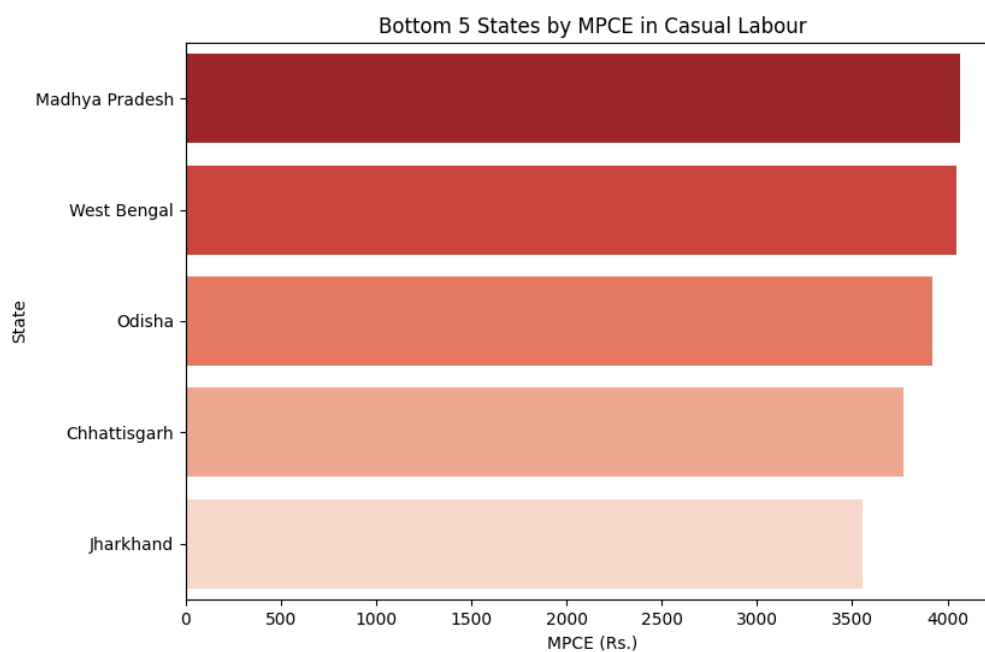


Figure 58: Bottom 5 States by MPCE in Casual Labour

16.8 Analysis of Top 5 States by MPCE in Others

- **Chart Overview:**

- This bar chart represents the Monthly Per Capita Expenditure (MPCE) for the "Others" category in the top five states of India.
- The states listed are: **Telangana, Rajasthan, Haryana, Maharashtra, and Karnataka.**
- The x-axis displays MPCE in Indian Rupees (**Rs.**), while the y-axis lists the states in descending order of expenditure.

- **Key Observations:**

- **Highest MPCE (Telangana):**
 - * Telangana leads the list with the highest MPCE, showcasing strong expenditure in this category.
- **Lowest MPCE among the Top 5 (Karnataka):**
 - * Karnataka ranks fifth but still exhibits significant MPCE, reflecting robust spending capacity.
- **Regional Patterns:**
 - * The inclusion of both southern states (e.g., Telangana and Karnataka) and northern/western states (e.g., Rajasthan and Haryana) highlights regional diversity in the "Others" category.

Policy Implications

- **Leveraging High-Performing States:**

- Telangana's strong performance in this category could be studied for replicable strategies to boost MPCE in other states.

- **Encouraging Balanced Growth:**

- States like Rajasthan and Haryana, which perform well in this category, could focus on translating these strengths into improved overall economic indicators.

- **Strengthening the "Others" Category Nationwide:**

- Investment in niche sectors or consumption categories falling under "Others" could spur economic growth across regions.

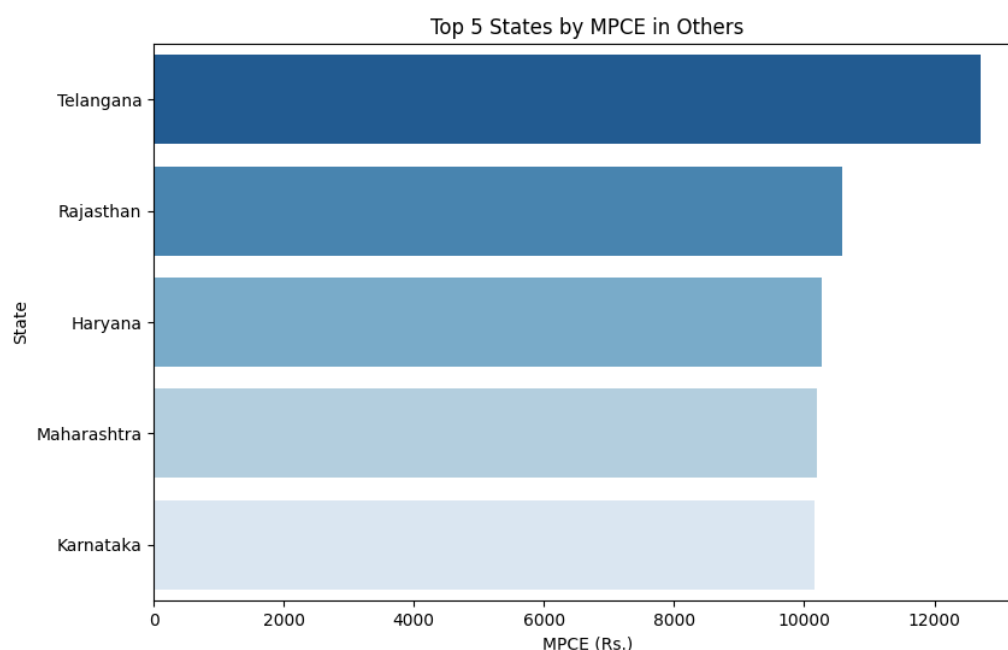


Figure 59: Top 5 States by MPCE in Others

16.9 Analysis of Bottom 5 States by MPCE in Others

- **Chart Overview:**

- The horizontal bar chart showcases the Monthly Per Capita Expenditure (MPCE) for the bottom five states in the "Others" category.
- States are listed from **Madhya Pradesh** to **Chhattisgarh**, with the x-axis representing MPCE in Indian Rupees (Rs.) from 0 to 8000.
- The y-axis labels the states, while the bar colors range from dark to light red, indicating expenditure levels.

- **Key Observations:**

- **Highest MPCE among Bottom 5 (Madhya Pradesh):**
 - * Madhya Pradesh leads this group with an MPCE close to 8000 Rs., suggesting relatively better expenditure conditions.
- **Lowest MPCE (Chhattisgarh):**
 - * Chhattisgarh registers the lowest MPCE, just above 6000 Rs., indicating significant expenditure challenges.
- **Regional Trends:**
 - * Northern and central states (e.g., Madhya Pradesh and Uttar Pradesh) exhibit higher MPCE within this group.
 - * Eastern states, such as Odisha and Bihar, reflect consistent low MPCE values, highlighting regional disparities.

Policy Implications

- **Targeted Economic Policies:**

- States like Chhattisgarh and Bihar need focused welfare programs and income stabilization initiatives to uplift MPCE in the "Others" category.

- **Infrastructure Investments:**

- Enhancing urban and rural infrastructure can provide equitable access to opportunities, increasing expenditure capacity.

- **Promoting Regional Equity:**

- Addressing regional disparities through economic development programs in low-MPCE states can ensure balanced growth.

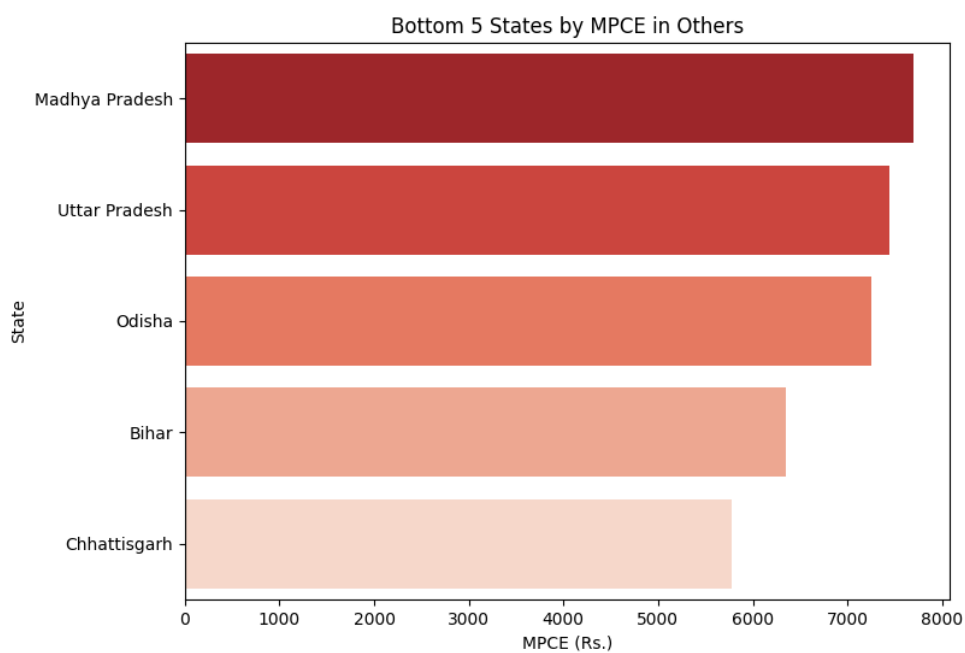


Figure 60: Bottom 5 States by MPCE in Others

16.10 Analysis of Top 5 States by MPCE in All

- **Chart Overview:**

- The chart highlights the Monthly Per Capita Expenditure (MPCE) across the top 5 states for all household types combined.
- States included are **Telangana, Haryana, Tamil Nadu, Karnataka, and Kerala**, listed in descending order based on MPCE values.
- The x-axis shows MPCE in Indian Rupees (Rs.), while the bars are shaded in varying tones of blue to represent expenditure levels.

- **Key Observations:**

- **Highest MPCE (Telangana):**
 - * Telangana has the highest MPCE, exceeding 8500 Rs., reflecting strong expenditure trends across all household categories.
- **Lowest MPCE (Kerala):**
 - * Kerala ranks fifth, with an MPCE below 7500 Rs., indicating stable but slightly lower spending compared to other top-performing states.
- **Regional Insights:**
 - * Southern states like Telangana, Tamil Nadu, Karnataka, and Kerala dominate the list, highlighting consistent economic advantages in the region.
 - * Haryana represents a strong contender from northern India, showcasing robust expenditure trends.

Policy Implications

- ****Encouraging High-MPCE States**:**
 - States with high MPCE, such as Telangana, could be studied for strategies to enhance economic growth across other regions.
- ****Fostering Regional Equality**:**
 - Bridging the gap between high-MPCE states and others can be achieved through targeted economic development and infrastructure investments.
- ****Strengthening Household Expenditure Nationwide**:**
 - Promoting balanced expenditure across regions and household types can ensure sustainable economic growth.

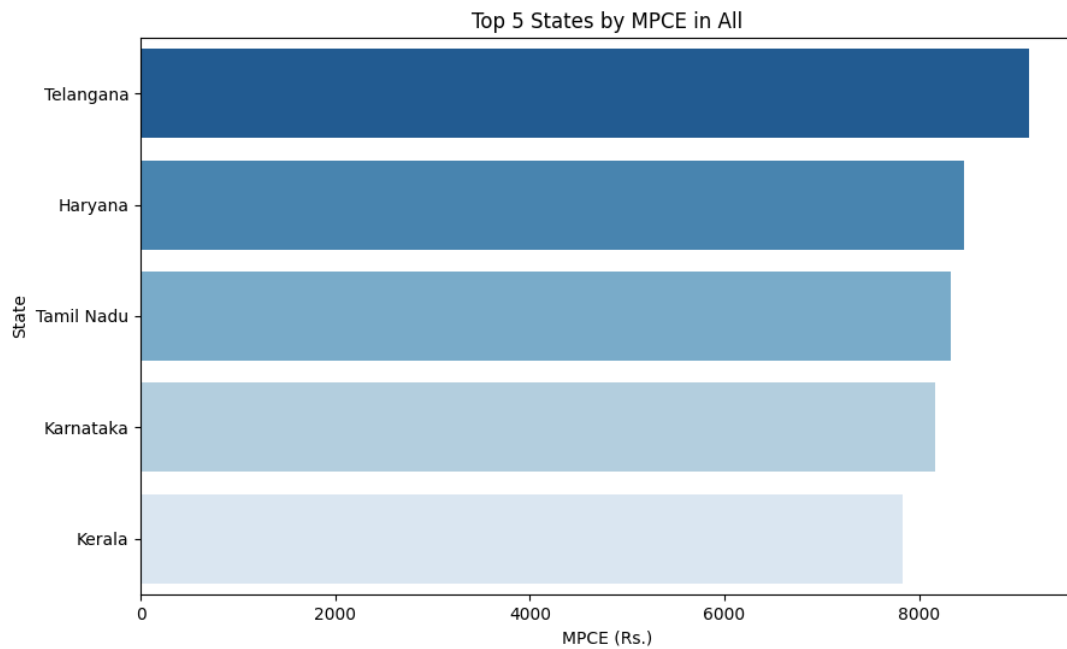


Figure 61: Top 5 States by MPCE in All

16.11 Analysis of Top 5 States by MPCE in All

- **Chart Overview:**

- The chart highlights the Monthly Per Capita Expenditure (MPCE) across the top 5 states for all household types combined.
- States included are **Telangana, Haryana, Tamil Nadu, Karnataka, and Kerala**, listed in descending order based on MPCE values.
- The x-axis shows MPCE in Indian Rupees (Rs.), while the bars are shaded in varying tones of blue to represent expenditure levels.

- **Key Observations:**

- **Highest MPCE (Telangana):**
 - * Telangana has the highest MPCE, exceeding 8500 Rs., reflecting strong expenditure trends across all household categories.
- **Lowest MPCE (Kerala):**
 - * Kerala ranks fifth, with an MPCE below 7500 Rs., indicating stable but slightly lower spending compared to other top-performing states.
- **Regional Insights:**
 - * Southern states like Telangana, Tamil Nadu, Karnataka, and Kerala dominate the list, highlighting consistent economic advantages in the region.
 - * Haryana represents a strong contender from northern India, showcasing robust expenditure trends.

Policy Implications

- ****Encouraging High-MPCE States**:**

- States with high MPCE, such as Telangana, could be studied for strategies to enhance economic growth across other regions.

- ****Fostering Regional Equality****:
 - Bridging the gap between high-MPCE states and others can be achieved through targeted economic development and infrastructure investments.
- ****Strengthening Household Expenditure Nationwide****:
 - Promoting balanced expenditure across regions and household types can ensure sustainable economic growth.

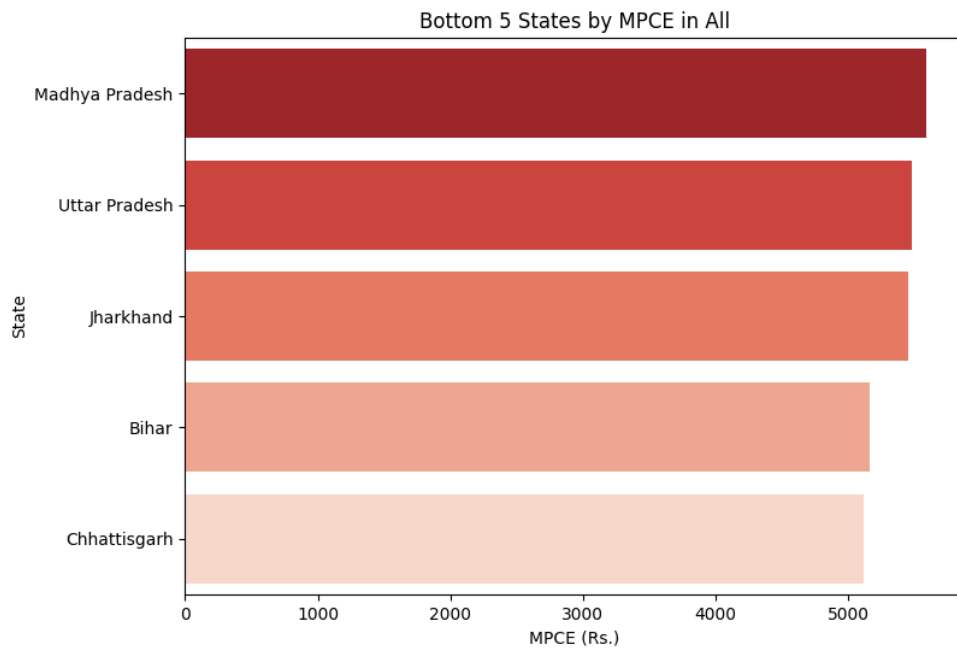


Figure 62: Top 5 States by MPCE in All

17 Analysis of Rural and Urban Gini Coefficients by State Source (Table 3.15: Gini coefficient of total consumption expenditure in 2011-12, 2022-23 and 2023-24, All-India) (Table 3.16: Gini coefficient of total consumption expenditure in 2023-24, major states)

- **Higher Urban Gini Coefficients:**

- Urban Gini coefficients (orange bars) exceed rural coefficients (purple bars) in most states, suggesting greater income inequality in urban areas.
- States like **Kerala** and **Punjab** display the largest urban-rural inequality gaps, highlighting significant economic disparities within these regions.

- **Relatively Equal States:**

- **Chhattisgarh** and **Rajasthan** exhibit smaller differences between rural and urban Gini coefficients, reflecting more balanced income distributions.

- **All-India Trend:**

- At the national level, urban areas generally report higher inequality compared to rural areas, as reflected in the larger Gini coefficients for urban settings.

Key Trends and Insights

- **Highest Urban Gini Coefficients:**

- States like **Kerala**, **Punjab**, and **Haryana** show the highest urban inequality, reflecting concentrated wealth disparities in urban areas.

- **Lowest Rural Gini Coefficients:**

- **Chhattisgarh**, **Assam**, and **Bihar** exhibit the least income inequality in rural regions, highlighting more equitable rural income distributions.

Policy Implications

- **Urban Challenges:**

- Address urban inequality through equitable economic opportunities, access to education, and affordable housing policies in high-inequality states like **Kerala** and **Punjab**.

- **Rural Stability:**

- Despite relatively low rural inequality, investment in rural infrastructure, skills development, and industries can further reduce poverty and improve quality of life.

- **Balanced Growth:**

- States like **Rajasthan** and **Chhattisgarh** can serve as models for achieving balanced urban-rural development, promoting inclusive policies nationwide.

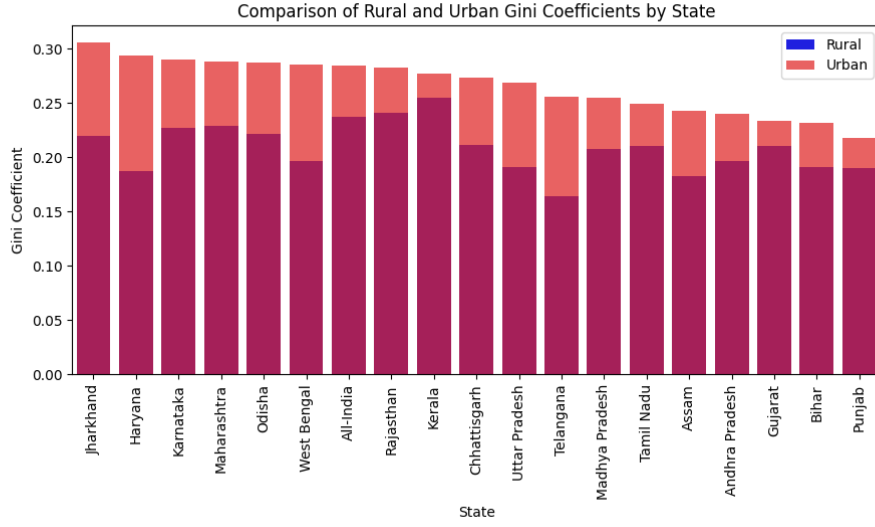


Figure 63: Analysis of Rural and Urban Gini Coefficients by State

Gini Coefficient Formula

$$G = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2\mu} \quad (1)$$

Where:

- G : Gini coefficient.
- n : Total number of observations.
- x_i and x_j : Individual income or expenditure observations.
- μ : Mean income or expenditure.

18 Analysis of Urban Non-Food Expenditure Distribution Across Indian States

• Categories of Non-Food Expenditure:

- The expenditure is categorized into:
 - * **Fuel & Light:** Spending on energy resources and utilities.
 - * **Conveyance:** Transportation expenses.
 - * **Education:** Costs for schooling, higher education, and resources.
 - * **Medical:** Healthcare expenses including medicines and treatments.
 - * **Miscellaneous Goods & Entertainment:** Leisure and general items.
 - * **Clothing, Bedding & Footwear:** Apparel and similar purchases.
 - * **Durable Goods:** Long-term assets like appliances and furniture.
 - * **Others:** Residual expenditures.

• Key State-Level Insights:

- **High-Spending States:**
 - * **Kerala & Tamil Nadu:** Significant allocation to **Education** and **Medical** expenses, reflecting high development indicators.

Urban Non-Food Expenditure Distribution

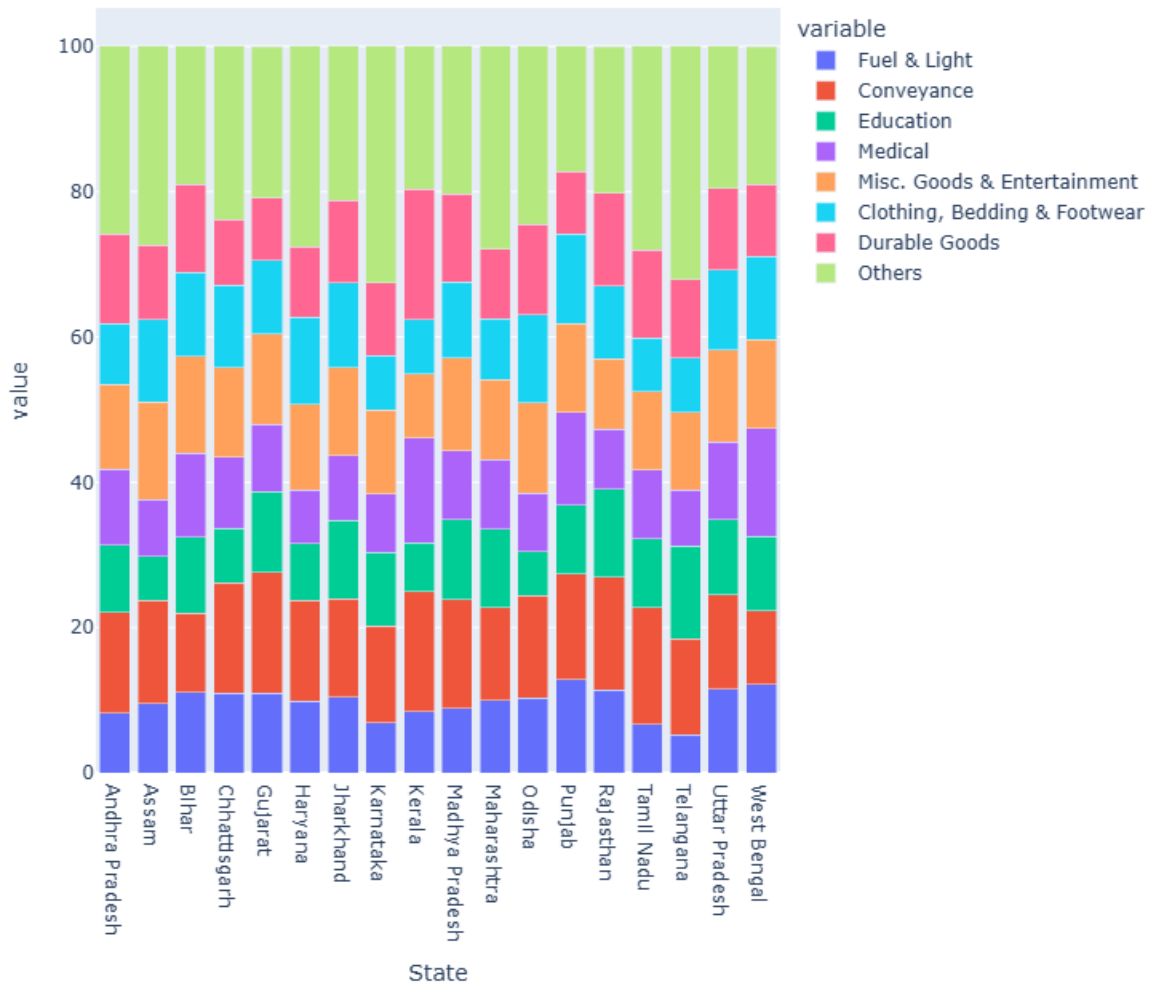


Figure 64: Urban Non-Food Expenditure Distribution Across Indian States

- * **Gujarat & Haryana:** Higher spending on **Miscellaneous Goods & Entertainment**, showcasing lifestyle-driven consumption patterns.
- **Low-Spending States:**
 - * **Bihar, Odisha, & Jharkhand:** Greater focus on **Fuel & Light**, highlighting basic living needs.
 - * **Chhattisgarh & Madhya Pradesh:** Spending leans toward healthcare and basic survival categories.
- **All-India Trends:**
 - **Fuel & Light:** A substantial share across all states, indicating the importance of energy and utilities.
 - **Education:** Higher spending in developed states like Kerala, while lagging in regions like Bihar.
 - **Medical:** Rising healthcare costs dominate household budgets universally.
 - **Durable Goods:** Significant variations, with advanced states allocating more compared to underdeveloped ones.

Policy Implications

- **Address Regional Disparities:**
 - Eastern states like Bihar and Odisha need targeted development in education and durable goods spending.
- **Foster Human Development:**
 - Public investment in **education** and **healthcare** can reduce inequalities and improve economic outcomes.
- **Encourage Balanced Growth:**
 - Southern states serve as models for balancing essential and lifestyle expenditures.

18.1 Analysis of Gini Coefficient Disparity (Urban - Rural)

- **Chart Overview:**
 - The bar chart represents the urban-rural Gini gap for various Indian states, quantifying disparities in income inequality.
 - The x-axis lists states such as Haryana, Telangana, West Bengal, Jharkhand, Uttar Pradesh, Odisha, and others.
 - The y-axis indicates the Gini coefficient difference, ranging from 0.00 to 0.10.
- **Observations:**
 - **Highest Urban-Rural Gap (Haryana):**
 - * Haryana has the largest Gini disparity (approximately 0.10), highlighting significant inequality in urban areas compared to rural ones.
 - **Lowest Urban-Rural Gap (Kerala):**
 - * Kerala shows the smallest Gini gap (slightly above 0.00), indicating relative equality between urban and rural income distributions.
 - * States like Punjab and Gujarat exhibit moderate urban-rural inequality gaps.
 - **All-India Average:**
 - * The national average urban-rural gap falls between 0.04 and 0.05, reinforcing the general trend of greater urban inequality.
- **Key Trends**
 - **Urban Dominance:**
 - * States with higher urban Gini coefficients face challenges in managing wealth concentration and access to resources.
 - **Rural Stability:**
 - * Rural regions generally display lower inequality due to smaller income ranges and subsistence-based economies.
- **Policy Implications**
 - **Targeted Programs for High-Gap States:**
 - * Initiatives like equitable urban development and skill-building programs could alleviate disparities in states like Haryana and West Bengal.
 - **Focus on Low-Gap States:**
 - * States like Kerala can maintain their balanced growth model, serving as a benchmark for equitable development.
 - **National-Level Strategies:**
 - * Policies fostering inclusive growth, such as investments in education, health, and infrastructure, can reduce the urban-rural divide across India.

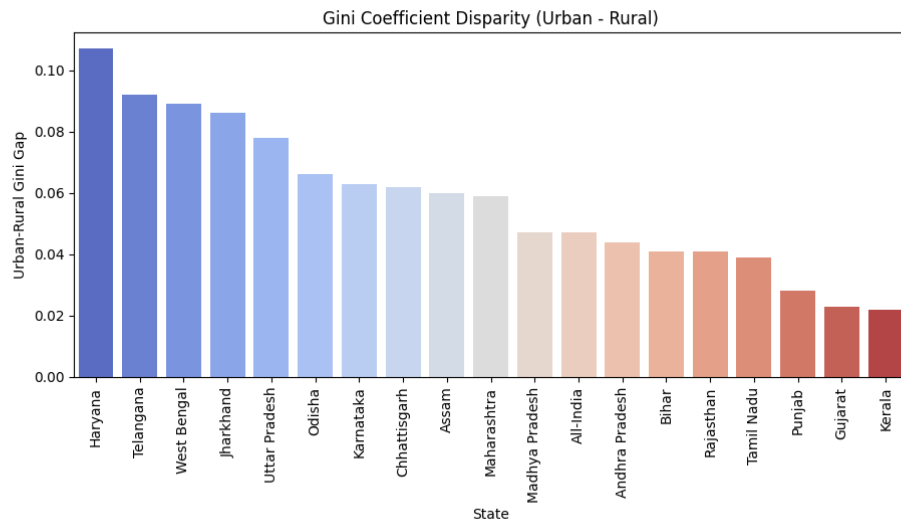


Figure 65: Analysis of Rural and Urban Gini Coefficients by State

18.2 Rural vs Urban Gini Coefficients by State

Plot Overview

- The scatter plot compares Rural Gini Coefficients (x-axis) with Urban Gini Coefficients (y-axis) for Indian states.
- The data points are color-coded to indicate different states, with labels for clarity.

General Observations

- Urban Gini coefficients are consistently higher than rural coefficients, reinforcing the trend that income inequality is greater in urban areas.
- States like **Kerala** and **Punjab** show higher overall Gini coefficients, indicating significant economic disparity in both rural and urban settings.
- **Chhattisgarh** and **Bihar** exhibit relatively low Gini coefficients, suggesting more equitable income distribution in these states.

Correlation

- A positive correlation is observed, meaning states with higher rural inequality also tend to have higher urban inequality.
- However, the disparity is more pronounced in urban areas as seen from the wider spread of data along the y-axis.

Key Insights

High Inequality States

- Kerala and Punjab show prominent disparities and require targeted policies to address inequality in both urban and rural areas.

Equitable Distribution States

- States like Chhattisgarh and Bihar may serve as examples for designing equitable income distribution policies, despite their lower overall economic output.

Urban Focus

- The consistently higher urban Gini coefficients indicate a need for better income redistribution mechanisms and inclusive urban development strategies.

Policy Implications

- **Focus on High-Inequality States:**
 - States with high Gini coefficients require targeted interventions, such as investment in education, healthcare, and job creation, to reduce income disparities.
- **Urban Development Policies:**
 - Developing affordable housing, skill training programs, and employment opportunities can help tackle urban inequality.
- **Addressing Rural Gaps:**
 - Promoting sustainable agricultural development and rural infrastructure can reduce rural income disparities.

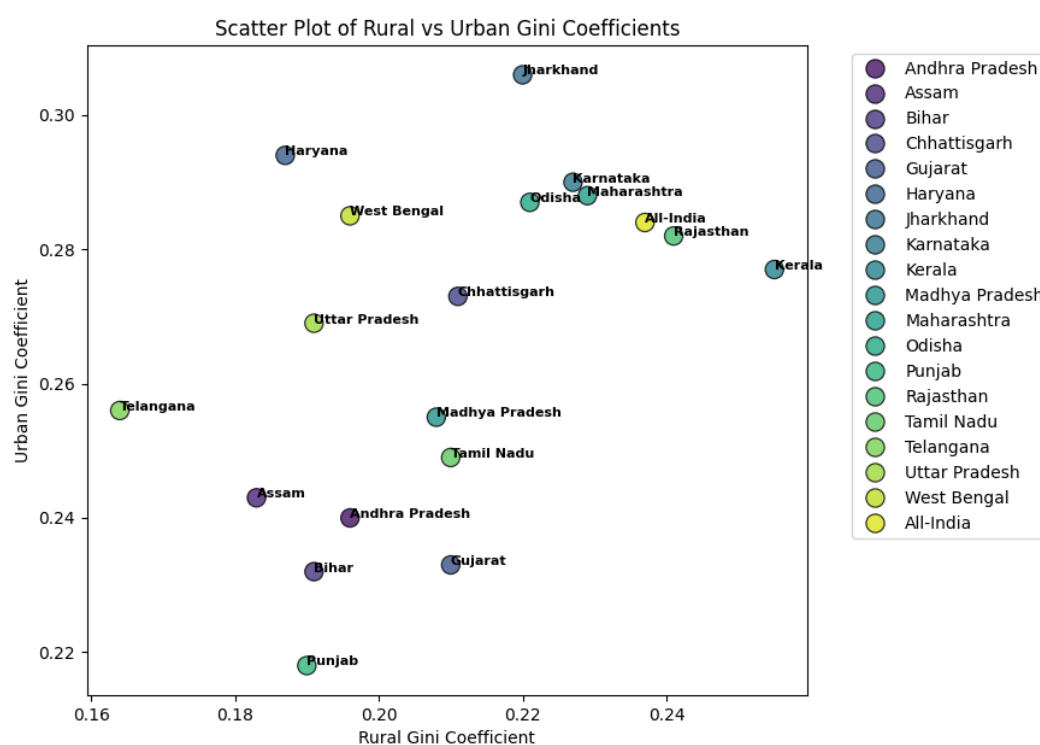


Figure 66: Rural vs Urban Gini Coefficients by State

18.3 Analysis of Urban Gini Coefficients

18.4 Top 5 States with Highest Urban Gini (Left Chart)

- Jharkhand, Haryana, Karnataka, Maharashtra, and Odisha exhibit the highest Urban Gini coefficients.
- These states show significant income inequality in urban areas, reflecting uneven wealth distribution and access to resources.

18.5 Bottom 5 States with Lowest Urban Gini (Right Chart)

- Punjab, Bihar, Gujarat, Andhra Pradesh, and Assam have the lowest Urban Gini coefficients.
- These states demonstrate more equitable income distribution in urban areas compared to others.

19 Key Observations

- **Inequality in High-Gini States:**
 - Urban income disparities in states like Jharkhand and Haryana may stem from uneven development, limited infrastructure, and disparities in job opportunities.
- **Equality in Low-Gini States:**
 - States such as Punjab and Assam reflect more balanced economic conditions, possibly due to inclusive policies and evenly distributed urban infrastructure.

20 Policy Implications

- **Address High Inequality:**
 - High-Gini states may benefit from programs targeting wealth redistribution, improved access to education, and economic development in urban areas.
- **Encourage Low-Gini States:**
 - Low-Gini states could serve as models for fostering inclusive economic policies that promote equity in urban settings.

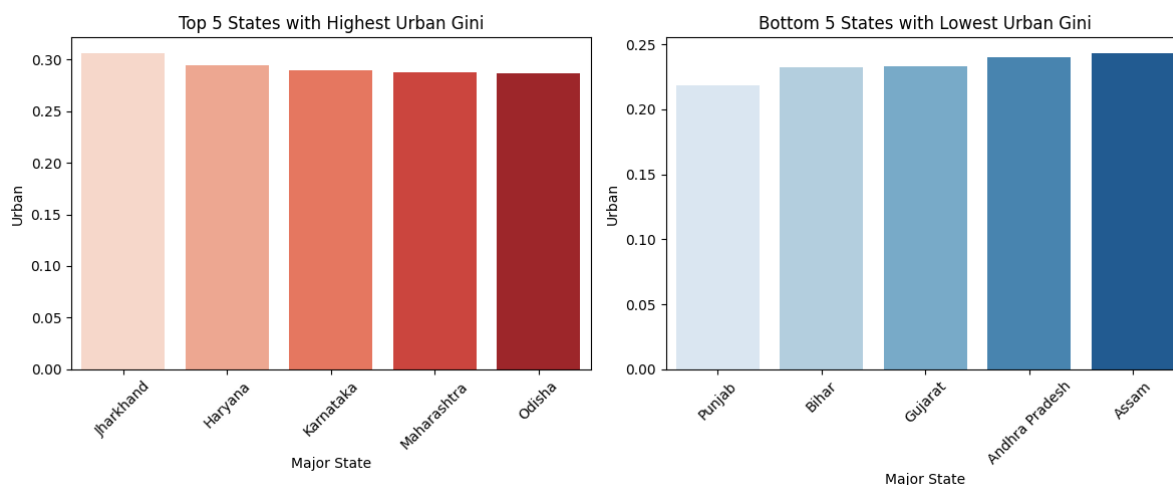


Figure 67: Urban Gini Coefficients in Top 5 and Bottom 5 States

21 Electricity Equity in Action: Sectoral Zero-Deprivation Achievements

India's commitment to providing universal, inclusive, and reliable electricity access has led to remarkable progress in several states and union territories (UTs). Despite this, electricity deprivation remains a challenge in certain regions, with varying degrees of intensity across rural (sector 1) and urban (sector 2) areas. While many states and UTs have made notable strides, there are still regions where a significant portion of households are deprived of electricity.

Rural Electrification: Reaching the Unreachable

States such as **Himachal Pradesh, Nagaland, Mizoram, Telangana, Goa, Ladakh, and Dadra & Nagar Haveli and Daman & Diu** have reported no rural households facing electricity deprivation. This achievement reflects substantial progress in overcoming geographical challenges through innovative, decentralized energy solutions, such as solar micro-grids and stand-alone systems.

However, states like **Meghalaya, Jharkhand, and Uttar Pradesh** continue to experience higher deprivation rates (approximately 4.35%, 1.75%, and 1.21%, respectively). These disparities highlight the need for targeted interventions to address localized barriers, such as remote geography, inadequate infrastructure, and socio-economic exclusion.

Urban Electrification: High Coverage with Room for Equity

Urban areas in **Delhi, Chandigarh, Sikkim, Jharkhand, Karnataka, and Puduchery** have achieved near-total electricity access (100%), reflecting the success of urban infrastructure development, grid connectivity, and efficient administrative outreach.

Nonetheless, deprivation persists in certain urban regions, with states like **Maharashtra, Bihar, and West Bengal** reporting modest deprivation rates (0.82%, 0.41%, and 0.85%, respectively) in some urban sectors. This suggests the presence of intra-urban inequality, particularly in slums, informal settlements, and peri-urban areas. These gaps often stem from issues such as land tenure security, bureaucratic challenges, and insufficient targeted urban policies.

Therefore, there is an urgent need for **equity-focused urban policies**, addressing the unique needs of underserved slum areas, informal settlements, and marginalized urban sectors. Key strategies should include ensuring formal electricity access, expanding last-mile connectivity, and addressing tenure security in informal housing.

Remote Regions Leading by Example

Smaller, geographically isolated regions such as **Lakshadweep, Arunachal Pradesh, and Ladakh** have successfully overcome geographic isolation to achieve near-universal electricity access. These regions serve as examples of how **tailored electrification policies**, including renewable energy solutions and community-based electrification models, can address access gaps even in the most remote contexts.

However, regions like **Madhya Pradesh (0.12%), Uttarakhand (0.18%), and Punjab (0.19%)** still report modest deprivation rates in rural areas, indicating the need for sustained and targeted efforts to address remaining gaps in electricity access, particularly in isolated or sparsely populated communities.

Policy Implications

- **Strengthen Rural Electrification in High-Deprivation States:** States such as **Meghalaya, Uttar Pradesh, and Jharkhand**—with rural deprivation rates exceeding 1%—should prioritize improving grid infrastructure and last-mile connectivity. Expanding investments in **solar micro-grids** and other renewable energy solutions suited to local conditions is essential to bridging the access divide.
- **Targeted Urban Policies to Address Intra-Urban Deprivation:** Although urban areas like **Delhi** and **Chandigarh** have reached near-total coverage, states like **Maharashtra, Bihar, and West Bengal** still face pockets of electricity deprivation. Addressing intra-urban inequalities will require **targeted urban policies**, with a focus on slums, informal settlements, and underserved

Table 10: State-wise Household Electricity Deprivation (HCES 2023-24)
Rural=1 and Urban=2

State	Sector	Total Households	Deprived Households	Deprivation Rate (%)
Meghalaya	1	563 295	24 510	4.35
Jharkhand	1	5 709 801	99 983	1.75
Uttar Pradesh	1	30 398 615	368 267	1.21
Tripura	1	804 896	9710	1.21
Gujarat	1	7 071 053	80 261	1.14
Chhattisgarh	1	4 591 188	49 844	1.09
Odisha	1	9 384 147	98 502	1.05
Assam	1	6 248 139	62 332	0.998
Rajasthan	1	10 936 974	96 940	0.886
West Bengal	1	16 976 256	144 533	0.851
Maharashtra	2	12 562 136	103 050	0.820
Manipur	2	183 691	1467	0.799
Andaman & N. Island	1	76 438	600	0.785
Manipur	1	499 367	3833	0.768
Tripura	2	201 873	1033	0.512
Andhra Pradesh	2	4 515 617	22 659	0.502
Maharashtra	1	14 768 694	70 186	0.475
Meghalaya	2	110 882	510	0.460
Haryana	2	2 554 196	11 289	0.442
Tamilnadu	2	10 686 204	44 057	0.412
Bihar	2	2 167 306	8855	0.409
Madhya Pradesh	2	4 484 483	16 287	0.363
Tamilnadu	1	11 176 960	39 823	0.356
West Bengal	2	8 004 246	27 761	0.347
Andaman & N. Island	2	50 536	156	0.309
Uttar Pradesh	2	9 075 344	27 668	0.305
D & N. Haveli & Daman & Diu	2	131 873	381	0.289
Puduchery	1	106 244	295	0.277
Gujarat	2	6 551 886	17 418	0.266
Assam	2	981 145	2360	0.241
Punjab	1	3 804 113	7127	0.187
Jammu & Kashmir	1	1 799 259	3312	0.184
Uttarakhand	1	1 856 434	3244	0.175
Arunachal Pradesh	1	217 653	377	0.173
Andhra Pradesh	1	9 468 709	16 205	0.171
Himachal Pradesh	2	243 984	380	0.156
Bihar	1	20 151 145	30 937	0.154
Telangana	2	5 407 536	7771	0.144
Odisha	2	2 193 468	2870	0.131
Rajasthan	2	4 247 935	5512	0.130
Madhya Pradesh	1	11 266 164	13 161	0.117
Delhi	2	3 643 667	3332	0.091
Kerala	1	4 886 446	4000	0.082
Sikkim	1	120 725	95	0.078
Haryana	1	3 627 657	2724	0.075
Chhattisgarh	2	1 332 890	959	0.072
Karnataka	1	9 300 573	4904	0.053
Kerala	2	4 925 091	2560	0.052
Punjab	2	2 744 353	815	0.030

Table 11: States and Sectors with Zero Electricity-Deprived Households

State	Sector	Total Households	Deprived Households
Jammu & Kashmir	2	566668	0
Himachal Pradesh	1	1557010	0
Chandigarh	1	11313	0
Chandigarh	2	242733	0
Uttarakhand	2	754955	0
Delhi	1	89607	0
Sikkim	2	57498	0
Arunachal Pradesh	2	62305	0
Nagaland	1	276945	0
Nagaland	2	120338	0
Mizoram	1	112840	0
Mizoram	2	102746	0
Jharkhand	2	1615380	0
D & N. Haveli & Daman & Diu	1	54438	0
Karnataka	2	6880675	0
Goa	1	134511	0
Goa	2	223872	0
Lakshadweep	1	3017	0
Lakshadweep	2	9344	0
Puduchery	2	240645	0
Telangana	1	5689698	0
Ladakh	1	43519	0
Ladakh	2	9736	0

urban communities. Policies must also address tenure security and the need for formal electricity access in these marginalized areas.

- **Leverage Renewable Energy Solutions for Remote Areas:** The successful electrification of remote regions like **Arunachal Pradesh**, **Lakshadweep**, and **Ladakh** highlights the potential of **decentralized renewable energy systems**. These models should be scaled up and adapted to other underserved, remote regions to ensure **universal access** to electricity, even in the most geographically challenging areas.
- **Ensure Equity in Electrification:** While coverage rates have improved, **equity** remains a crucial consideration. It is imperative that the government focus on reducing electricity deprivation rates in both rural and urban sectors, ensuring that **no household**—particularly in economically disadvantaged areas—is left behind. Achieving universal electricity access should be framed not only in terms of coverage but also in terms of ensuring **equal access** for all citizens, regardless of their socio-economic status or geographic location.

21.1 Electricity Deprivation: Mapping State- and Sector-Level Disparities

Table 12: State-wise and Sector-wise Distribution of Households with No Lighting Arrangement

State Name	Sector	Estimated Total HHs	Estimated No Lighting HHs	No Lighting Rate (%)
Jharkhand	1	5,709,801	76,700	1.34
Rajasthan	1	10,936,974	55,760	0.510
Manipur	2	183,691	770	0.419
Gujarat	1	7,071,053	25,120	0.355
Bihar	2	2,167,306	7,229	0.334
Haryana	2	2,554,196	7,486	0.293
Meghalaya	2	110,882	316	0.285
Odisha	1	9,384,147	25,217	0.269
Tamilnadu	2	10,686,204	20,760	0.194
Punjab	1	3,804,113	7,127	0.187
West Bengal	2	8,004,246	13,590	0.170
Uttar Pradesh	1	30,398,615	49,911	0.164
Maharashtra	1	14,768,694	22,016	0.149
Chhattisgarh	1	4,591,188	6,371	0.139
Gujarat	2	6,551,886	8,894	0.136
Uttar Pradesh	2	9,075,344	10,266	0.113
Madhya Pradesh	2	4,484,483	5,015	0.112
Jammu & Kashmir	1	1,799,259	1,656	0.0920
Tripura	2	201,873	178	0.0884
Andhra Pradesh	2	4,515,617	3,817	0.0845
Andhra Pradesh	1	9,468,709	7,537	0.0796
Sikkim	1	120,725	94.7	0.0784
Maharashtra	2	12,562,136	9,126	0.0726
Meghalaya	1	563,295	357	0.0634
Tripura	1	804,896	492	0.0611
Rajasthan	2	4,247,935	1,968	0.0463
Karnataka	1	9,300,573	3,078	0.0331
West Bengal	1	16,976,256	5,594	0.0330
Delhi	2	3,643,667	1,096	0.0301
Tamilnadu	1	11,176,960	2,596	0.0232
Kerala	1	4,886,446	993	0.0203
Bihar	1	20,151,145	2,574	0.0128
Telangana	2	5,407,536	633	0.0117
Madhya Pradesh	1	11,266,164	527	0.00468

Despite India’s remarkable achievements in expanding electricity access, a small yet significant number of households continue to experience extreme energy deprivation, as evidenced by the absence of any lighting arrangement. The classification of households with “*no lighting arrangement*” represents the most severe form of electricity exclusion—indicating a complete lack of access to both conventional and alternative sources of lighting. This indicator serves as a critical benchmark for identifying the deepest pockets of energy poverty, often concentrated in geographically remote or socio-economically marginalized regions. Addressing this residual deprivation is essential for realizing the goal of universal, equitable, and inclusive electrification.

Persistent Deprivation in Rural Areas of Key States

As shown in Table ??, rural sectors (Sector 1) in several states still report considerable shares of households with no lighting arrangement. Jharkhand stands out with a rural deprivation rate of 1.34%, followed by Rajasthan (0.51%), Gujarat (0.36%), and Odisha (0.27%). These figures indicate persistent structural gaps in infrastructure access, particularly in remote hamlets, forested regions, and tribal belts.

While national programs such as Saubhagya have expanded grid connectivity, these findings suggest that grid access alone does not guarantee service delivery. Barriers such as household-level affordability constraints, poor maintenance of infrastructure, or limited awareness may continue to obstruct meaningful access.

Urban Deprivation: Low but Unequal

Though overall urban deprivation is significantly lower, some urban areas still report measurable levels of electricity exclusion. For example, Bihar (urban deprivation: 0.33%), Haryana (0.29%), and West Bengal (0.17%) reflect urban inequality, possibly rooted in the presence of slums, unrecognized settlements, or undocumented migrant households without formal grid access. Maharashtra, despite its industrialized status, reports a rural deprivation rate of 0.15% and urban deprivation of 0.07%, highlighting intra-state disparity.

The persistence of electricity deprivation in urban areas calls for more inclusive planning frameworks that target tenure insecurity, bureaucratic bottlenecks in utility provisioning, and infrastructure deficits in informal settlements.

Zero Deprivation Regions: Models of Success

Several states and UTs—such as Himachal Pradesh, Chandigarh, Kerala, Sikkim, Nagaland, Arunachal Pradesh, and the Andaman & Nicobar Islands—report zero deprivation in both rural and urban sectors. These regions demonstrate that universal electrification is attainable even in geographically challenging contexts, when supported by localized energy planning and sustained public investment.

Their success often stems from a combination of factors: decentralized renewable solutions (e.g., solar micro-grids), strong local governance, and community-based models of energy service delivery.

Key Policy Insights

- **Targeted Rural Investment:** States with rural deprivation rates above 0.25%—notably Jharkhand, Rajasthan, Gujarat, and Odisha—require granular, household-level interventions. Last-mile electrification efforts should integrate renewable energy micro-solutions and leverage Gram Panchayat-level data to identify pockets of exclusion.
- **Addressing Urban Informality:** Urban electricity deprivation, though numerically smaller, often affects marginalized populations with weak formal entitlements. Policy focus must shift toward integrating slum areas into formal electricity networks, using innovative delivery mechanisms such as prepaid meters and community billing systems.
- **Replication of Success Models:** Zero-deprivation states provide replicable governance and technology models that can be adapted in high-deprivation regions. Cross-state knowledge transfers and central coordination could accelerate this process.
- **Equity as the New Metric:** Electrification success should not be measured merely by connectivity rates but by equitable and inclusive access. Households with no lighting arrangement represent the most severe form of energy poverty and must be prioritized in all future energy policies.

22 Evaluating the Pradhan Mantri Garib Kalyan Yojana (PMGKY): Coverage Rates and Regional Implications

State Name	Sector	Estimated Total Households	Estimated Total PMGKY Households	PMGKY_Coverage_Rate
Top 5				
Nagaland	1	276944.77	265184.1	95.75
Andhra Pradesh	1	9468708.6	8570151.7	90.51
Telangana	1	5689698.13	5080737.15	89.30
Chhattisgarh	1	4591188.06	4000220.03	87.13
Assam	1	6248139.01	4794848.88	76.74
Bottom 5				
Delhi	1	89607.17	0	0.00
Chandigarh	2	242732.94	960.18	0.40
Chandigarh	1	11312.84	56.10	0.50
Delhi	2	3643666.77	33303.68	0.91
Andaman & N. Island	2	50536.00	583.12	1.15

Table 13: Top 5 and Bottom 5 PMGKY Coverage Rates by State

The table presents the **Top 5** and **Bottom 5** states in terms of the **PMGKY (Pradhan Mantri Garib Kalyan Yojana) Coverage Rate**, which reflects the proportion of households benefiting from the PMGKY scheme relative to the total weighted households in each state. The coverage rate is a critical indicator of the effectiveness of the scheme in reaching the intended beneficiaries, particularly in the context of welfare distribution.

Top 5 States by PMGKY Coverage Rate

- **Nagaland** stands out with the highest coverage rate of **95.75%**, indicating that almost the entire targeted population in the state has received the benefits of the PMGKY scheme. This suggests a strong implementation and effective reach in this region.
- **Andhra Pradesh** follows closely with a coverage rate of **90.51%**, reflecting a highly successful implementation of the scheme in this state.
- **Telangana** and **Chhattisgarh** also show high coverage rates of **89.30%** and **87.13%**, respectively, demonstrating the substantial outreach of the PMGKY program in these states.
- **Assam**, with a coverage rate of **76.74%**, rounds out the top five, indicating that a significant portion of the population in Assam has benefited from the program, although there is still room for further improvements.

These states reflect successful implementation and high outreach, with most of the eligible households being included in the scheme.

Bottom 5 States by PMGKY Coverage Rate

- **Delhi** records the lowest coverage rate at **0.00%**. This likely points to either incomplete or absent implementation of the PMGKY scheme for a significant portion of Delhi's population, which warrants further investigation into the barriers to coverage in the national capital.
- **Chandigarh** (both Sector 1 and Sector 2) follows with coverage rates of **0.50%** and **0.40%** respectively, suggesting very limited reach of the scheme in the region. The stark contrast between Chandigarh's performance and other states highlights a need for targeted intervention to improve coverage in the union territory.
- **Andaman & Nicobar Islands**, with a coverage rate of **1.15%**, represents another region where PMGKY's reach appears minimal. Although slightly higher than Delhi and Chandigarh, it still suggests a lack of widespread access.
- The relatively lower coverage rates in these states (Delhi, Chandigarh, Andaman & Nicobar Islands) signal the necessity for enhancing program implementation and outreach, particularly in urban and remote regions where logistical challenges might be more prominent.

Policy Implications

The stark contrast in the **PMGKY Coverage Rates** across states highlights significant regional disparities in the implementation of the scheme. The states with higher coverage rates have likely benefitted from better infrastructure, governance, and administrative efficiency, whereas those with lower rates might be facing logistical challenges, such as insufficient distribution mechanisms or barriers to beneficiary identification.

To ensure more equitable distribution of welfare benefits, it is crucial for policymakers to focus on:

- **Strengthening outreach efforts** in states with lower coverage, particularly Delhi, Chandigarh, and Andaman & Nicobar Islands.
- **Improving administrative mechanisms**, such as targeting vulnerable populations and overcoming barriers to scheme accessibility.
- **Monitoring and evaluation** to assess the effectiveness of the program, ensuring that the benefits are reaching the intended recipients across all states.

22.1 No Cooking Arrangement Rates by State

Table 14: No Cooking Arrangement Rate for Major States

State Name	Sector	Est Total Households	Est PMGKY Households	No Cooking Arrangement Rate(%)
Top 5				
Telangana	2	5407536	1453080	26.87
Odisha	2	2193468	390149	17.79
Himachal Pradesh	2	243984	36306	14.88
Karnataka	2	6880675	990654	14.4
Tamilnadu	2	10686204	1176535	11.01
Bottom 5				
Jammu & Kashmir	1	1799259	0	0
Chandigarh	1	11313	0	0
Haryana	1	3627657	0	0
Delhi	1	89607	0	0
Arunachal Pradesh	1	217653	0	0

- **Telangana** (26.87%)
- **Odisha** (17.79%)
- **Himachal Pradesh** (14.88%)
- **Karnataka** (14.4%)
- **Tamil Nadu** (11.01%)

The high No Cooking Arrangement Rates in these states may indicate a significant portion of households either lack or are not utilizing cooking arrangements, which could be reflective of underlying socio-economic conditions. These findings warrant further investigation into factors like access to cooking fuel, affordability, and the effectiveness of government schemes in these regions.

For instance, **Telangana** stands out with the highest rate of 26.87%, suggesting a particular area of concern that may require targeted intervention to ensure households have access to basic cooking facilities. Similarly, **Odisha** and **Himachal Pradesh**, though with lower rates, show relatively high proportions of households without cooking arrangements, pointing to potential gaps in service delivery or availability of essential resources.

Bottom 5 States with the Lowest No Cooking Arrangement Rate

On the other hand, the states with the lowest No Cooking Arrangement Rates, indicating relatively better access to cooking arrangements, are:

- **Jammu & Kashmir** (0%)
- **Chandigarh** (0%)

- **Haryana** (0%)
- **Delhi** (0%)
- **Arunachal Pradesh** (0%)

Notably, these states have recorded a 0% rate, suggesting that, at least within the context of the data analyzed, there is no significant population without access to cooking arrangements. This could point to better infrastructure, targeted policy interventions, or more favorable socio-economic conditions in these regions. However, it is essential to verify if the zero rates are a reflection of accurate reporting or potential underreporting. Further investigation into local data collection methods and government schemes may be necessary to confirm these findings.

22.2 State-wise Rural Dependence on Firewood and Chips as Primary Cooking Energy Source in India

Table 15: State-wise Rural Usage of Firewood and Chips as Primary Cooking Energy Source

State Name	Sector	Estimated Households	Estimated Total Households	Percentage Share
Chhattisgarh	1	3,814,579	4,591,188	83.1
Meghalaya	1	438,670	563,295	77.9
West Bengal	1	13,108,669	16,976,256	77.2
Rajasthan	1	8,290,335	10,936,974	75.8
Odisha	1	7,105,336	9,384,147	75.7
Jharkhand	1	4,317,514	5,709,801	75.6
Tripura	1	578,654	804,896	71.9
Madhya Pradesh	1	7,964,373	11,266,164	70.7
Assam	1	3,977,712	6,248,139	63.7
Nagaland	1	175,068	276,945	63.2
Himachal Pradesh	1	918,412	1,557,010	59.0
Uttar Pradesh	1	14,802,041	30,398,615	48.7

A substantial proportion of rural households in several Indian states continue to rely on traditional biomass specifically firewood and chips as their primary source of cooking energy. States such as **Chhattisgarh (83.1%)**, **Meghalaya (77.9%)**, **West Bengal (77.2%)**, and **Rajasthan (75.8%)** report some of the highest usage rates, with over three-fourths of households in these regions dependent on firewood-based cooking.

Such reliance suggests enduring challenges in clean energy accessibility, despite the roll-out of national initiatives like the ***Pradhan Mantri Ujjwala Yojana (PMUY)***. Particularly in central and eastern India, the data points to regional disparities in the availability and affordability of LPG connections, as well as infrastructural limitations in last-mile delivery.

The use of firewood and chips is not only an indicator of energy poverty but also carries significant implications for public health, environmental sustainability, and gender equity. Exposure to indoor air pollution from biomass fuels increases the burden of respiratory diseases, while the time spent collecting firewood disproportionately affects women and children, limiting opportunities for education and employment.

These findings underscore the need for targeted, region-specific policy responses. State-level interventions such as enhanced LPG subsidy mechanisms, community-based awareness campaigns, and decentralized energy solutions could serve as effective levers in transitioning rural households toward cleaner cooking alternatives.

22.3 State-wise Estimation of LPG Coverage: Top and Bottom Performers in Household Access

The table presents the **Estimated LPG Coverage Rates** for the top and bottom five states in India, highlighting the proportion of households with access to liquefied petroleum gas (LPG) across different sectors (rural and urban). The **top five states**, namely **Chandigarh, Mizoram, Ladakh, Goa, and Delhi**, exhibit near-perfect LPG coverage rates, with all but Ladakh achieving 100% LPG coverage.

Table 16: Top and Bottom 5 States by Estimated LPG Coverage Rates

State Name	Sector	Total Households	LPG Households	Coverage (%)
Top 5 States				
Chandigarh	1	11312.84	11312.84	100.00
Mizoram	2	102745.61	102745.61	100.00
Ladakh	2	9736.46	9736.46	100.00
Ladakh	1	43307.49	43519.41	99.51
Goa	2	222301.59	223872.07	99.30
Bottom 5 States				
Jharkhand	1	979300.63	5709801.22	17.15
Odisha	1	1580619.08	9384147.19	16.84
Chhattisgarh	1	701863.28	4591188.06	15.29
West Bengal	1	3477749.54	16976255.74	20.49
Rajasthan	1	2505953.94	10936973.91	22.91

This suggests that these regions have effectively implemented policies or programs that ensure universal access to LPG, aligning with national objectives to promote cleaner cooking fuels.

Conversely, the **bottom five states—Jharkhand, Odisha, Chhattisgarh, West Bengal, and Rajasthan**—display significantly lower LPG coverage, ranging from 15.29% to 22.91%. This disparity underscores potential challenges in these regions, such as infrastructure limitations, affordability issues, or insufficient program outreach. These states may require targeted interventions to improve LPG penetration, especially in rural areas, in line with the **Pradhan Mantri Ujjwala Yojana** and other government initiatives aimed at increasing access to clean cooking fuel.

By highlighting the extremes in coverage, this analysis emphasizes the need for region-specific policy measures that focus on accessibility, affordability, and sustainability of LPG distribution systems. Further research could explore the socioeconomic and demographic factors that contribute to these disparities and propose strategies for enhancing coverage in lagging states.

- **Top performers:** 100% LPG coverage indicates successful implementation and access.
- **Bottom performers:** Highlight gaps in infrastructure and access that need attention.
- **Policy Implication:** Need for tailored interventions in lagging states.
- **National Impact:** Aligns with government efforts to provide cleaner cooking fuels for all.

Provides a critical insight into India's progress in ensuring equitable access to LPG, relevant for understanding national policy and the socio-economic implications of clean energy access.

22.4 Analysis of Ration Card Distribution and Coverage Gaps Sector and State-Level Examination

The data from HCES 2023–24 reveals a critical blind spot in the coverage of India’s food security infrastructure. An analysis of the percentage of households not possessing any form of ration card — including Antyodaya Anna Yojana (AAY), Below Poverty Line (BPL), Priority Households (PHH), State Food Security Scheme (SFSS), and others — highlights states and sectors where systemic exclusion from the Public Distribution System (PDS) remains alarmingly high.

The top five state–sector combinations with the highest percentage of uncarded households are

Table 17: Top 5 States by Proportion of Households Without a Health Card

State Name	Sector	(%) Households Without Ration Card
Chandigarh	Rural	81.1%
Chandigarh	Urban	79.8%
Delhi	Urban	74.2%
Meghalaya	Urban	66.4%
Delhi	Rural	65.0%

Chandigarh emerges as the most affected region, indicating near-total exclusion from the Public Distribution System (PDS), irrespective of settlement type. Key figures include

- 81.1% of rural households without any ration card
- 79.8% of urban households without any ration card

These alarming statistics highlight serious lapses in identification, outreach, and enrolment processes.

In Delhi, the situation is comparably dire, despite being the National Capital Territory with expectedly robust administrative mechanisms

- 74.2% of urban households report having no ration card
- 65.0% of rural households report having no ration card

This systemic undercoverage may be attributed to factors such as large-scale in-migration, urban informality, and documentation gaps among slum-dwellers and tenants.

Meghalaya’s urban sector also presents significant concerns, with 66.4% of households lacking ration cards. Potential impediments include topographical barriers, limited bureaucratic penetration, and digital connectivity constraints that hinder effective beneficiary registration and service delivery.

Note: A detailed table presenting the percentage of households without any ration card across all States and Union Territories, disaggregated by rural and urban sectors, is provided in the Appendix.

22.5 State-Level Dependence on State Food Security Schemes (SFSS)

The table presents the top five states with the highest Percentage of households dependent on State Food Security Schemes (SFSS), based on weighted estimates from the HCES 2023–24.

Table 18: Top 5 States by Household Dependence on State Food Security Schemes (SFSS)

State Name	Estimated SFSS Households	Estimated Total Households	SFSS Share (%)
West Bengal	6,637,859	24,980,502	26.60%
Telangana	1,679,019	11,097,234	15.10%
Punjab	826,673	6,548,466	12.60%
Uttarakhand	251,691	2,611,389	9.64%
Odisha	417,648	11,577,615	3.61%

- **West Bengal** records the highest household dependence on State Food Security Schemes (SFSS), with over **26.6%** of households accessing rations through state-funded provisions. This substantial figure translating to more than **6.6 million households** indicates that the state government is

playing a dominant role in ensuring food security, shouldering a responsibility that ideally should be more equitably distributed between central and state institutions.

- **Telangana** and **Punjab** follow with **15.1%** and **12.6%** coverage, respectively. These levels of dependence suggest that large sections of the population in these states may be underserved by centrally provided entitlements, compelling state governments to fill critical welfare gaps through independent schemes.
- In **Uttarakhand**, where **9.64%** of households are enrolled in SFSS, the burden on the state may stem from geographic isolation, logistical challenges, and the need for customised distribution networks, especially in rural and hilly regions.
- **Odisha's** relatively lower SFSS coverage of **3.61%** still amounts to a significant absolute burden when viewed in the context of the state's broader welfare portfolio and fiscal constraints.

23 Insights and Policy Implications

The findings from the analysis reveal significant variations in household non-food consumption patterns across Indian states, shaped by diverse economic and social factors. Key determinants such as income disparities, regional economic development, government interventions, and access to essential services significantly influence expenditure trends.

For high-performing states like Kerala and Tamil Nadu, the analysis emphasizes robust investments in education and healthcare, reflecting these states' commitment to fostering human capital and infrastructure. Their expenditure distribution indicates a focus on long-term socio-economic well-being. Conversely, states like Bihar, Odisha, and Jharkhand exhibit a larger allocation towards basic necessities such as fuel and light, which underscores the economic challenges faced by households in these regions.

Nationally, the rising healthcare expenditure and variations in spending on durable goods and education highlight the dual nature of household priorities: survival-driven spending versus aspirational consumption. Developed states appear to allocate resources toward lifestyle-oriented categories, while less-developed regions prioritize basic sustenance.

These insights provide a foundation for targeted policy recommendations aimed at improving economic welfare and household resilience. Addressing regional disparities through enhanced infrastructure, skill development, and inclusive growth strategies can uplift underperforming states. Moreover, sustained investment in education and healthcare across all regions can build resilience and foster equitable urban growth.

By addressing these factors holistically, policymakers can balance household consumption patterns, ensuring both immediate needs and long-term aspirations are met.

24 Conclusion and Recommendations

Conclusion: The analysis underscores the diverse expenditure patterns across rural and urban households, reflecting economic disparities and regional inequalities. Rural areas tend to allocate a larger share of their income to basic necessities like fuel and light, while urban households show a broader distribution, with significant spending on education, healthcare, and durable goods in more developed states. Inflation emerges as a critical factor influencing consumption behavior, especially in low-income groups where rising costs constrain spending on non-essential categories. Additionally, the role of government interventions in shaping household financial stability cannot be overstated. Well-designed policies have the potential to alleviate disparities and promote inclusive growth.

Recommendations:

- **Address Rural-Urban Disparities:**
 - Implement targeted welfare programs aimed at improving access to education and healthcare in rural areas.
 - Invest in rural infrastructure to reduce the dependency on basic necessity expenditure, freeing resources for long-term investments.
- **Mitigate the Effects of Inflation:**

- Strengthen price regulation mechanisms to protect lower-income groups from the adverse impacts of rising costs.
- Introduce subsidies for essential goods and services to reduce the financial burden on vulnerable households.

- **Promote Balanced Regional Development:**

- Encourage investments in education and skill development in underperforming states like Bihar, Jharkhand, and Odisha.
- Foster urbanization in a sustainable manner to ensure equitable access to opportunities and services.

- **Strengthen Government Policies:**

- Design policies that prioritize inclusive growth by addressing structural barriers to economic participation.
- Regularly assess and adapt policies to respond to changing economic conditions and household needs.

By implementing these recommendations, policymakers can bridge the gap between rural and urban expenditure patterns, enhance household resilience to inflation, and create a framework for equitable and sustainable economic growth.

25 Policy Recommendations

The findings from the analysis provide valuable insights into the socio-economic dynamics shaping household expenditure patterns across India. Based on these observations, the following policy recommendations are proposed to address key challenges such as income disparities, inflation impacts, and infrastructure gaps, while promoting inclusivity and sustainable development.

25.1 Strengthening Welfare Programs

The disparities in household expenditure patterns, particularly between rural and urban areas, highlight the need for targeted welfare initiatives to support lower-income households. Policies can focus on:

- **Direct Financial Assistance:** Cash transfer schemes can ensure immediate relief to vulnerable households, enabling them to meet essential expenditure needs such as food, healthcare, and education.
- **Subsidized Essential Services:** Providing subsidies for energy, education, and healthcare can alleviate economic burdens and enhance access to critical resources.
- **Job Creation Programs:** Initiatives focusing on skill-building, vocational training, and entrepreneurship development can empower low-income groups to secure sustainable livelihoods.

By strengthening welfare programs, policymakers can bridge income gaps and promote financial resilience across diverse demographics.

25.2 Monitoring Inflation Impacts

Inflation is a critical factor influencing household consumption behavior, particularly in lower-income groups where rising costs can disproportionately affect spending on non-essential categories. Key measures include:

- **Price Regulation Mechanisms:** Establishing monitoring systems to track inflation trends in essential commodity markets can help implement effective price controls.
- **Affordable Pricing Frameworks:** Regulatory measures ensuring stable prices for food, fuel, and medical supplies can safeguard vulnerable households against economic shocks.
- **Public Awareness Campaigns:** Educating citizens about inflation-related developments and providing guidance on budgeting and financial planning can enhance household resilience.

By addressing the impacts of inflation, policymakers can create a more stable economic environment conducive to balanced consumption patterns.

25.3 Enhancing Digital Payment Infrastructure

Digital payment systems have become integral to modern economies, offering convenience, transparency, and security in financial transactions. Expanding this infrastructure can promote financial inclusion and efficiency. Recommended actions include:

- **Expanding Access to Digital Platforms:** Investment in digital payment systems across urban and rural areas can ensure ease of access for all households, reducing dependency on cash transactions.
- **Financial Literacy Programs:** Providing educational resources on using digital payment platforms can empower households to transition to a cashless economy.
- **Collaborations with Financial Institutions:** Partnering with banks and technology providers to improve the reliability, security, and user experience of digital payment systems can accelerate adoption rates.

Enhanced digital infrastructure can simplify financial transactions, reduce transaction costs, and foster greater economic participation.

25.4 Addressing Regional Disparities

The stark differences in expenditure patterns between developed and underdeveloped states necessitate targeted interventions to promote balanced regional growth. Policymakers can focus on:

- **Investments in Infrastructure:** Enhancing rural and urban infrastructure in underperforming states like Bihar and Odisha can improve access to education, healthcare, and employment opportunities.
- **Skill Development and Education:** Introducing skill-building programs and expanding educational facilities in economically disadvantaged regions can foster upward mobility.
- **Promoting Entrepreneurship:** Creating a supportive ecosystem for micro-enterprises and small businesses can boost local economies and reduce dependency on external aid.

Addressing regional disparities through these initiatives can bridge the gap between high-performing and underperforming states.

25.5 Fostering Sustainable Urban Development

Urban areas often exhibit higher expenditure levels in categories such as healthcare, education, and lifestyle-driven goods. To ensure this growth is inclusive and sustainable, policymakers can implement:

- **Affordable Housing Programs:** Expanding access to affordable housing can mitigate the cost of living in urban centers and improve financial stability for low-income households.
- **Enhanced Public Services:** Investments in urban infrastructure, including public transportation, healthcare facilities, and educational institutions, can support balanced urban growth.
- **Smart Urban Planning:** Leveraging technology and data-driven approaches to plan urban development can ensure equitable access to resources and services.

Sustainable urban development strategies can enhance the quality of life for all urban residents while minimizing environmental and social risks.

25.6 Strengthening Government Policies

Effective government policies play a pivotal role in shaping household financial stability and consumption patterns. To maximize their impact, policymakers should consider:

- **Inclusive Growth Strategies:** Designing policies that address structural barriers to economic participation, such as discrimination and unequal access to resources, can foster inclusivity.
- **Periodic Policy Assessments:** Regular evaluations of policy effectiveness can help adapt interventions to evolving economic conditions and household needs.

- **Collaborative Policy Design:** Engaging stakeholders, including local governments, NGOs, and community representatives, in policy formulation can ensure initiatives are aligned with ground realities.

Strong government policies can lay the foundation for long-term economic stability and social well-being.

26 Appendix

26.1 Estimated Total Households PMGKY Coverage Rates by State

Table 19: Estimated Total Households PMGKY Coverage Rates by State

State Name	Sector	Estimated Total Households	Estimated PMGKY Households	PMGKY Coverage Rate
Delhi	1	89607.17	0	0.00
Chandigarh	2	242732.94	960.18	0.40
Chandigarh	1	11312.84	56.10	0.50
Delhi	2	3643666.77	33303.68	0.91
Andaman & N. Island	2	50536.00	583.12	1.15
Himachal Pradesh	2	243984.11	6504.44	2.67
Mizoram	2	102745.61	4237.84	4.12
Mizoram	1	112840.32	5784.21	5.13
Himachal Pradesh	1	1557009.68	163361.80	10.49
Punjab	2	2744353.4	304035.77	11.08
Andaman & N. Island	1	76437.63	8504.69	11.13
Meghalaya	2	110882.15	13117.21	11.83
Jammu & Kashmir	2	566667.93	68092.14	12.02
D & N. Haveli & Daman & Diu	2	131872.64	20967.97	15.90
Haryana	2	2554195.86	415768.63	16.28
Ladakh	2	9736.46	1728.85	17.76
Maharashtra	2	12562136.12	2377529.01	18.93
Gujarat	2	6551886.4	1288341.34	19.66
Arunachal Pradesh	2	62305.06	14115.17	22.65
Sikkim	2	57497.87	13695.65	23.82
Uttarakhand	2	754954.96	182106.8	24.12
Punjab	1	3804112.58	943878.26	24.81
Rajasthan	2	4247934.51	1119608.87	26.36
Ladakh	1	43519.41	11600.28	26.66
Goa	2	223872.07	63983.15	28.58
Puduchery	2	240644.51	71882.81	29.87
Madhya Pradesh	2	4484482.51	1427289.05	31.83
Jharkhand	2	1615379.51	517381.91	32.03
Assam	2	981145.46	314833.53	32.09
Jammu & Kashmir	1	1799259.39	603183.97	33.52
Karnataka	2	6880674.79	2338623.25	33.99
Manipur	2	183690.58	63298.12	34.46
Haryana	1	3627657.33	1251326.07	34.49
Kerala	2	4925091.17	1770124.57	35.94
Tamilnadu	2	10686203.56	3865701.57	36.17
Odisha	2	2193467.79	798064.55	36.38
Lakshadweep	1	3016.77	1139.68	37.78
West Bengal	2	8004246.47	3036925.05	37.94
Lakshadweep	2	9343.77	3823.12	40.92
Puduchery	1	106244.13	43590.3	41.03
Uttar Pradesh	2	9075343.67	3908988.67	43.07
Tamilnadu	1	11176960.22	4857665.36	43.46
Kerala	1	4886446.27	2198356.7	44.99
Bihar	2	2167306.01	986182.09	45.50
Manipur	1	499367.07	228367.02	45.73
Meghalaya	1	563295.34	263967.62	46.86
Tripura	2	201872.51	98071.52	48.58
Telangana	2	5407535.9	2722447.07	50.35
Uttarakhand	1	1856433.92	998905.64	53.81
Goa	1	134511.13	72384.36	53.81
Chhattisgarh	2	1332889.98	776042.61	58.22
Arunachal Pradesh	1	217652.92	129196.71	59.36
Gujarat	1	7071052.71	4333705.83	61.29
Madhya Pradesh	1	11266164.09	7111959.84	63.13
Maharashtra	1	14768693.59	9347707.35	63.29
West Bengal	1	16976255.74	10866517.28	64.01
Odisha	1	9384147.19	6047324.5	64.44
Rajasthan	1	10936973.91	7277146.14	66.54
Jharkhand	1	5709801.22	3827899.12	67.04
Sikkim	1	120725.43	82332.13	68.20
D & N. Haveli & Daman & Diu	1	54438.21	37310.99	68.54
Tripura	1	804895.95	565920.69	70.31
Karnataka	1	9300572.68	6612606.94	71.10
Bihar	1	20151145.33	14419793.53	71.56
Andhra Pradesh	2	4515617.41	3243661.63	71.83
Uttar Pradesh	1	30398614.51	22013180.67	72.42
Nagaland	2	120338.14	87964.07	73.10
Assam	1	6248139.01	4794848.88	76.74
Chhattisgarh	1	4591188.06	4000220.03	87.13
Telangana	1	5689698.13	5080737.15	89.30
Andhra Pradesh	1	9468708.6	8570151.7	90.51
Nagaland	1	276944.77	265184.1	95.75

26.2 Estimated Total Households No Cooking Arrangement Rate by State

Table 20: Estimated Total Households No Cooking Arrangement Rate by State

State Name	Sector	Est Total Households	Est PMGKY Households	No Cooking Arrangement Rate
Telangana	2	5407536	1453080	26.87
Odisha	2	2193468	390149	17.79
Himachal Pradesh	2	243984	36306	14.88
Karnataka	2	6880675	990654	14.4
Tamilnadu	2	10686204	1176535	11.01
Andaman & N. Island	1	76438	6238	8.16
Andhra Pradesh	2	4515617	364725	8.08
Puduchery	2	240645	17957	7.46
Sikkim	2	57498	4113	7.15
Lakshadweep	2	9344	636	6.81
Odisha	1	9384147	599548	6.39
D & N. Haveli & Daman & Diu	2	131873	7727	5.86
Rajasthan	2	4247935	244859	5.76
Kerala	2	4925091	274733	5.58
Maharashtra	2	12562136	642970	5.12
Gujarat	2	6551886	322761	4.93
Chhattisgarh	2	1332890	54675	4.1
Lakshadweep	1	3017	108	3.58
D & N. Haveli & Daman & Diu	1	54438	1725	3.17
Andhra Pradesh	1	9468709	294600	3.11
West Bengal	2	8004246	239068	2.99
Jharkhand	2	1615380	44436	2.75
Telangana	1	5689698	145289	2.55
Meghalaya	2	110882	2417	2.18
Delhi	2	3643667	73018	2
Manipur	2	183691	3614	1.97
Haryana	2	2554196	49370	1.93
Tamilnadu	1	11176960	205642	1.84
Kerala	1	4886446	88277	1.81
Maharashtra	1	14768694	238181	1.61
Chandigarh	2	242733	3879	1.6
Uttarakhand	2	754955	11608	1.54
Jammu & Kashmir	2	566668	8251	1.46
Uttar Pradesh	2	9075344	131205	1.45
Karnataka	1	9300573	134053	1.44
Assam	2	981145	12011	1.22
Chhattisgarh	1	4591188	55232	1.2
Punjab	2	2744353	32355	1.18
Bihar	2	2167306	23368	1.08
Madhya Pradesh	2	4484483	47335	1.06
Puduchery	1	106244	1079	1.02
Uttarakhand	1	1856434	15881	0.86
Rajasthan	1	10936974	78872	0.72
Himachal Pradesh	1	1557010	10890	0.7
Jharkhand	1	5709801	32299	0.57
West Bengal	1	16976256	95077	0.56
Gujarat	1	7071053	39115	0.55
Goa	2	223872	1178	0.53
Punjab	1	3804113	19087	0.5
Sikkim	1	120725	350	0.29
Arunachal Pradesh	2	62305	171	0.27
Uttar Pradesh	1	30398615	39317	0.13
Goa	1	134511	179	0.13
Madhya Pradesh	1	11266164	3439	0.03
Assam	1	6248139	1213	0.02
Bihar	1	20151145	2564	0.01
Jammu & Kashmir	1	1799259	0	0
Chandigarh	1	11313	0	0
Haryana	1	3627657	0	0
Delhi	1	89607	0	0
Arunachal Pradesh	1	217653	0	0
Nagaland	1	276945	0	0
Nagaland	2	120338	0	0
Manipur	1	499367	0	0
Mizoram	1	112840	0	0
Mizoram	2	102746	0	0
Tripura	1	804896	0	0
Tripura	2	201873	0	0
Meghalaya	1	563295	0	0
Andaman & N. Island	2	50536	0	0
Ladakh	1	43519	0	0
Ladakh	2	9736	0	0

26.3 Estimated Total Households LPG Coverage Rates by State

Table 21: Estimated Total Households LPG Coverage Rates by State

State_Name	Sector	Estimated total Households	Estimated LPG Households	Percentage_Share
Chandigarh	1	11312.84	11312.84	100
Mizoram	2	102745.61	102745.61	100
Ladakh	2	9736.46	9736.46	100
Ladakh	1	43307.49	43519.41	99.51
Goa	2	222301.59	223872.07	99.3
Delhi	1	88914.49	89607.17	99.23
Chandigarh	2	237689.28	242732.94	97.92
Andaman & N. Island	2	49321.68	50536	97.6
Goa	1	130719.98	134511.13	97.18
Punjab	2	2664312.6	2744353.4	97.08
Arunachal Pradesh	2	60360.17	62305.06	96.88
Manipur	2	177480.82	183690.58	96.62
Telangana	1	5421699.89	5689698.13	95.29
Haryana	2	2419726.68	2554195.86	94.74
Uttarakhand	2	713697.93	754954.96	94.54
Assam	2	925233.65	981145.46	94.3
Jammu & Kashmir	2	532860.7	566667.93	94.03
Nagaland	2	112028.46	120338.14	93.09
Mizoram	1	105018.27	112840.32	93.07
Sikkim	2	53385.21	57497.87	92.85
Puduchery	2	221220.4	240644.51	91.93
Puduchery	1	96544.03	106244.13	90.87
Madhya Pradesh	2	4071363.87	4484482.51	90.79
Uttar Pradesh	2	8204801.52	9075343.67	90.41
Andhra Pradesh	2	4050122.78	4515617.41	89.69
D & N. Haveli & Daman & Diu	2	114753.56	131872.64	87.02
Andhra Pradesh	1	8228560.55	9468708.6	86.9
Bihar	2	1868683.16	2167306.01	86.22
Tamilnadu	2	9172547.74	10686203.56	85.84
Rajasthan	2	3636890.5	4247934.51	85.62
Maharashtra	2	10749947.44	12562136.12	85.57
Delhi	2	3097320.29	3643666.77	85.01
Punjab	1	3199396.36	3804112.58	84.1
Karnataka	2	5785882.63	6880674.79	84.09
Andaman & N. Island	1	64151.39	76437.63	83.93
West Bengal	2	6638326.08	8004246.47	82.94
Himachal Pradesh	2	200040.96	243984.11	81.99
Karnataka	1	7581340.27	9300572.68	81.51
Tamilnadu	1	9096379.58	11176960.22	81.39
Chhattisgarh	2	1083113.67	1332889.98	81.26
Sikkim	1	96583.73	120725.43	80
Meghalaya	2	86748.69	110882.15	78.24
Manipur	1	388489.92	499367.07	77.8
Jharkhand	2	1252909.7	1615379.51	77.56
Lakshadweep	2	7155.13	9343.77	76.58
Kerala	2	3647419.71	4925091.17	74.06
Telangana	2	3945185.09	5407535.9	72.96
Tripura	2	139581.87	201872.51	69.14
Maharashtra	1	10146540.03	14768693.59	68.7
Jammu & Kashmir	1	1188112.32	1799259.39	66.03
Gujarat	2	4289477.94	6551886.4	65.47
Odisha	2	1392734.45	2193467.79	63.49
Gujarat	1	4327019.73	7071052.71	61.19
Haryana	1	2156350.8	3627657.33	59.44
Kerala	1	2856582.56	4886446.27	58.46
Arunachal Pradesh	1	126248.95	217652.92	58
Uttarakhand	1	1065581.84	1856433.92	57.4
Lakshadweep	1	1705.35	3016.77	56.53
D & N. Haveli & Daman & Diu	1	27384	54438.21	50.3
Uttar Pradesh	1	13504784.46	30398614.51	44.43
Himachal Pradesh	1	621849.45	1557009.68	39.94
Bihar	1	7595293.67	20151145.33	37.69
Nagaland	1	101876.52	276944.77	36.79
Assam	1	2269213.79	6248139.01	36.32
Madhya Pradesh	1	3204506.69	11266164.09	28.44
Tripura	1	226241.59	804895.95	28.11
Rajasthan	1	2505953.94	10936973.91	22.91
West Bengal	1	3477749.54	16976255.74	20.49
Meghalaya	1	110745.93	563295.34	19.66
Jharkhand	1	979300.63	5709801.22	17.15
Odisha	1	1580619.08	9384147.19	16.84
Chhattisgarh	1	701863.28	4591188.06	15.29

Table 22: Percentage of Households Without Any type Ration Card by State and Sector

State Name	Sector	Percentage (%)
Chandigarh	1	81.1
Chandigarh	2	79.8
Delhi	2	74.2
Meghalaya	2	66.4
Delhi	1	65.0
Sikkim	2	59.1
Punjab	2	55.5
Assam	2	55.4
Odisha	2	53.2
D & N. Haveli & Daman & Diu	2	50.4
Madhya Pradesh	2	50.3
Jharkhand	2	49.6
Arunachal Pradesh	2	48.7
Haryana	2	47.5
Uttarakhand	2	47.2
Bihar	2	47.2
Uttar Pradesh	2	45.9
Himachal Pradesh	2	45.8
Karnataka	2	45.7
Telangana	2	39.8
Maharashtra	2	32.7
Goa	2	28.6
Meghalaya	1	27.8
Manipur	2	27.2
Punjab	1	26.7
Nagaland	2	23.2
Gujarat	2	23.1
Manipur	1	22.3
Andhra Pradesh	2	21.9
Bihar	1	21.8
Madhya Pradesh	1	19.5
Chhattisgarh	2	19.0
Tamilnadu	2	18.8
Arunachal Pradesh	1	18.7
Uttar Pradesh	1	17.0
D & N. Haveli & Daman & Diu	1	16.4
Haryana	1	15.8
Rajasthan	2	15.4
Odisha	1	15.2
West Bengal	2	14.5
Goa	1	13.9
Uttarakhand	1	13.1
Assam	1	12.1
Jharkhand	1	10.5
Maharashtra	1	10.3
Sikkim	1	10.2
Ladakh	2	8.86
Lakshadweep	2	8.36
Karnataka	1	7.4
Jammu & Kashmir	2	7.12
Kerala	2	6.61
Tamilnadu	1	5.58
Telangana	1	5.07
Himachal Pradesh	1	4.32

State Name	Sector	Precentage (%)
Kerala	1	3.46
Andhra Pradesh	1	3.37
Andaman & N. Island	2	3.04
Puduchery	2	2.95
Gujarat	1	2.91
Nagaland	1	2.87
Puduchery	1	2.60
Tripura	1	2.28
Rajasthan	1	2.23
Andaman & N. Island	1	2.19
Chhattisgarh	1	2.11
Tripura	2	1.78
Lakshadweep	1	1.69
Mizoram	2	1.64
Jammu & Kashmir	1	1.61
Mizoram	1	1.11
West Bengal	1	0.65
Ladakh	1	0.32

27 References

- **Household Consumer Expenditure Survey (HCES):** Ministry of Statistics and Programme Implementation (MOSPI). Official page available at: <https://mospi.gov.in/>
- **HCES Unit-Level Data:** Access and download the unit-level data for the Household Consumer Expenditure Survey (HCES) provided by MOSPI at: <https://microdata.gov.in/NADA/index.php/catalog/226>